Calculus Fundamentals: Differentiation

Lecture 4

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welcome to calculus

- calculus allows us to deal with continuity in a consistent and productive way
- differentiation is the process of taking a derivative

notation

- with one variable:

"f prime x" $f'(x)$

- with multiple variables:

"the derivative of f of x with respect to x" $\frac{df(x)}{dx}$ or $\frac{\partial f(x, y)}{\partial x}$

the paradox

Discrete change

- change between two measures of a concept at two distinct, discrete moments in time
  → first difference between two observations over a discrete interval

Instantaneous change

- change at a specific point in time
- derivative of function *f(x)* with respect to *x* tells us the **instantaneous rate of change** of the function at each point

(First) derivative

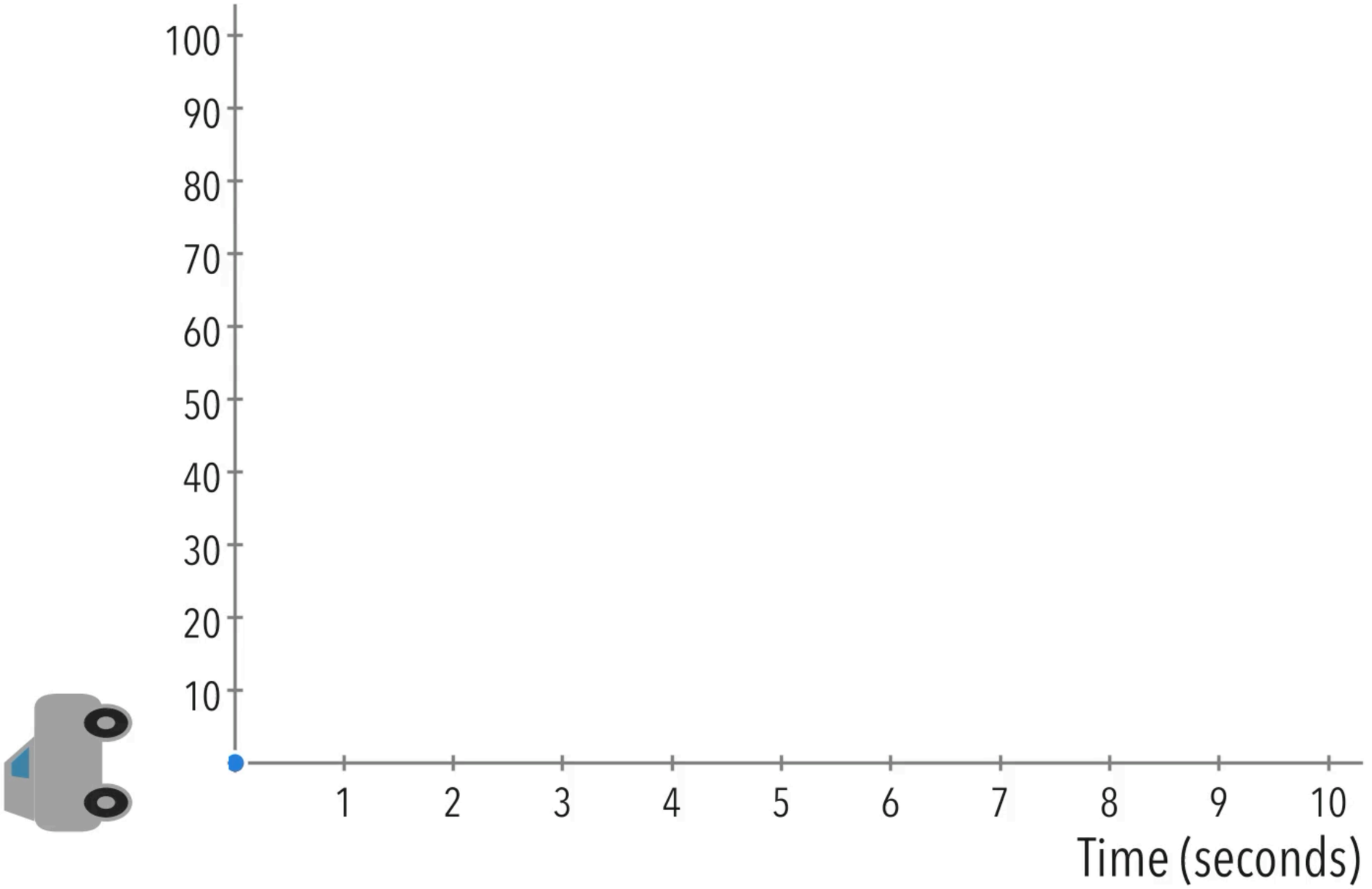
- describes reactivity to change in function's output based on input argument *x*

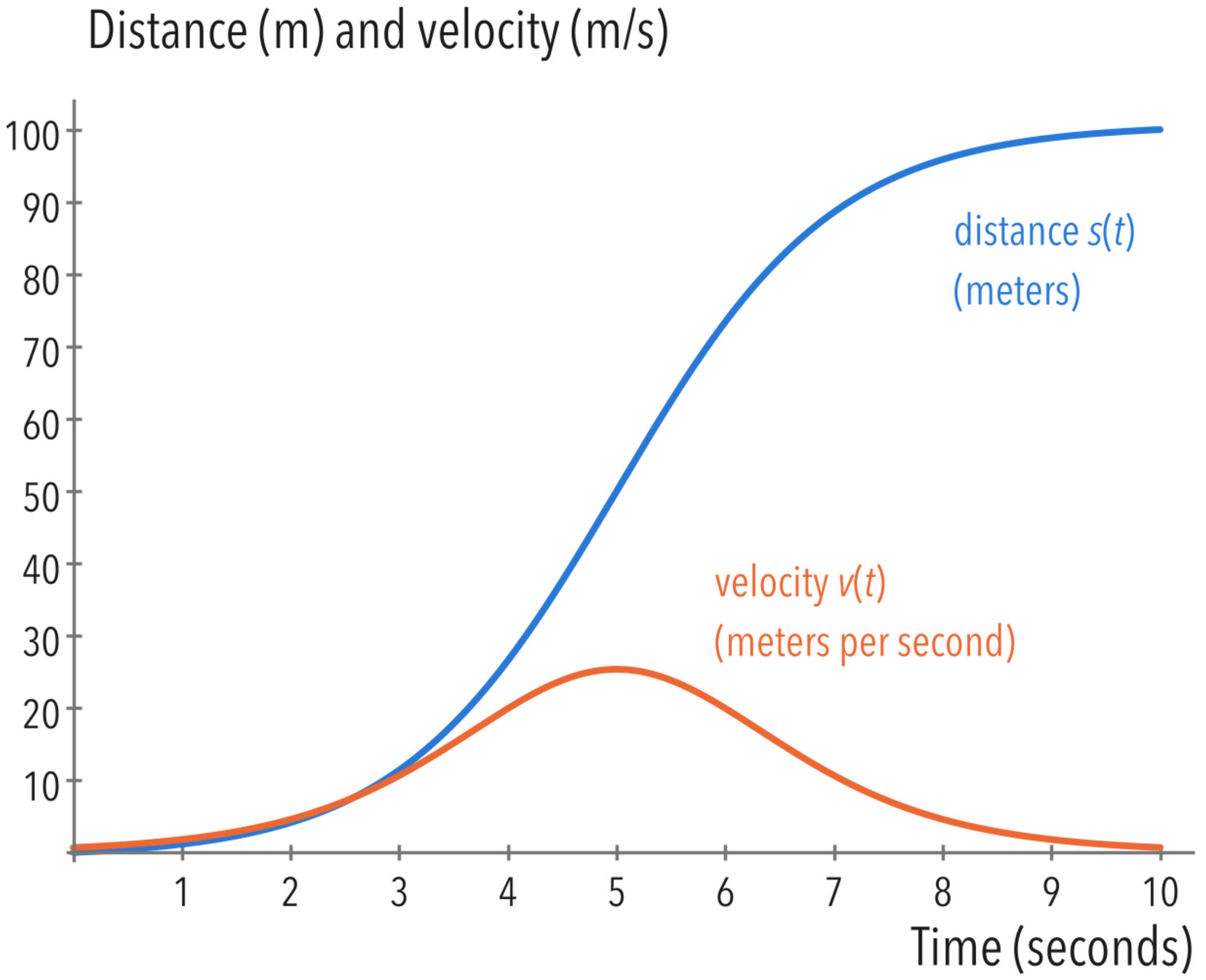
Time (in seconds): 0.0

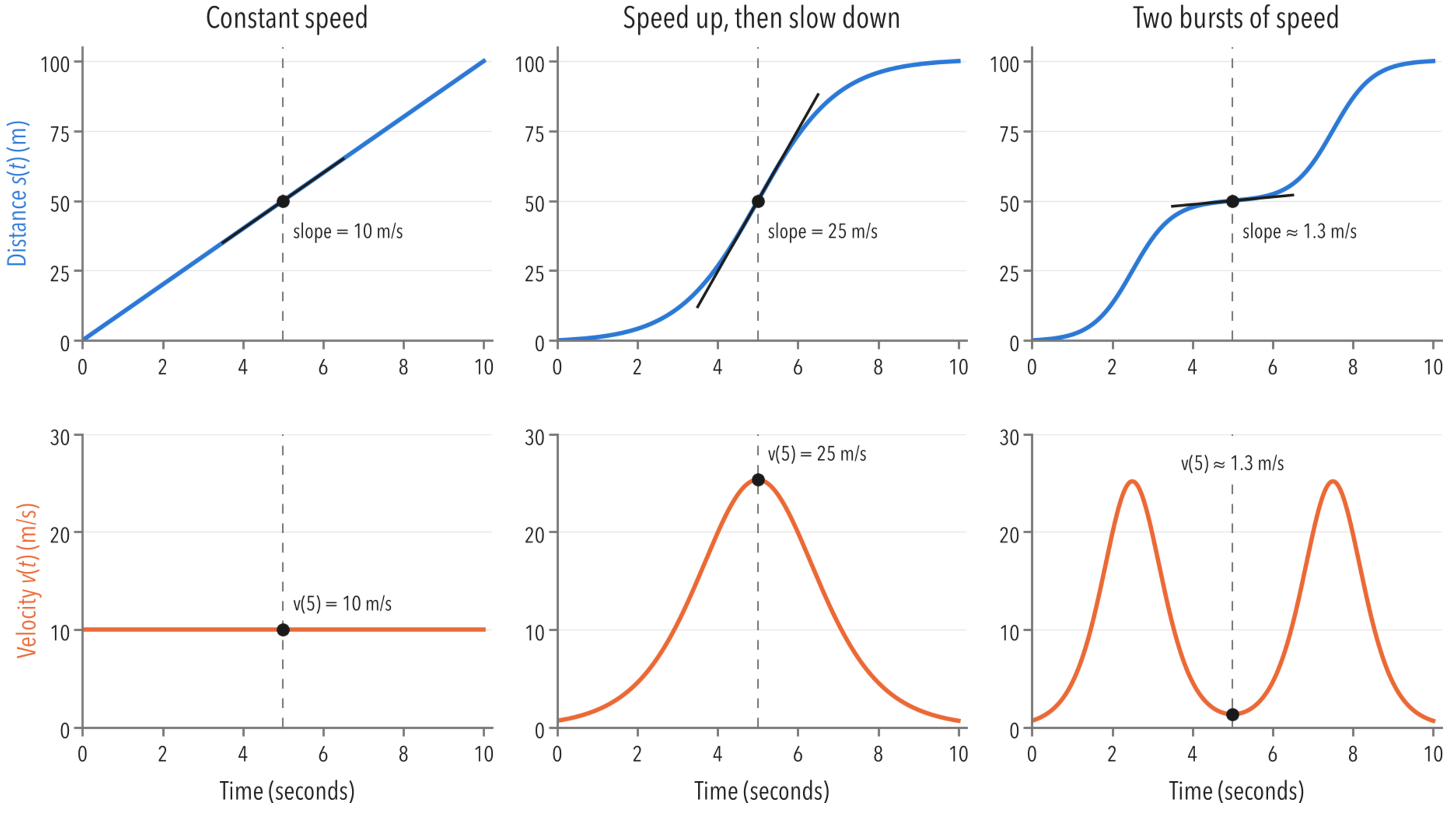


Distance traveled (meters)

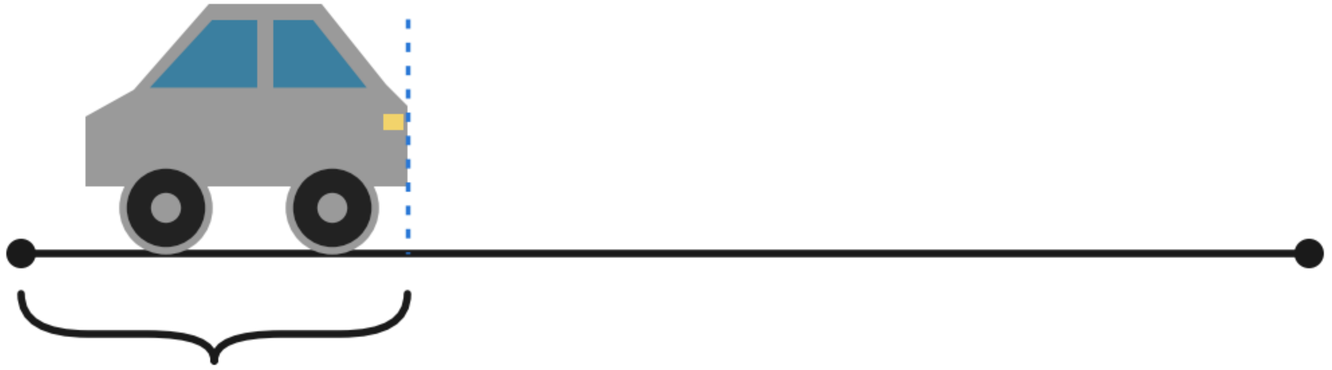
t = 0.0 s s = 0.0 m




Distance (m) and velocity (m/s)
100
90
80
70
60
50
40
30
20
10
distance s(t)
(meters)
velocity v(t)
(meters per second)
1
2
3
4
5
6
7
8
9
10
Time (seconds)

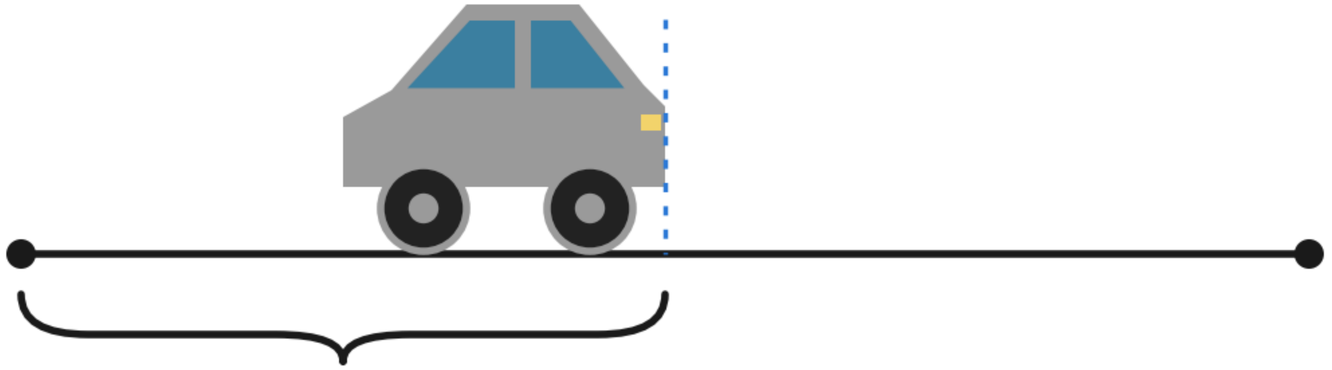

Constant speed
Speed up, then slow down
Two bursts of speed
Distance s(t) (m)
Velocity v(t) (m/s)
Time (seconds)
slope = 10 m/s
slope = 25 m/s
slope ≈ 1.3 m/s
v(5) = 10 m/s
v(5) = 25 m/s
v(5) ≈ 1.3 m/s
0
25
50
75
100
10
20
30
2
4
6
8
10

Time: 4 seconds



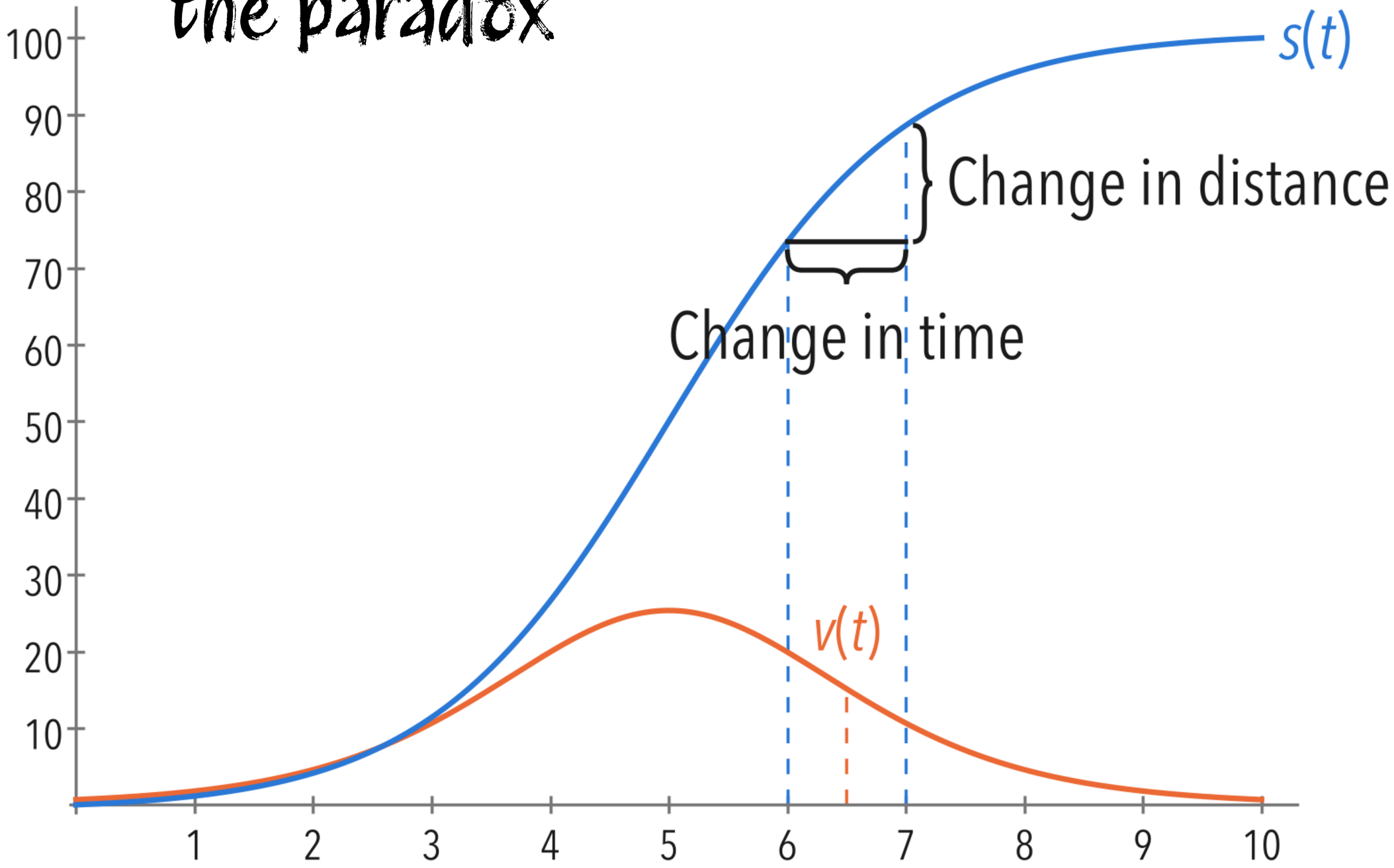
30 meters

Time: 5 seconds



50 meters

the paradox

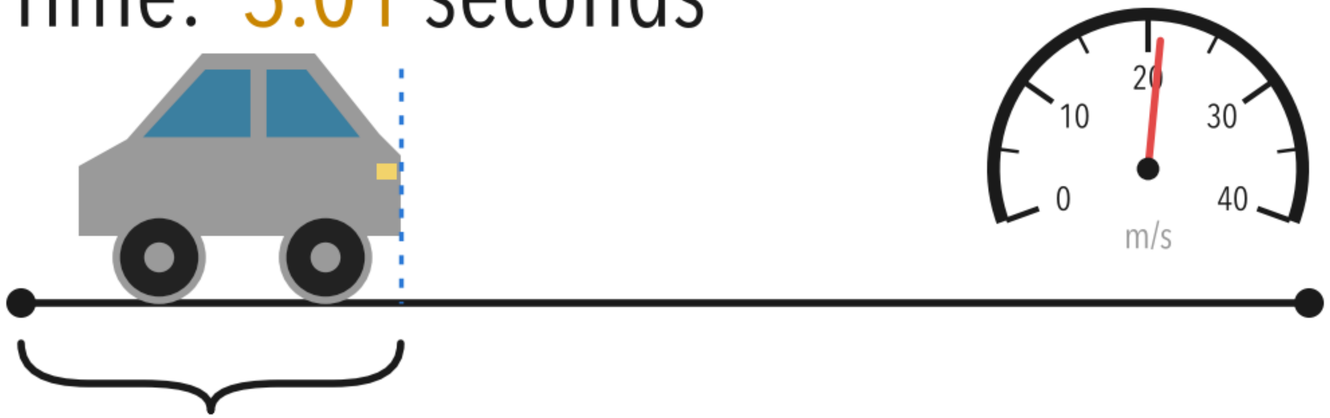

100
90
80
70
60
50
40
30
20
10
1
2
3
4
5
6
7
8
9
10
s(t)
Change in distance
Change in time
v(t)

Time: 3 seconds



20 meters

Time: 3.01 seconds



20.21 meters

Time: 3 seconds

20 meters



Velocity

$\Downarrow$

distance

in time

~~instantenous~~ rate of change

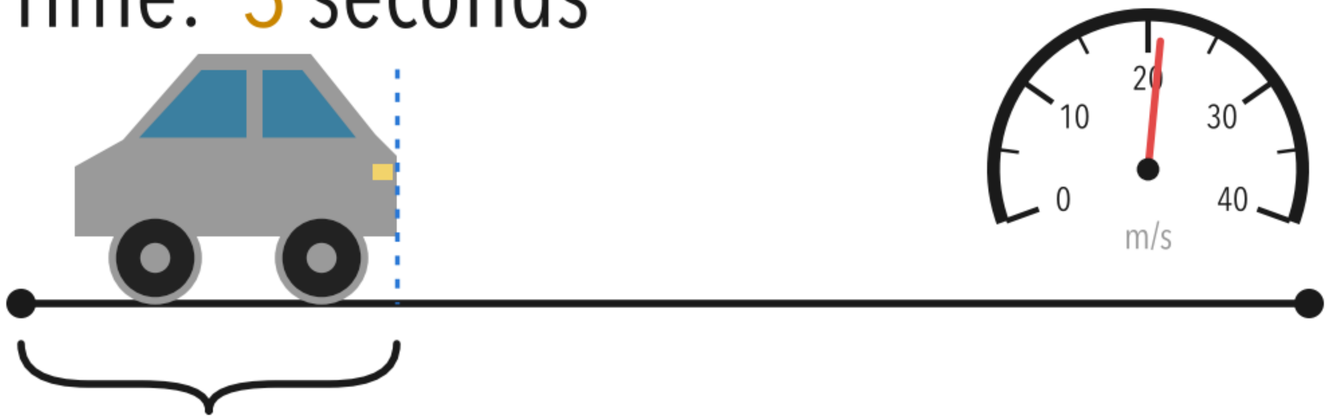
over a very small time interval

Time: 3.01 seconds

20.21 meters

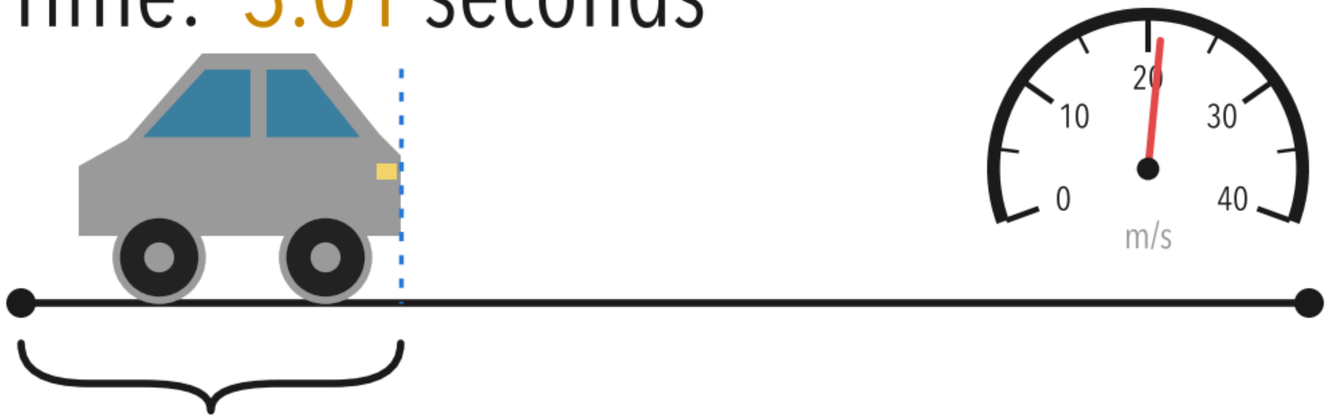
$$\frac{(20.21 - 20)\text{ meters}}{(3.01 - 3)\text{ seconds}}$$

Time: 3 seconds



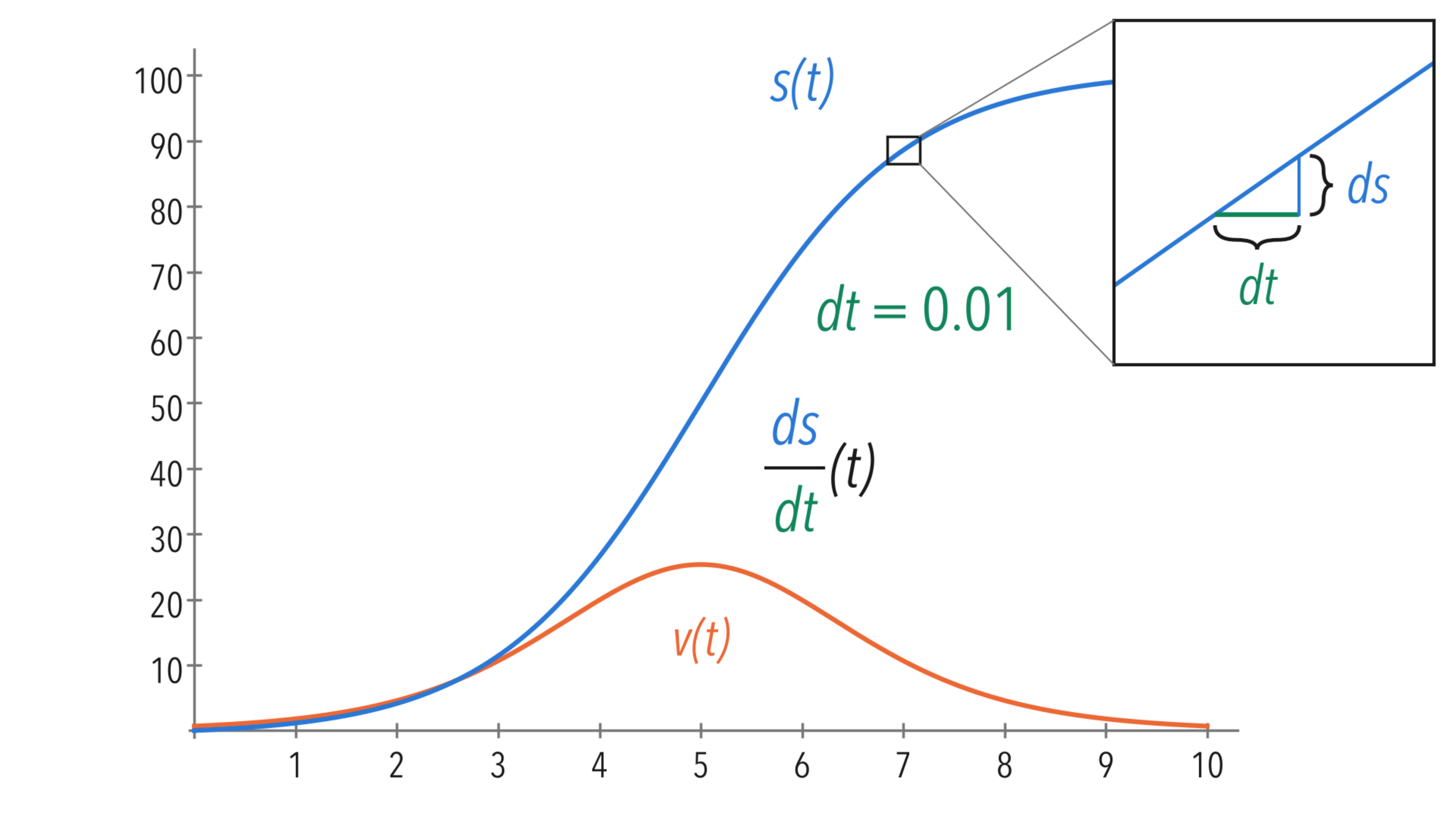
20 meters

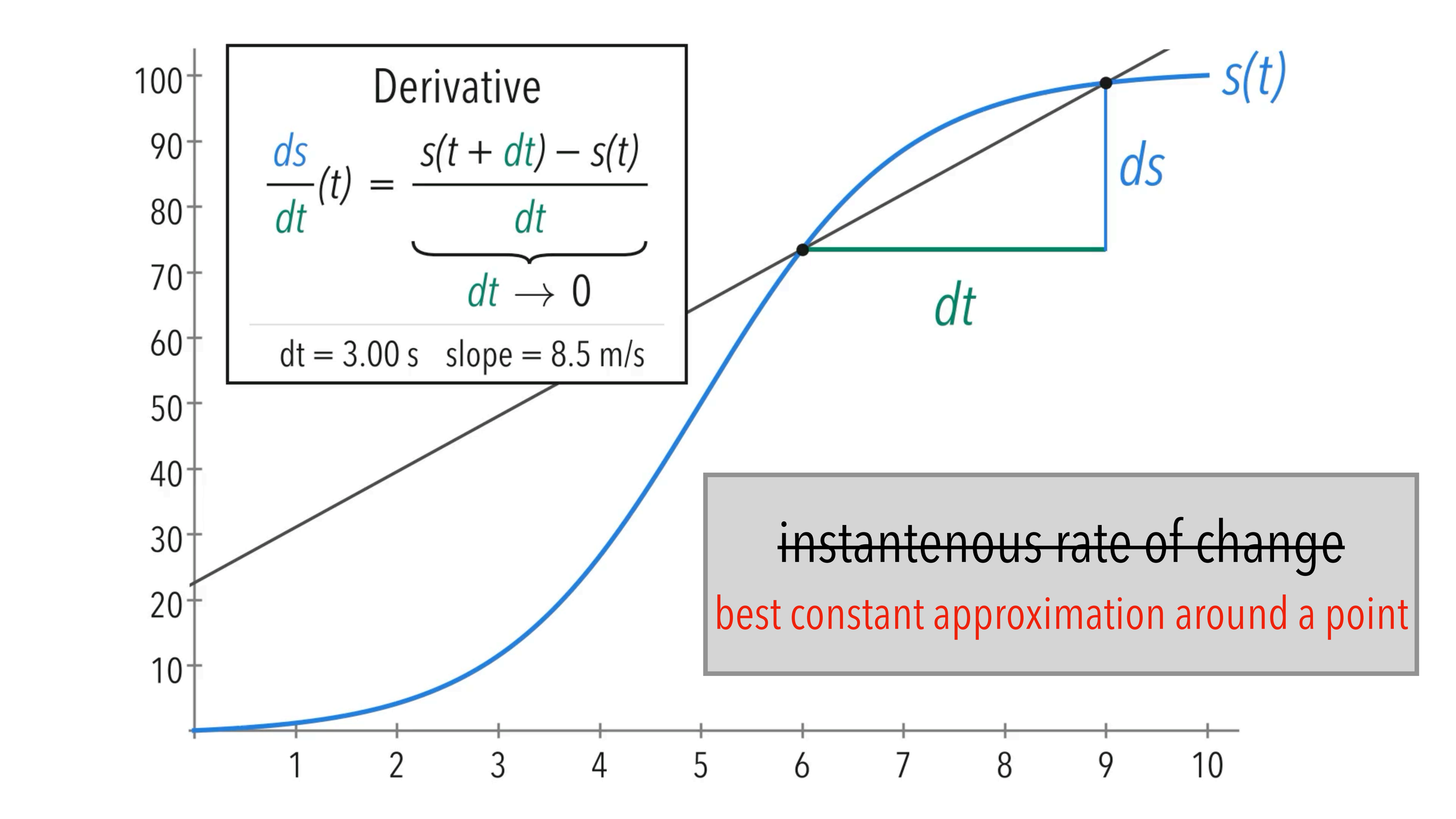
Time: 3.01 seconds

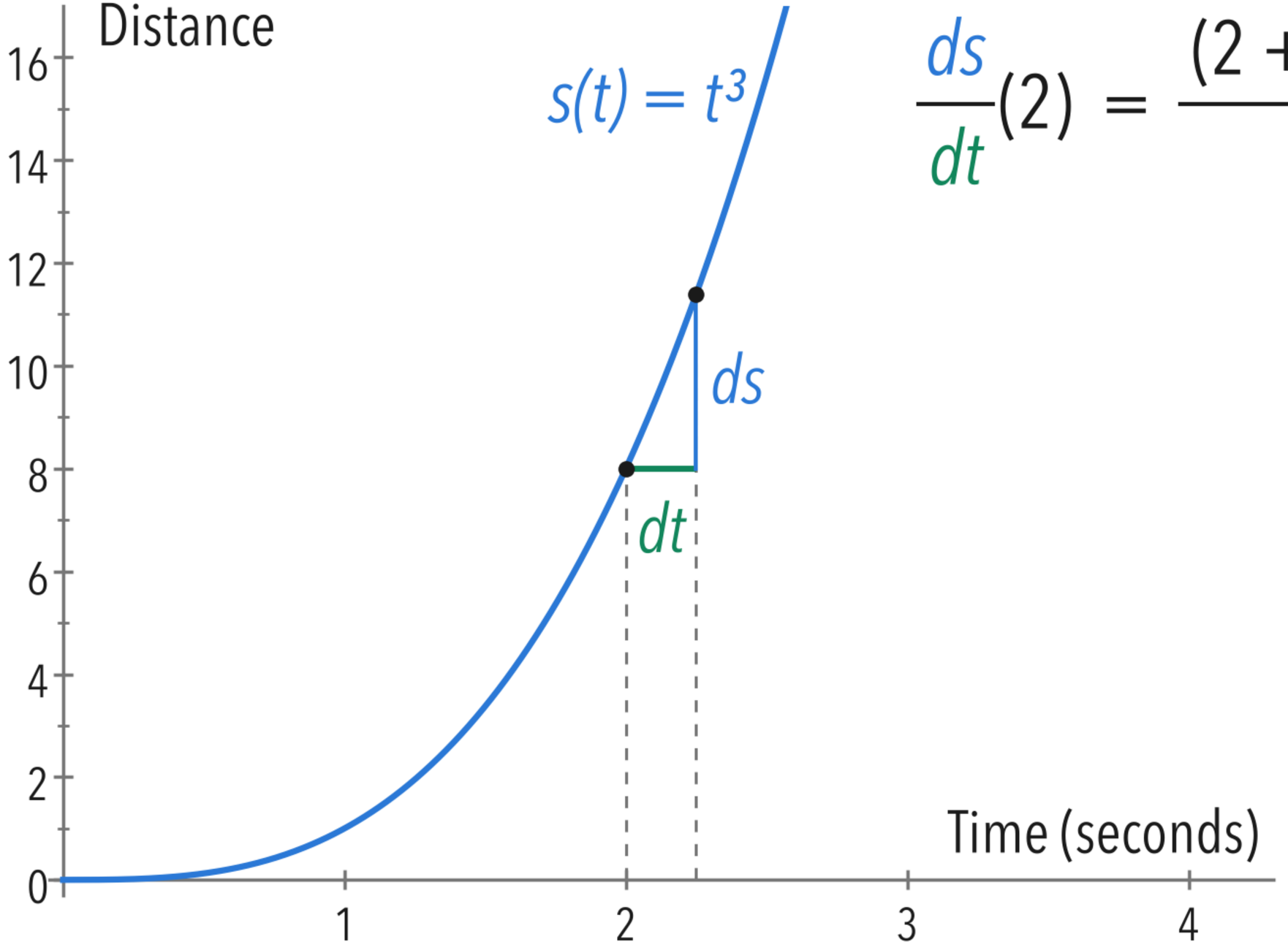


20.21 meters

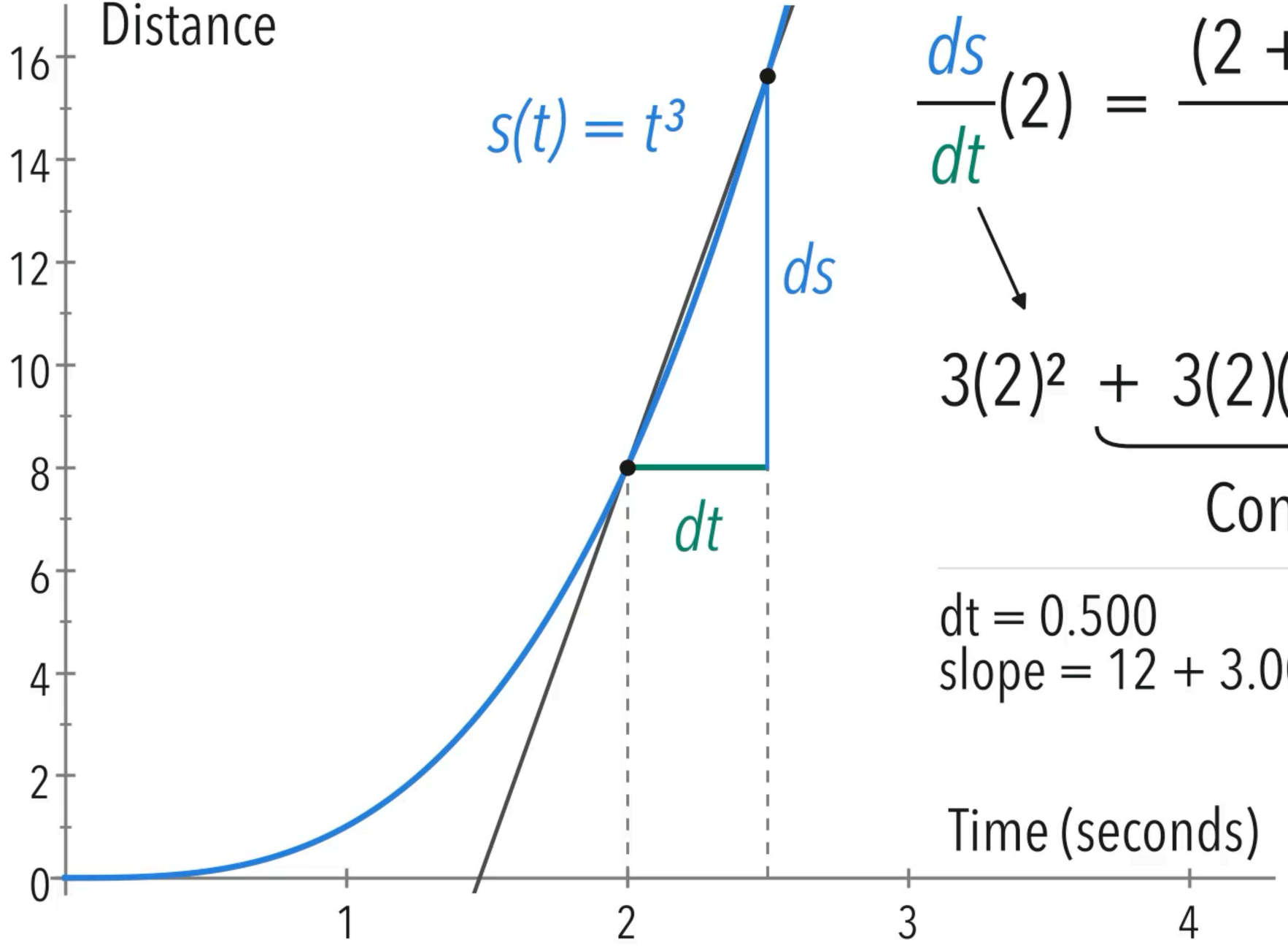
$$\frac{\overbrace{(20.21 - 20)}^{ds}\ \text{meters}}{(\underbrace{3.01 - 3}_{dt})\ \text{seconds}}$$


100
90
80
70
60
50
40
30
20
10
1
2
3
4
5
6
7
8
9
10
s(t)
dt = 0.01
ds
dt
ds/dt (t)
v(t)


Derivative
$\frac{ds}{dt}(t) = \underbrace{\frac{s(t+dt)-s(t)}{dt}}_{dt \to 0}$
dt = 3.00 s slope = 8.5 m/s
s(t)
ds
dt
100
90
80
70
60
50
40
30
20
10
1
2
3
4
5
6
7
8
9
10
~~instantenous rate of change~~
best constant approximation around a point



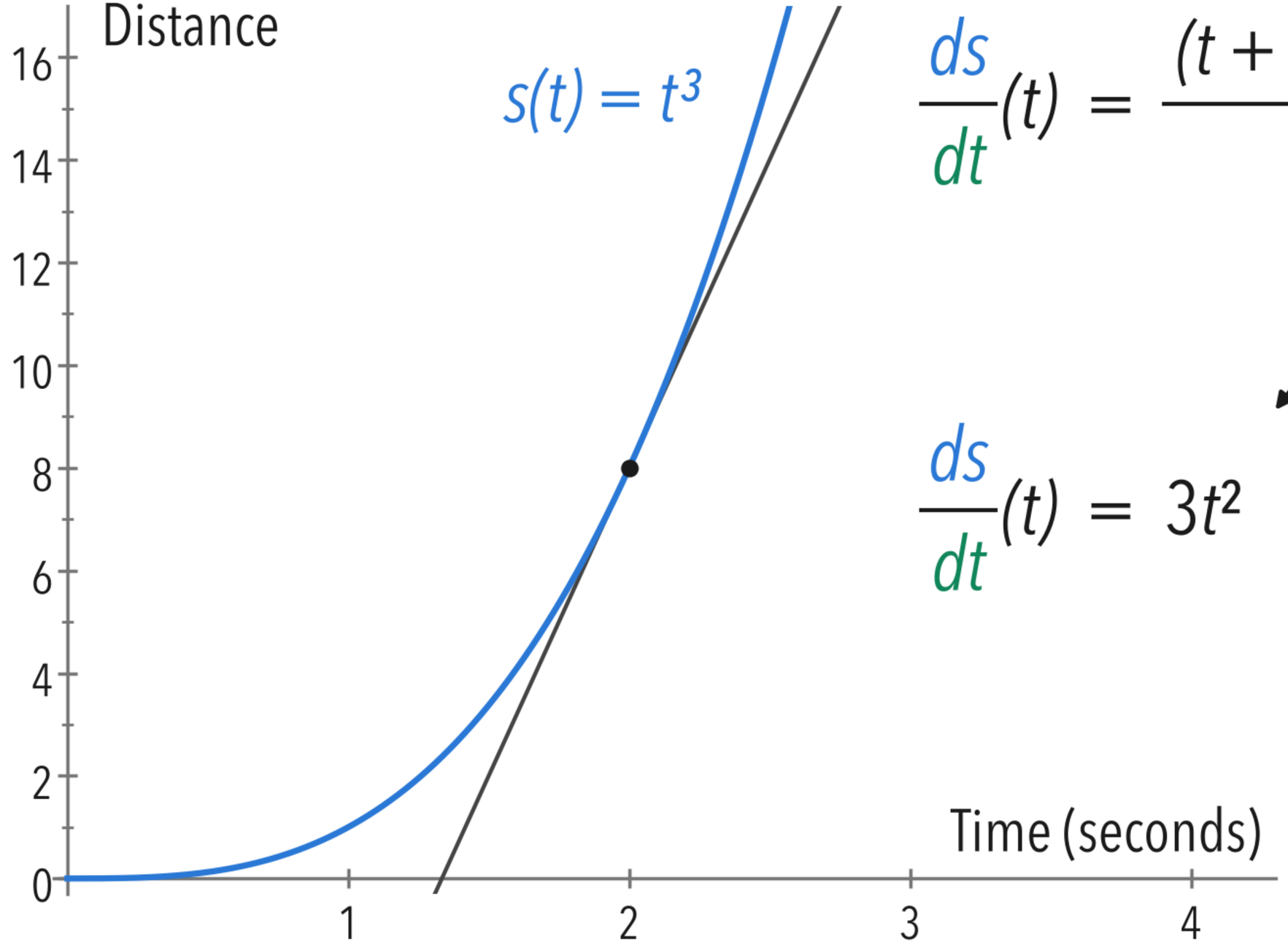
$$\frac{ds}{dt}(2) = \frac{(2 + dt)^3 - 2^3}{dt}$$



$$\frac{ds}{dt}(2) = \frac{(2 + dt)^3 - 2^3}{dt}$$

$$3(2)^2 \underbrace{+ 3(2)(dt) + (dt)^2}_{\text{Contains } dt}$$

dt = 0.500
slope = 12 + 3.000 + 0.250 = 15.250



$$\frac{ds}{dt}(t) = \frac{(t + dt)^3 - t^3}{dt}$$

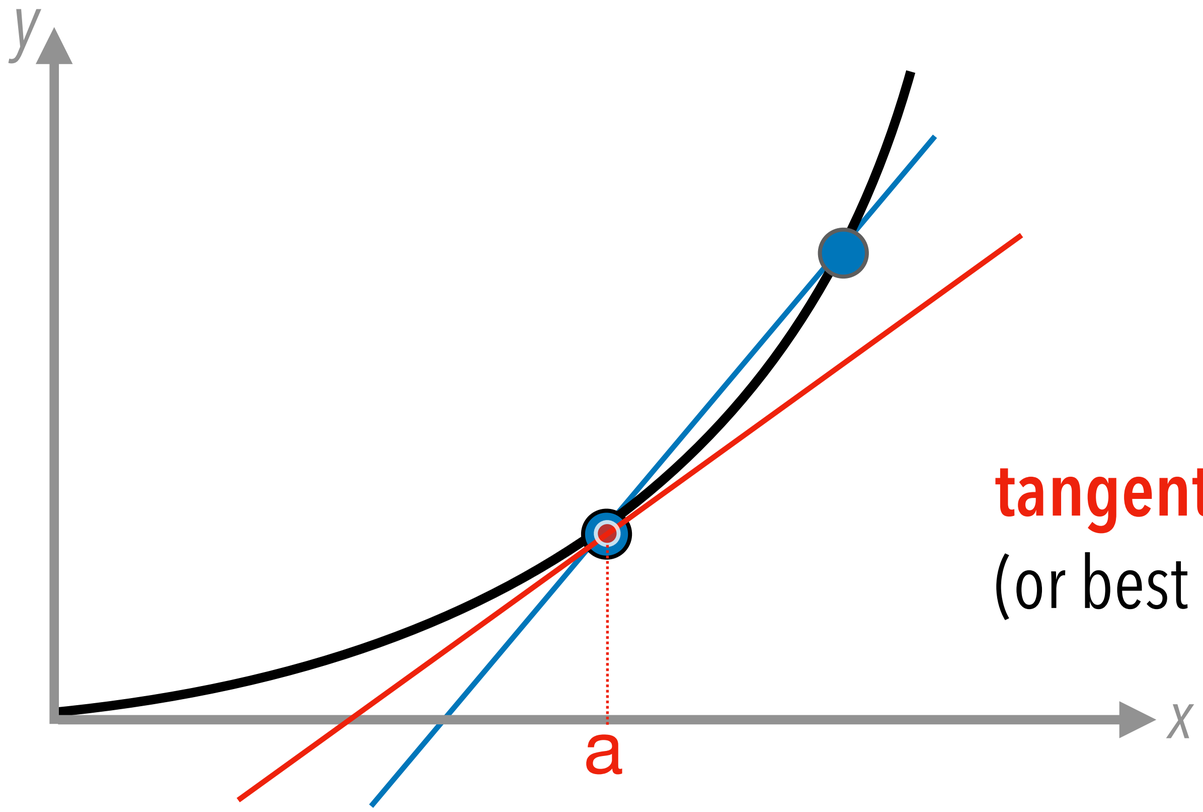
As $dt \to 0$

$$\frac{ds}{dt}(t) = 3t^2$$

let's formalize this in the language of math...

secants and tangents

secants graphically represent discrete change



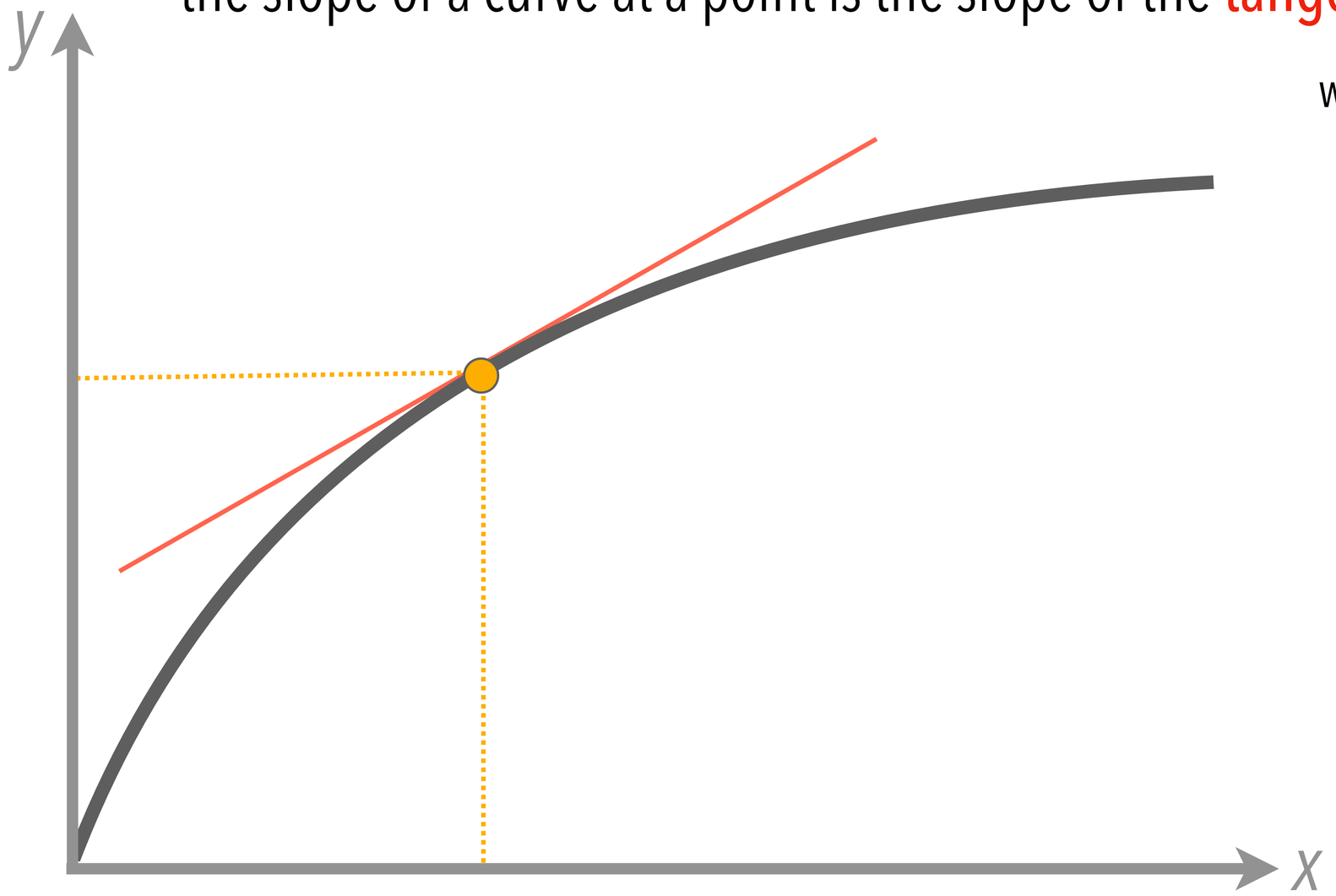
tangents graphically represent instantaneous change (or best constant approximation) at x=a

Note: "secare" means to intersect and "tangere" means to touch in Latin

tangents

the slope of a curve at a point is the slope of the tangent line at that point

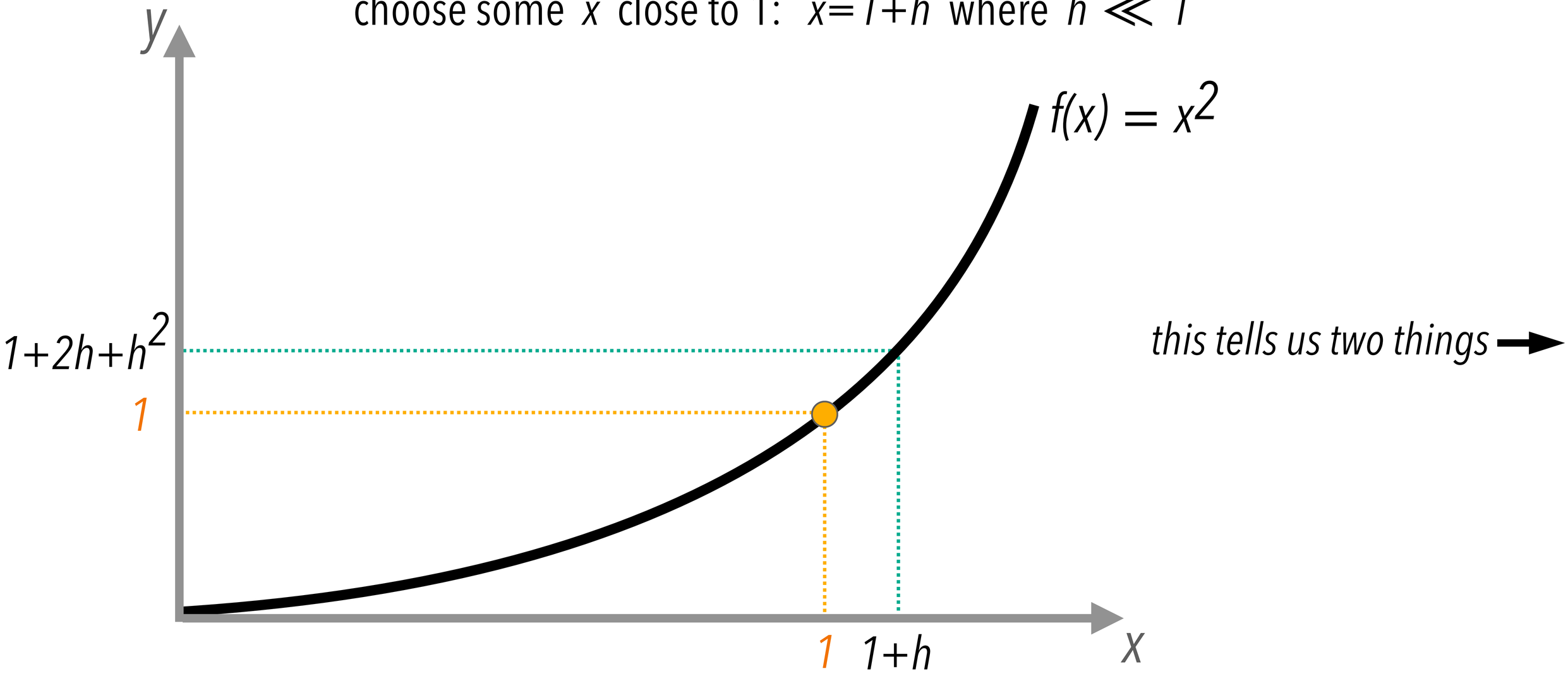
we find these slopes algebraically



finding the slope at a point

how does this function behaves when x is close to1?

choose some x close to 1: $x=1+h$ where $h \ll 1$

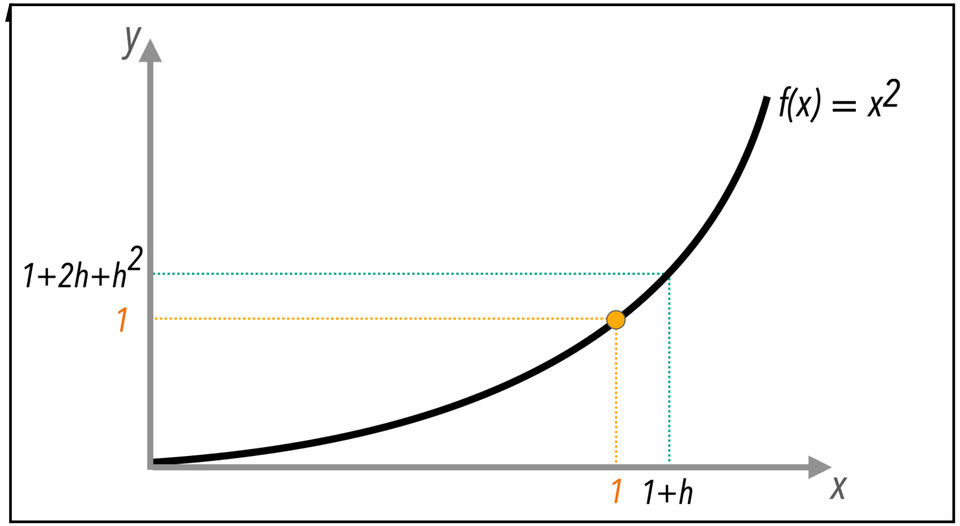


finding the slope at a point

1. Near *1* the value of $f(x)$ is also near *1*, since if h is small, then $2h+h^2$ is also small (this is the concept of continuity)
2. Increasing x from *1* to $1+h$ will increase the value of $f(x)$ from $f(1)$ to $f(1) + 2h + h^2$ which is an increase of approximately 2h

$\Longrightarrow$ the rate of change at *1*,
defined as the derivative of the function $f(x) = x^2$
is $f'(1)=2$

What about when $x=c$?



finding the slope at a point

What about when $x=c$?

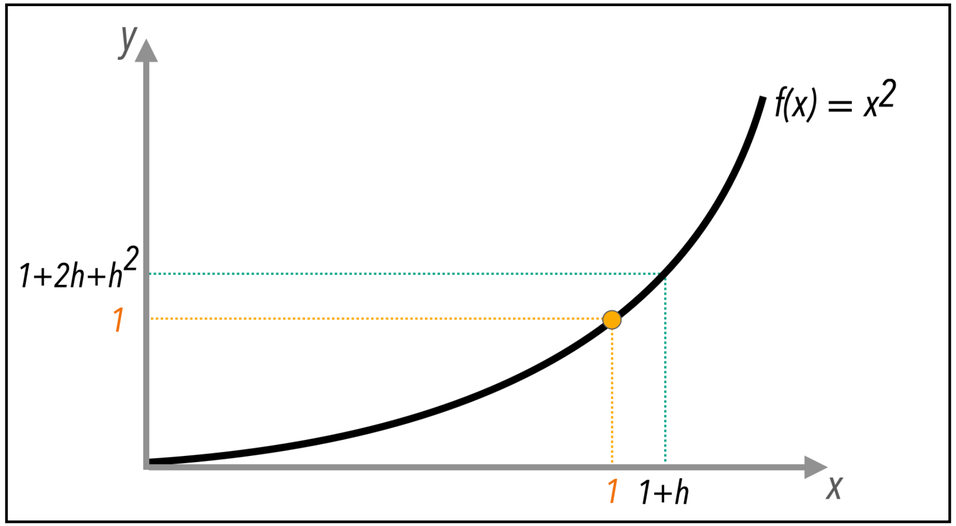
As x changes from c to $c+h$,

$f(x)=x^2$ changes from c^2 to $(c+h)^2 = c^2 + 2ch + h^2$

the change = the coefficient of h

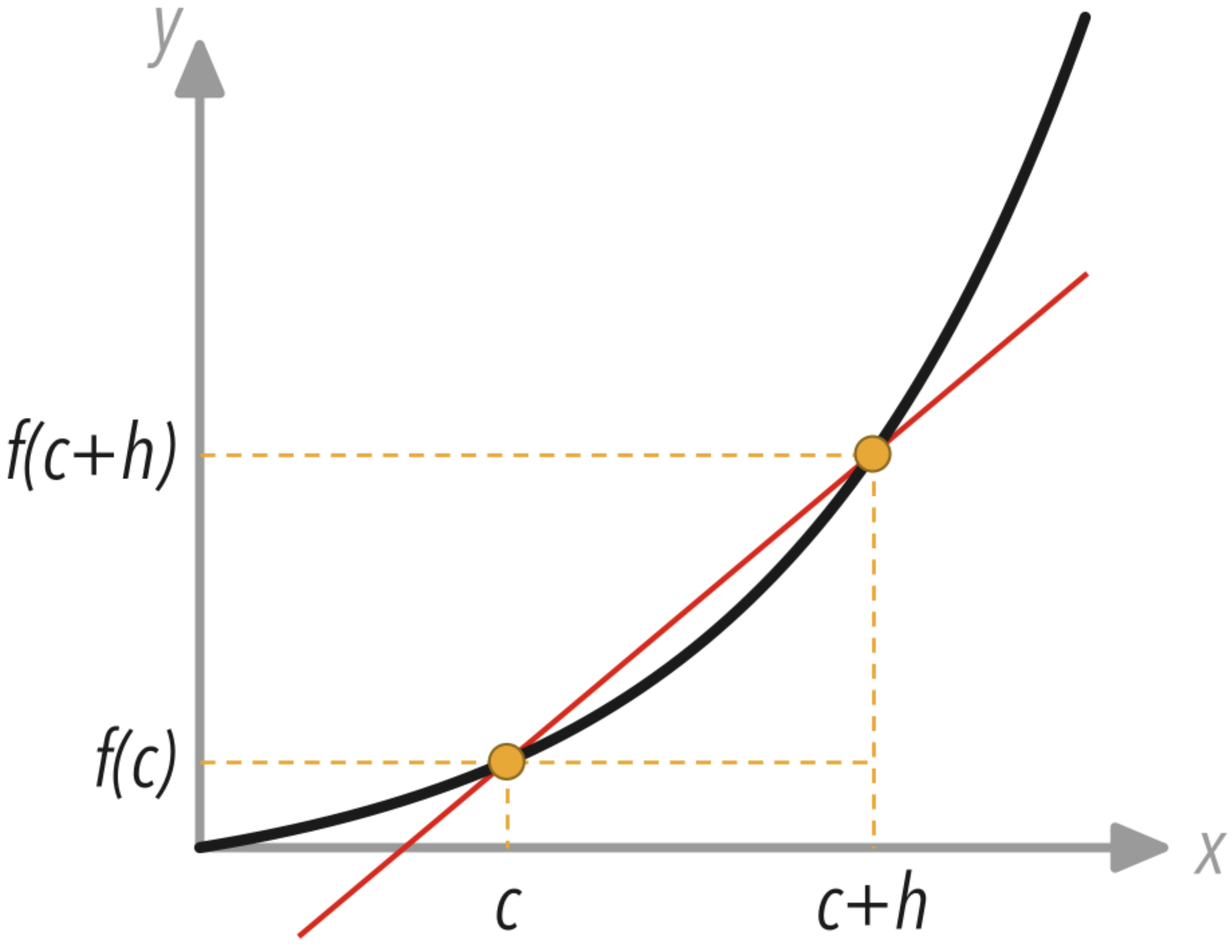
$\Longrightarrow$ the derivative of f at c is $2c$

and in general $f'(x)=2x$



secants

geometrically, we calculate the slope of the secant joining points on the curve:

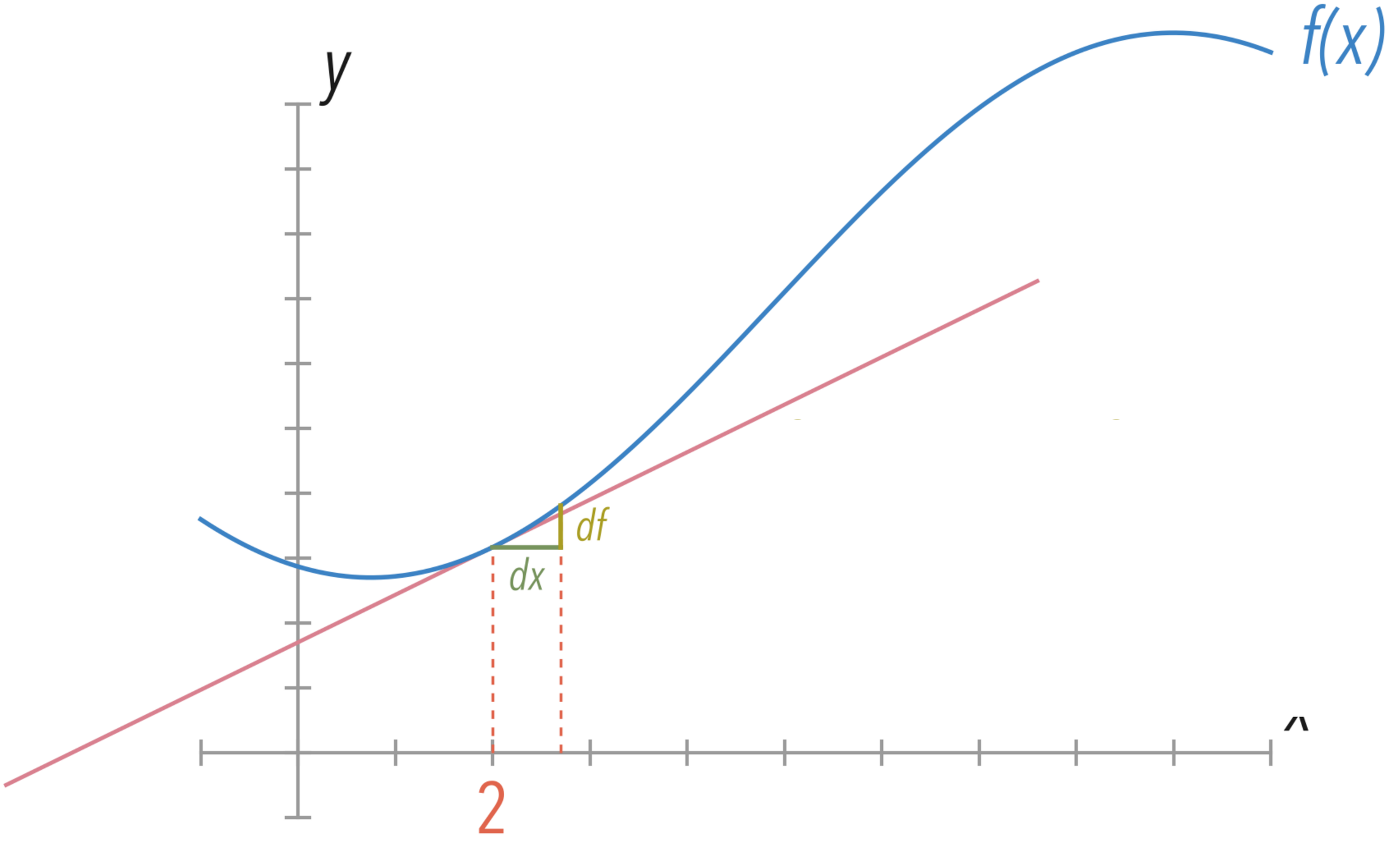


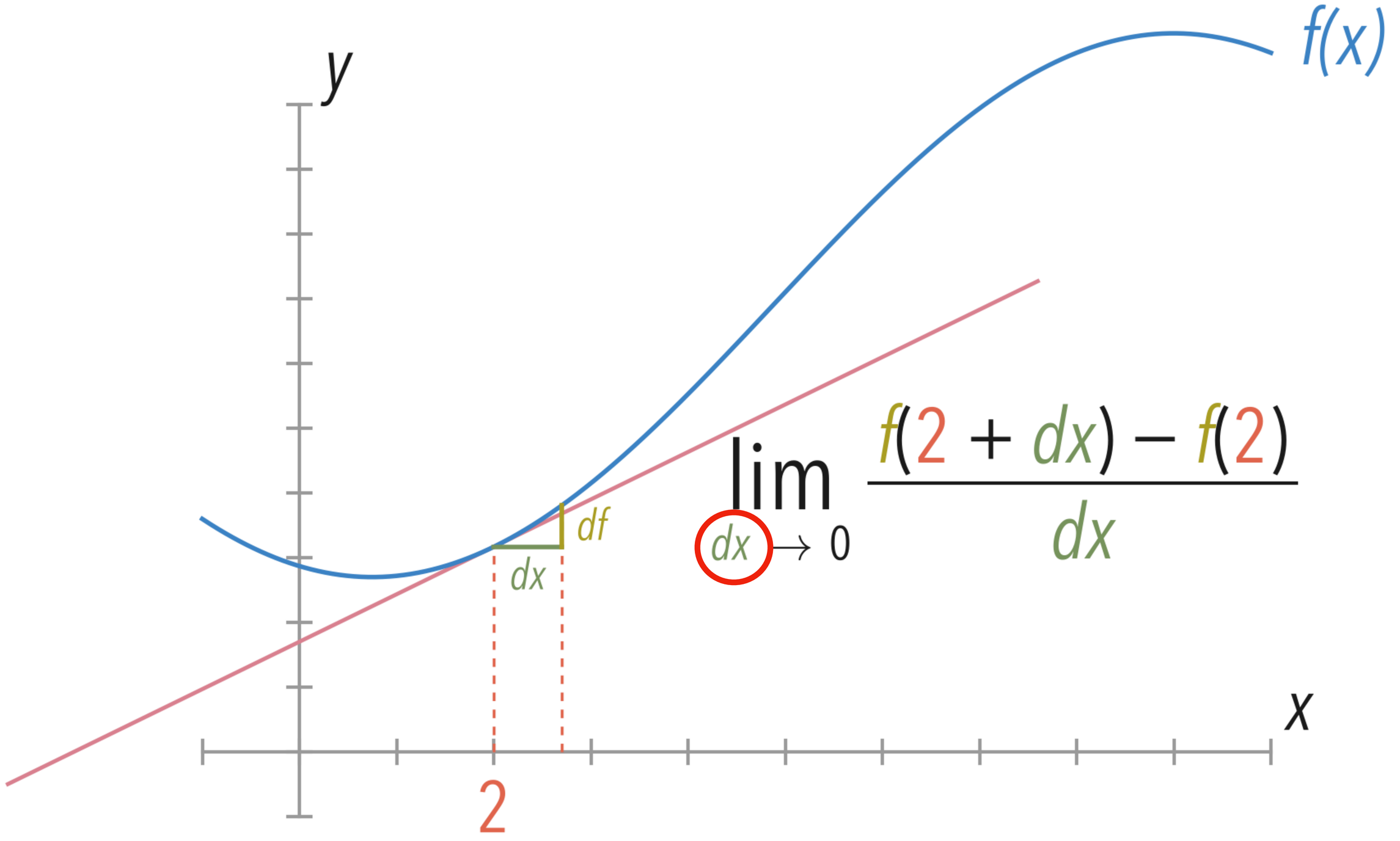
$$\frac{\text{change in } f}{\text{change in } x} = \frac{f(c+h) - f(c)}{(c+h) - c} = \frac{f(c+h) - f(c)}{h}$$

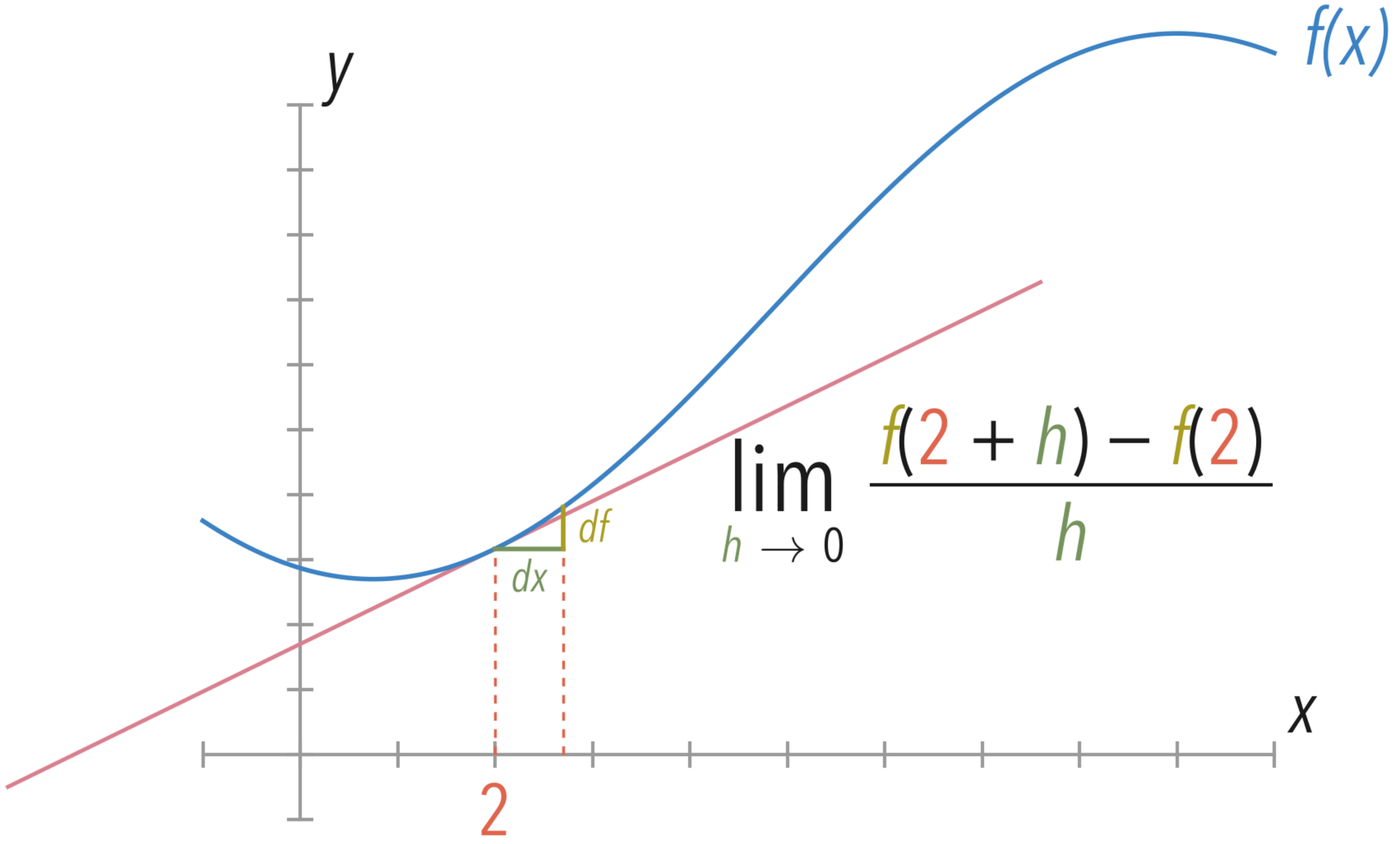
and examine what happens as h approaches 0

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$$

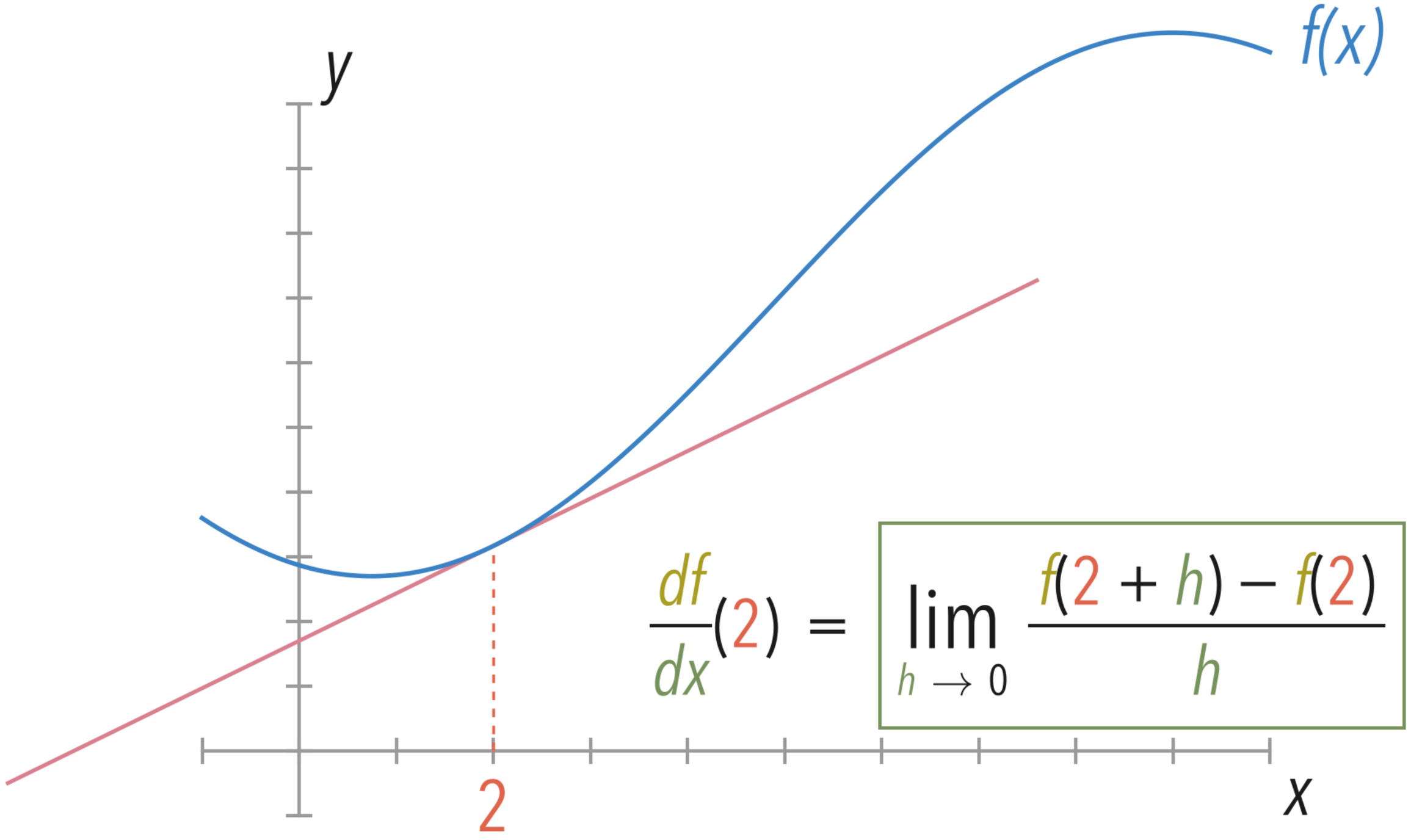
in the hopes that the slope of the secant will approach the slope of the tangent line


f(x)
y
df
dx
2


f(x)
y
df
dx
2
x
lim
dx → 0
f(2 + dx) − f(2)
dx



$$\lim_{h \to 0} \frac{f(2+h) - f(2)}{h}$$


f(x)
y
2
x
$\frac{df}{dx}(2) = \lim_{h \to 0} \frac{f(2+h) - f(2)}{h}$

derivative of a function

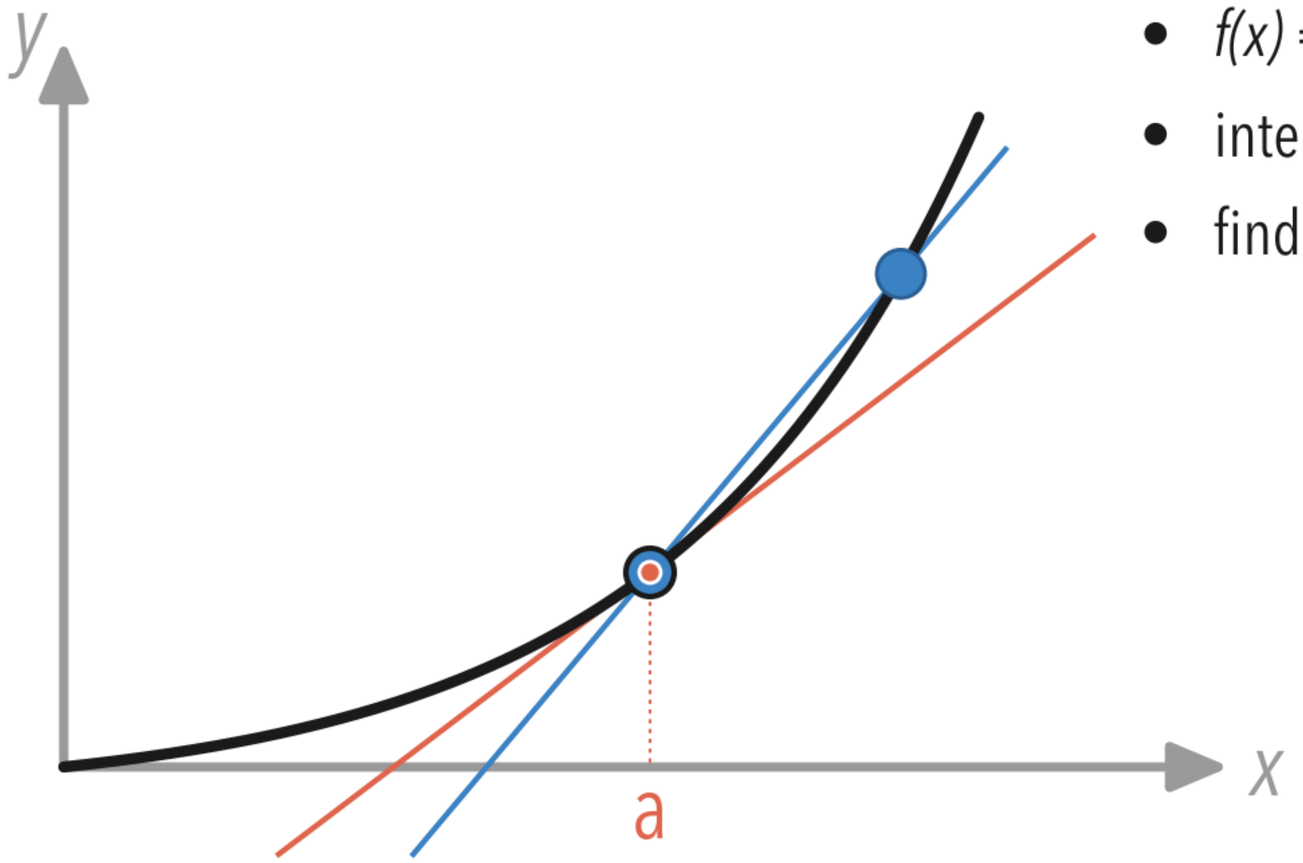
We define the derivative of a function f at x as

$$f'(x) = \frac{d}{dx} f(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

provided that the limit exists. If the limit exists, we say f is differentiable at x.

If we simply say f is differentiable, we mean f is differentiable at all values of x.
In this case, $f'(x)$ is also a function of x.

summary: secants and tangents



secants graphically represent discrete change

- $f(x) = mx + b$
- intercept b as point 1
- find mx to reach point 2

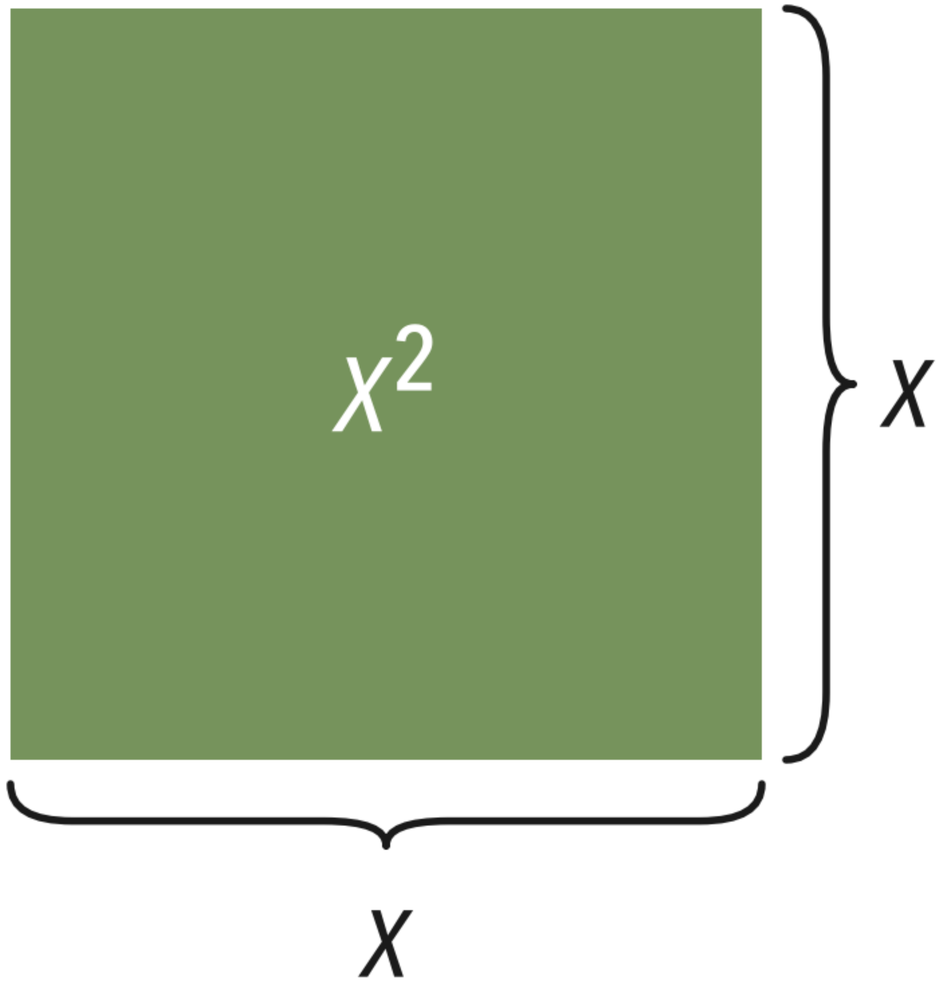
slope of the secant

$$\frac{f(x + h) - f(x)}{h}$$

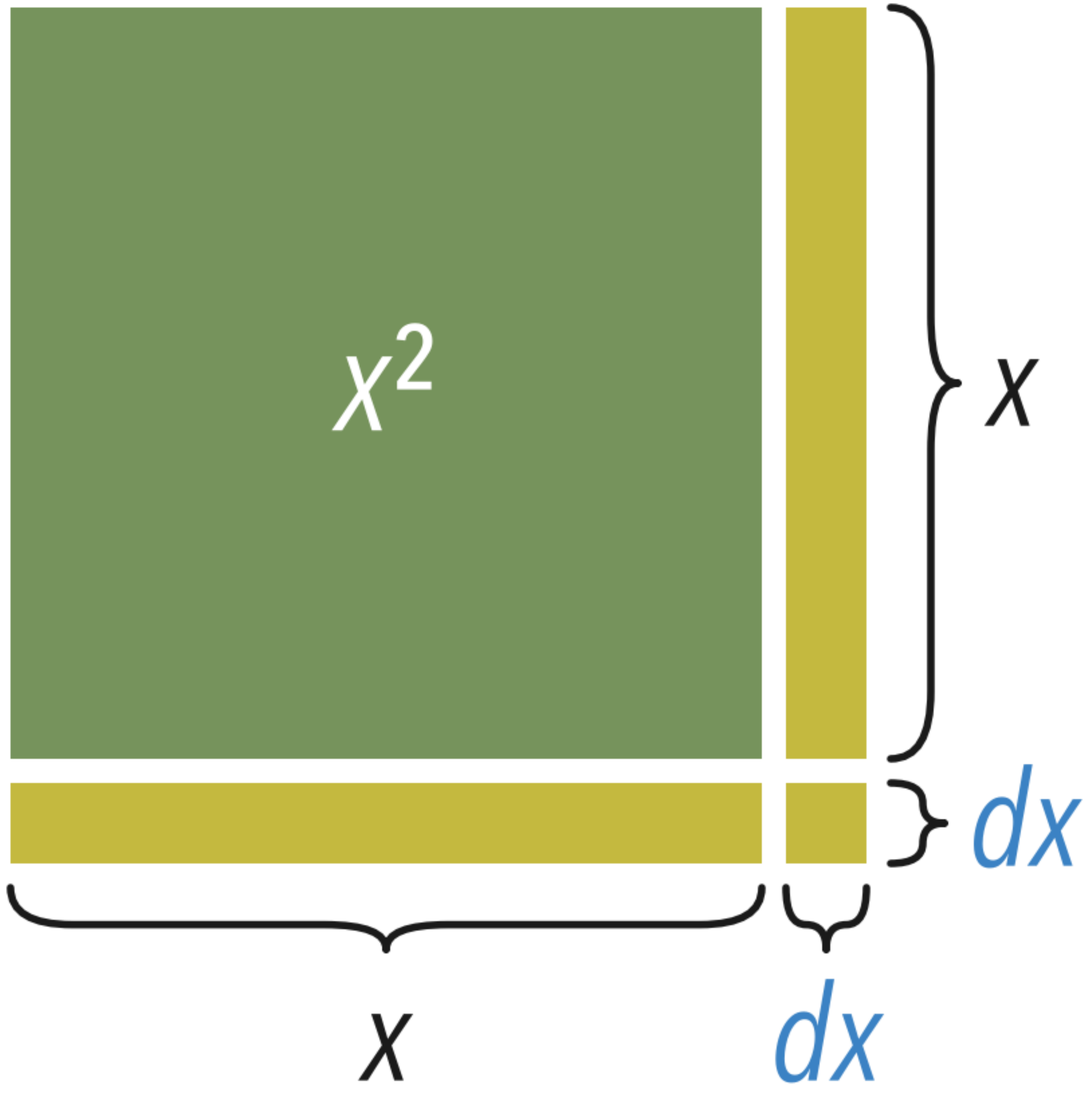
Note: "secare" means to intersect and "tangere" means to touch in Latin

a geometric view on the derivative...

$$f(x) = x^2$$

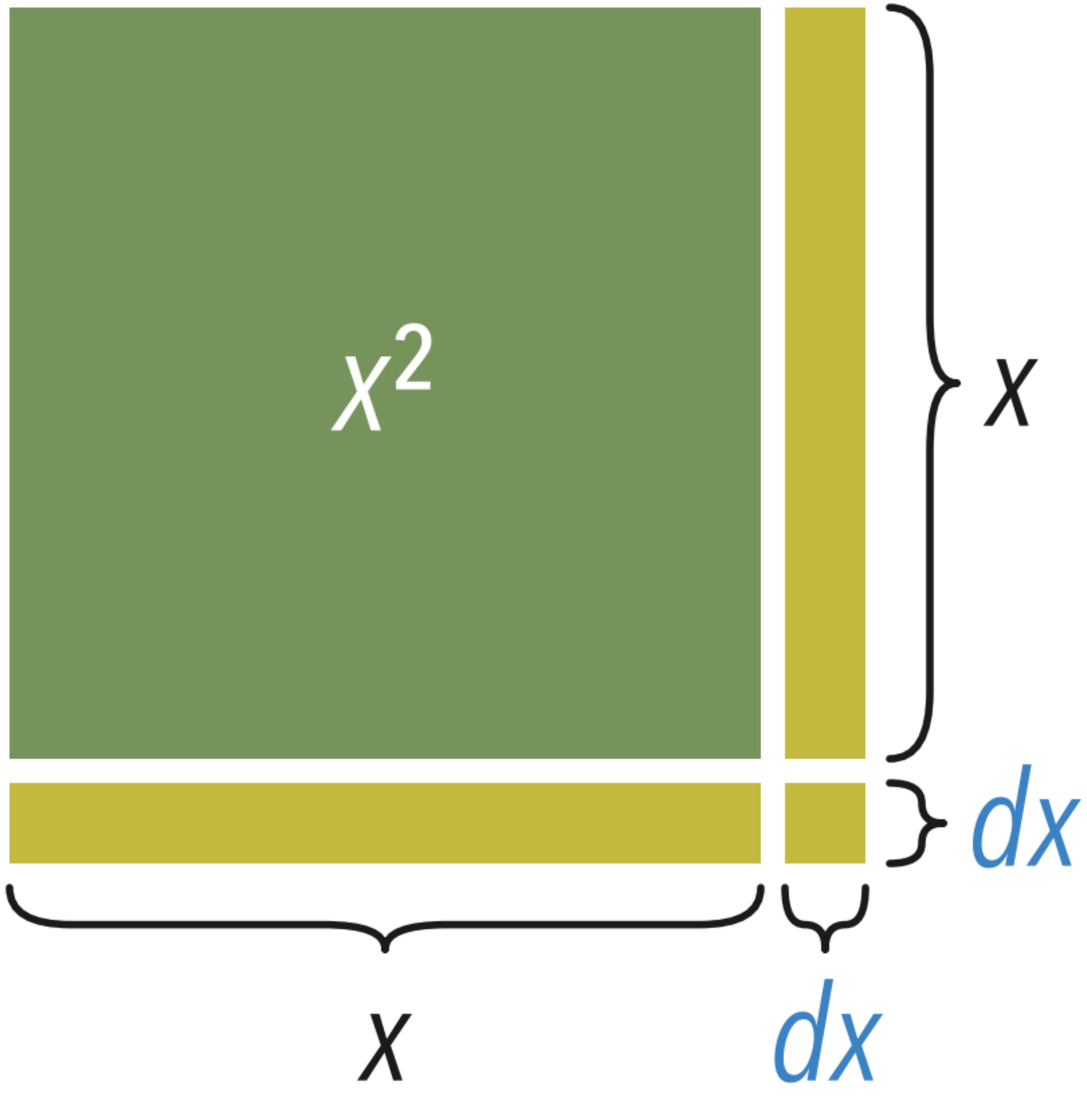


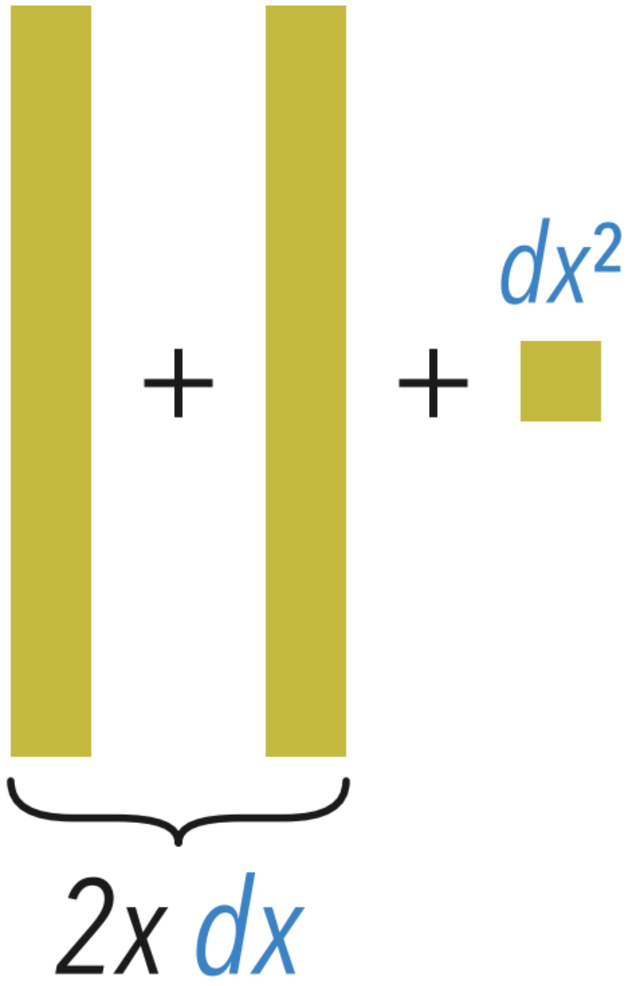
$$f(x) = x^2$$



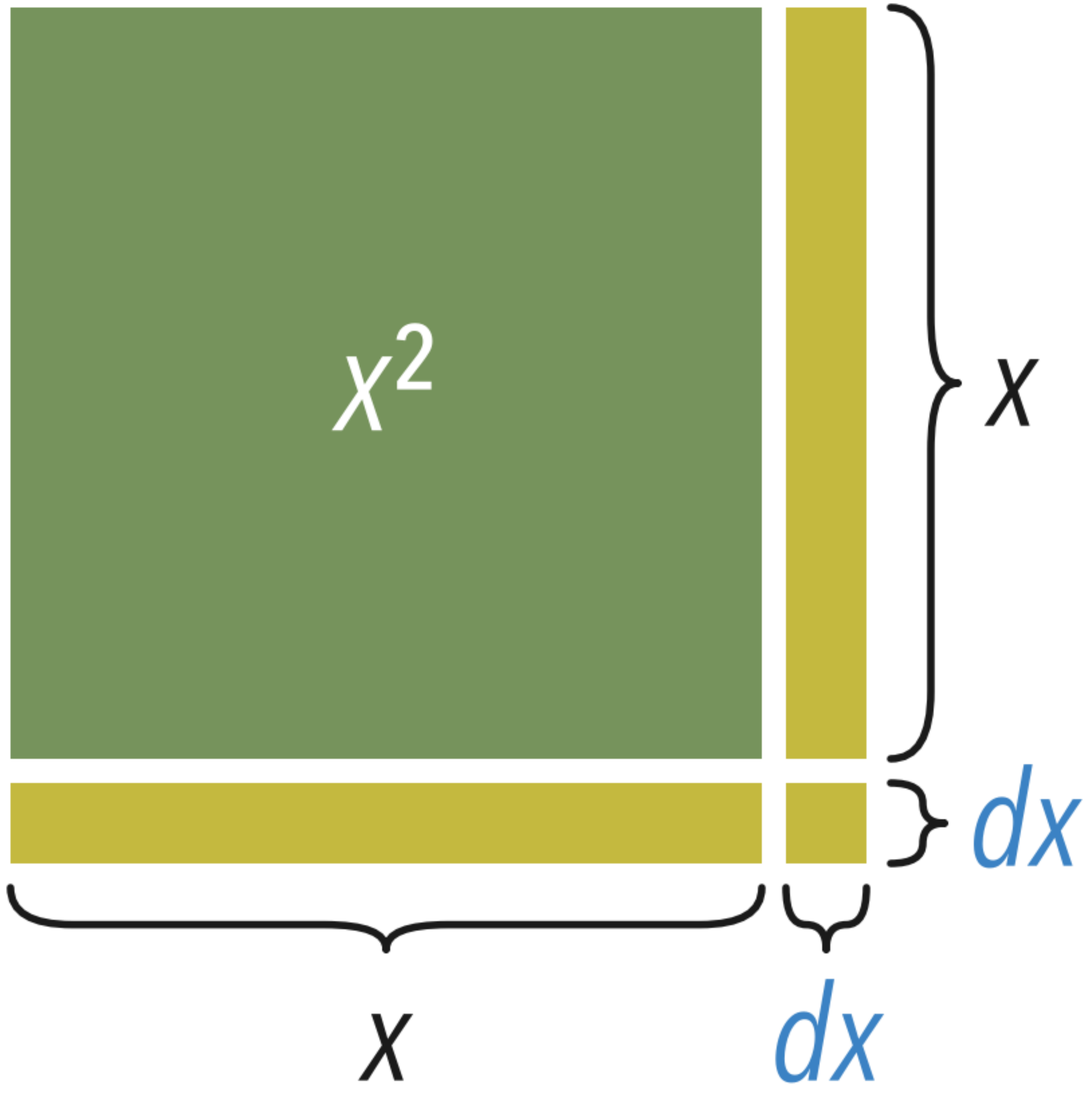
increase X by sa tiny amount,
what happens to the area of the square f?

$f(x) = x^2$



$df =$  $+ dx^2$

$f(x) = x^2$

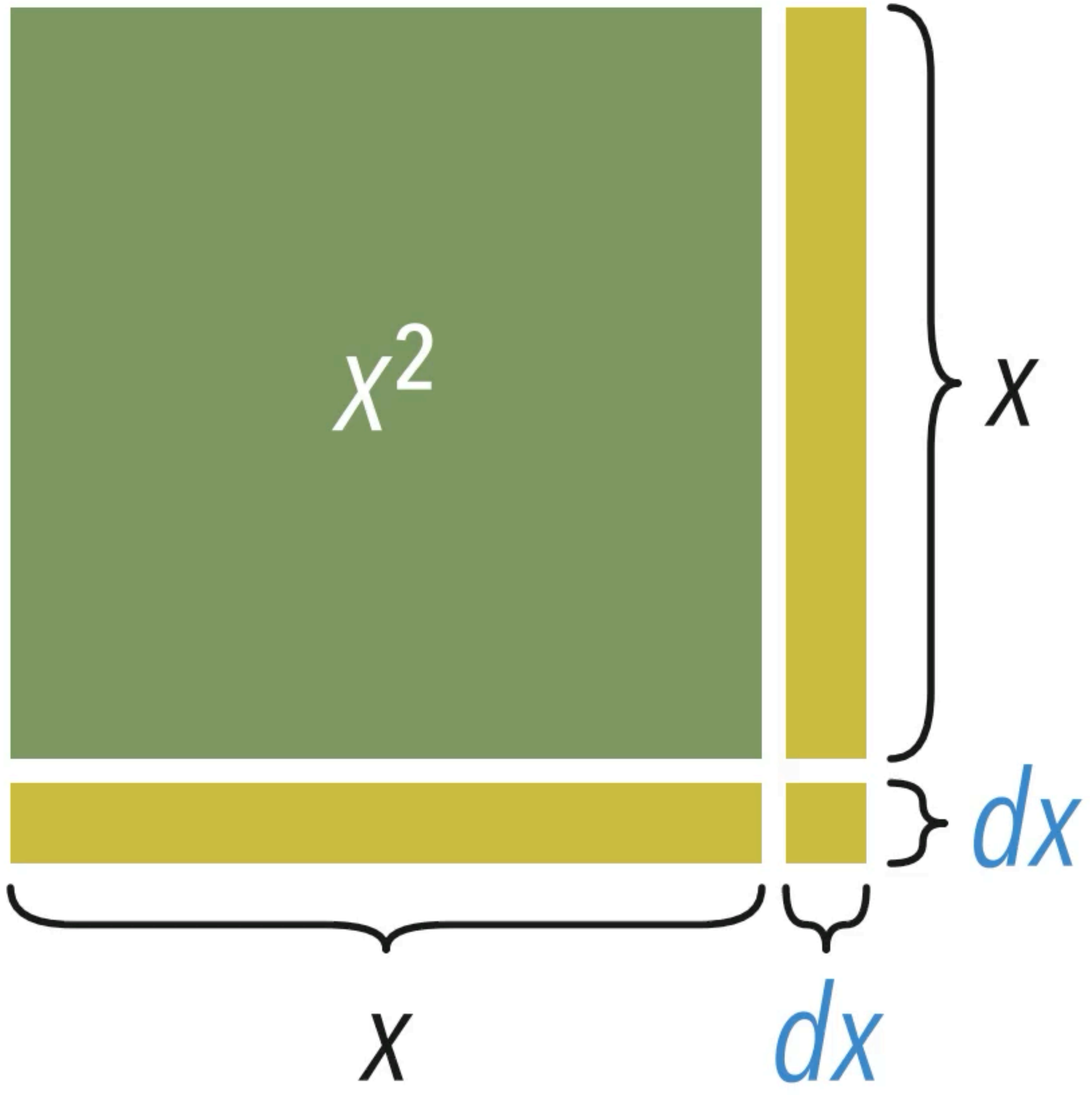


$$\frac{df}{dx} = \quad + \quad + \quad dx$$

$2x$

$$\frac{df}{dx} = 2x + dx$$

$$f(x) = x^2$$



$$\frac{df}{dx} = \square + \square + \square$$

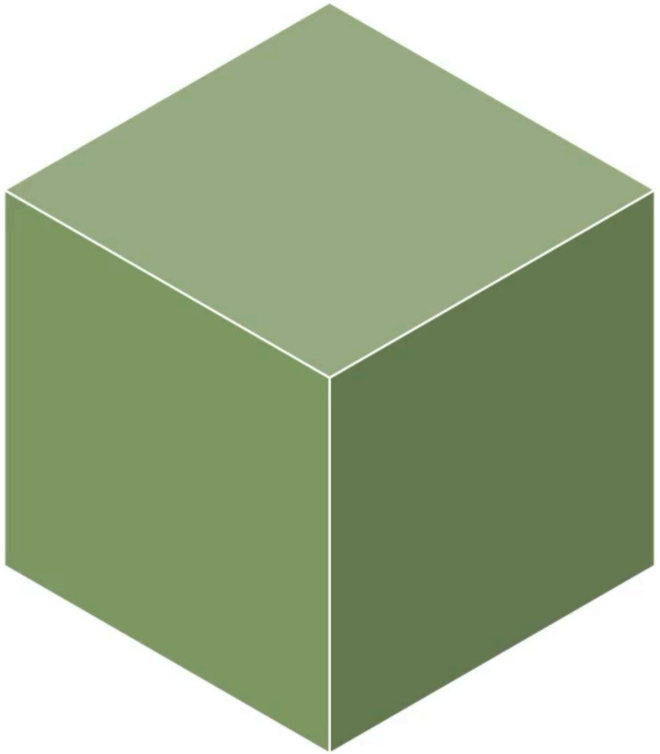
dx

$= 0.107$

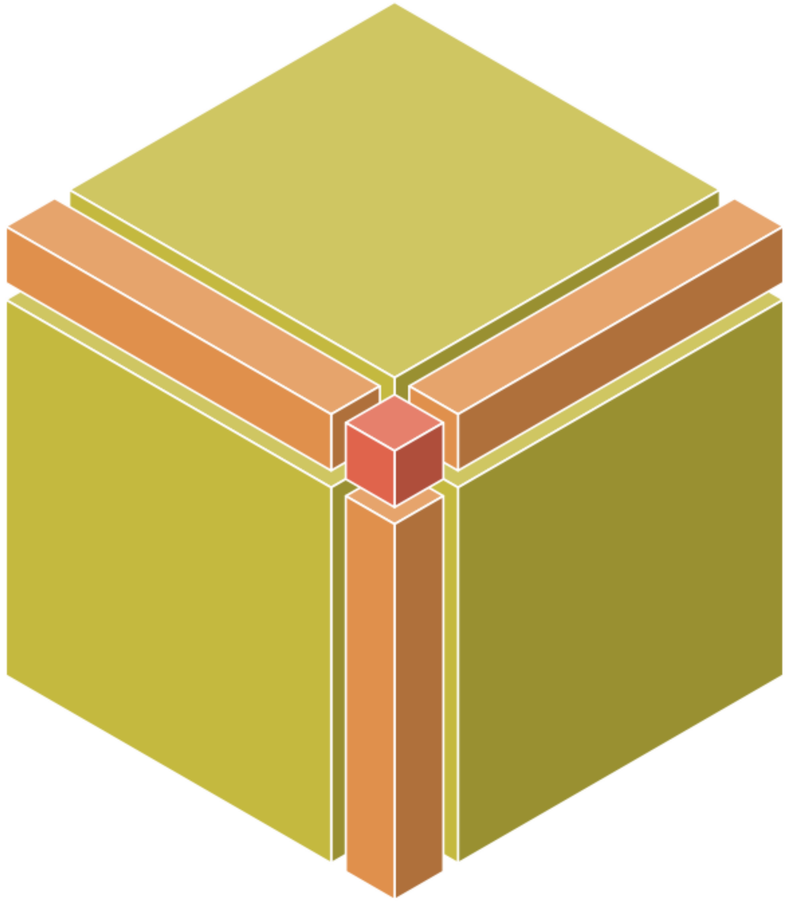
$2x$

$$\frac{df}{dx} = 2x + dx$$

$$f(x) = x^3$$



Cube with side x: volume x^3



Cube with side x: volume x^3

- 3 slabs: $x^2\,dx$ each
- 3 rods: $x\,dx^2$ each
- 1 corner: dx^3

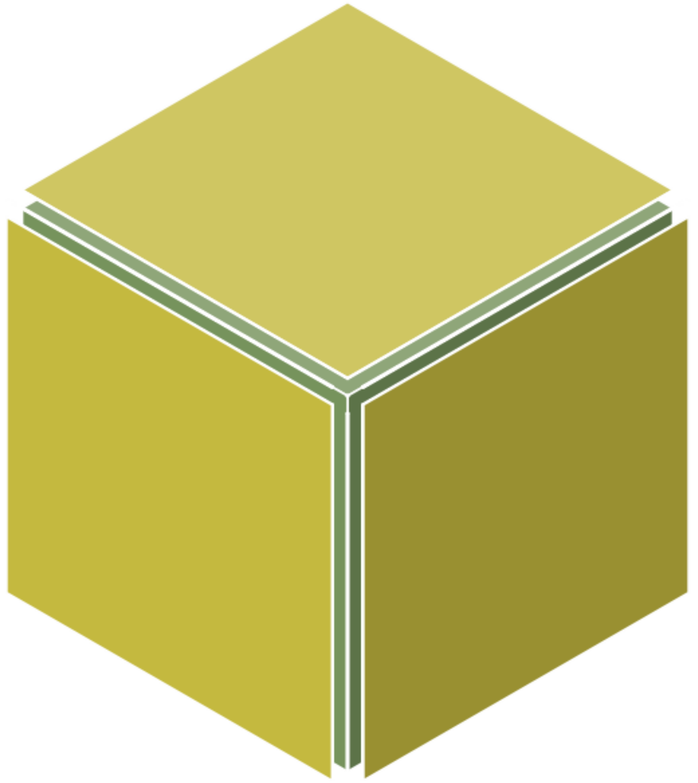
$$f(x) = x^3$$

$$\frac{df}{dx} = \underbrace{\square + \square + \square}_{3x^2} + \underbrace{\text{—} + \text{—} + \text{—} + \cdot}_{3x\,dx} + \underbrace{\cdot}_{dx^2}$$

$dx^2 = 0.0225$

$3x\,dx = 0.450$

$$\frac{df}{dx} = 3x^2 + 3x\,dx + dx^2$$



$$f(x) = x^3$$

$$\frac{df}{dx} = \underbrace{\square + \square + \square}_{3x^2}$$

Cube with side x: volume x^3

- 3 slabs: $x^2\,dx$ each
- 3 rods: $x\,dx^2$ each
- 1 corner: dx^3

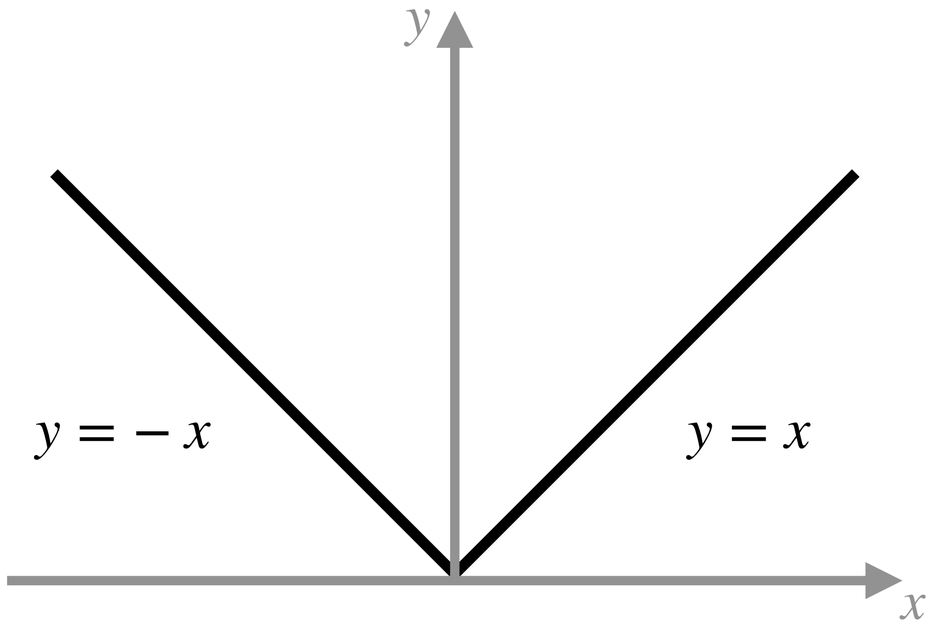
$$\frac{df}{dx} = 3x^2 + 3x\,dx + dx^2 \to 3x^2$$

as $dx \to 0$

derivative of a function

example

$$f(x) = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$



is not differentiable at a certain point because it is not continuous at (0,0), we can draw many many many tangent lines there. Also:

for $h > 0$, $\dfrac{f(0+h)-f(0)}{h} = \dfrac{h-0}{h} = 1$

for $h < 0$, $\dfrac{f(0+h)-f(0)}{h} = \dfrac{-h-0}{h} = -1$

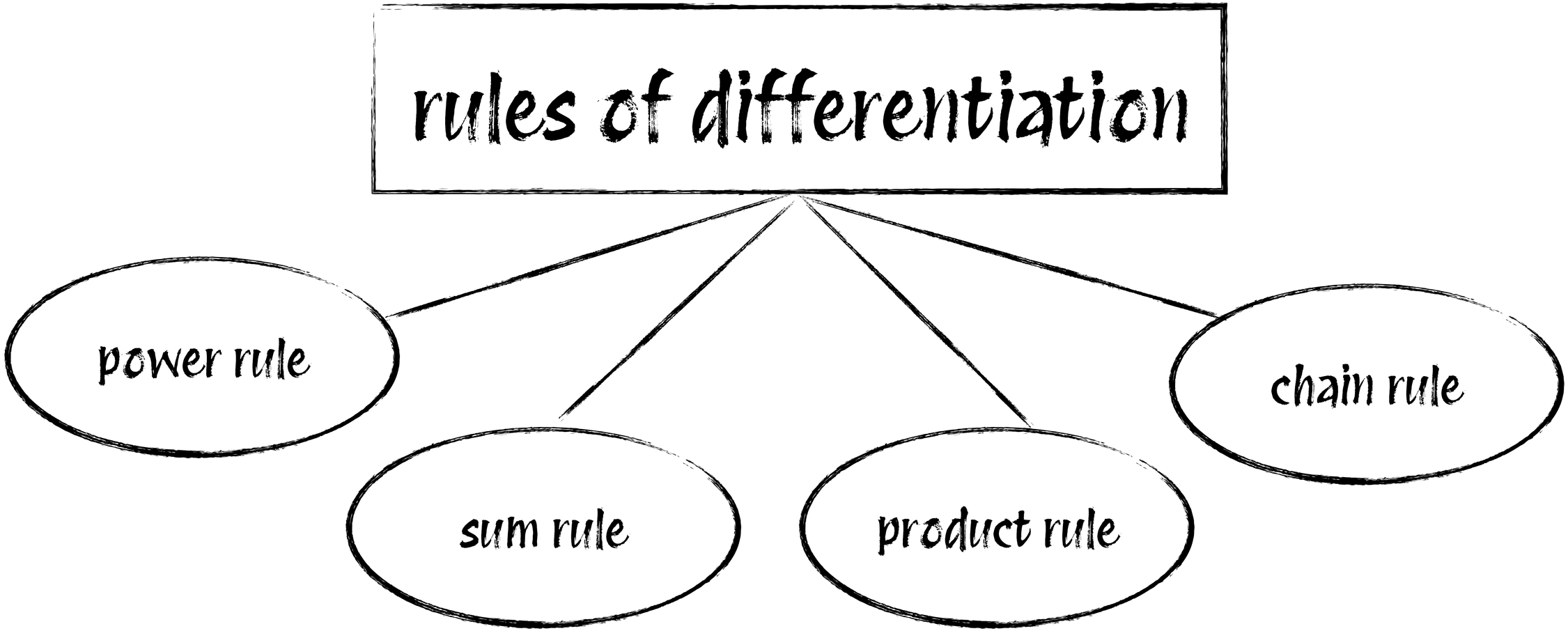
i.e. $\lim\limits_{h\to 0} \dfrac{f(0+h)-f(0)}{h}$ does not exist since taking the limit from both sides must give the same answer.

derivative of a function

exercise 1

Solve $\frac{d}{dx}3x^2$ using the definition of derivative $\frac{d}{dx}f(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

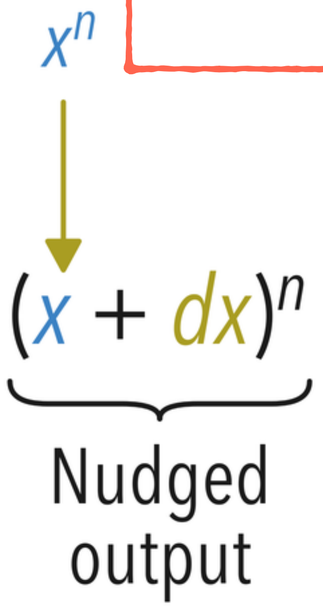
In theory, we could now go on and find derivatives of "all" algebraic functions by definition.
But that's too time consuming and impractical…



we build a “machine” with these three bits to help us to help us differentiate functions

the power rule

Let f be a function $f(x) = x^n$. Then, $\frac{d}{dx} f(x) = f'(x) = nx^{n-1}$



the power rule

Let f be a function $f(x) = x^n$. Then, $\dfrac{d}{dx} f(x) = f'(x) = nx^{n-1}$

x^n

$$\underbrace{(x + dx)^n}_{\text{Nudged output}} = \overbrace{(x + dx)(x + dx)(x + dx) \cdots (x + dx)}^{n \text{ times}}$$

$$= x^n + \underbrace{n\,x^{n-1}\,dx}$$

$$+ dx \cdot x \cdot x \cdots x$$
$$+ x \cdot dx \cdot x \cdots x$$
$$+ x \cdot x \cdot dx \cdots x$$
$$\vdots$$
$$+ x \cdot x \cdot x \cdots dx$$

the power rule

Let f be a function $f(x) = x^n$. Then, $\dfrac{d}{dx} f(x) = f'(x) = nx^{n-1}$

x^n

$$\underbrace{(x + dx)^n}_{\text{Nudged output}} = \overbrace{(x + dx)(x + dx)(x + dx) \cdots (x + dx)}^{n \text{ times}}$$

$$= \underbrace{x^n}_{\text{starting value}} + n\,x^{n-1}\,dx + (\text{Multiple of } dx^2)$$

$$+ dx \cdot x \cdot x \cdots x$$
$$+ x \cdot dx \cdot x \cdots x$$
$$+ x \cdot x \cdot dx \cdots x$$
$$\vdots$$
$$+ x \cdot x \cdot x \cdots dx$$

x^n is the output before the nudge, so it cancels:

$$df = (x + dx)^n - x^n$$
$$= n\,x^{n-1}\,dx + (\text{Multiple of } dx^2)$$
$$df/dx = n\,x^{n-1} + (\text{Multiple of } dx)$$
$$\to n\,x^{n-1} \text{ as } dx \to 0$$

the sum rule

The derivative of a sum of two functions is the sum of their derivatives

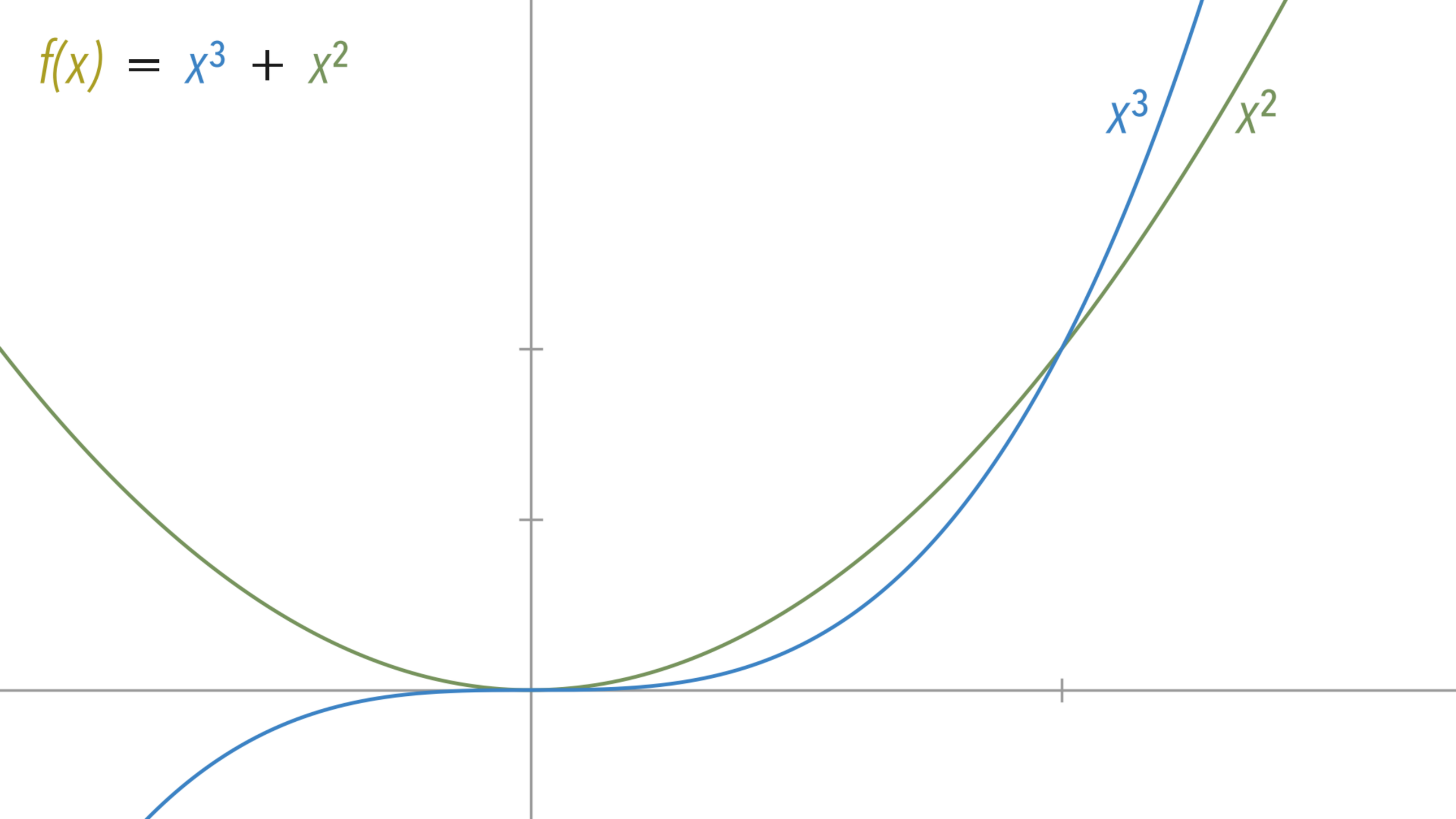
If g and h are differentiable, then

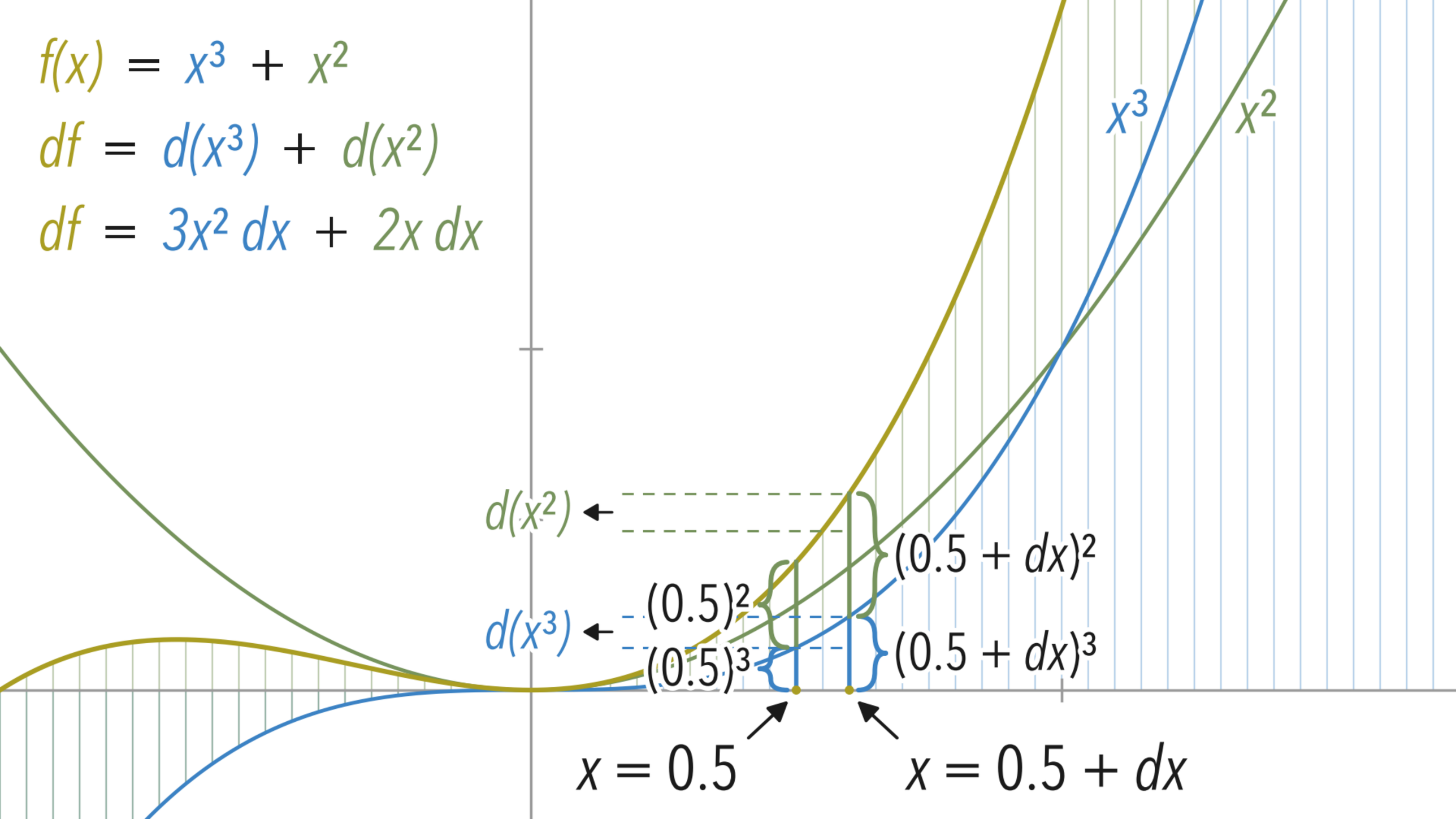
$$\frac{d}{dx}\big(g(x)+h(x)\big)=\frac{d}{dx}\big(g(x)\big)+\frac{d}{dx}\big(h(x)\big)=g'(x)+h'(x)$$

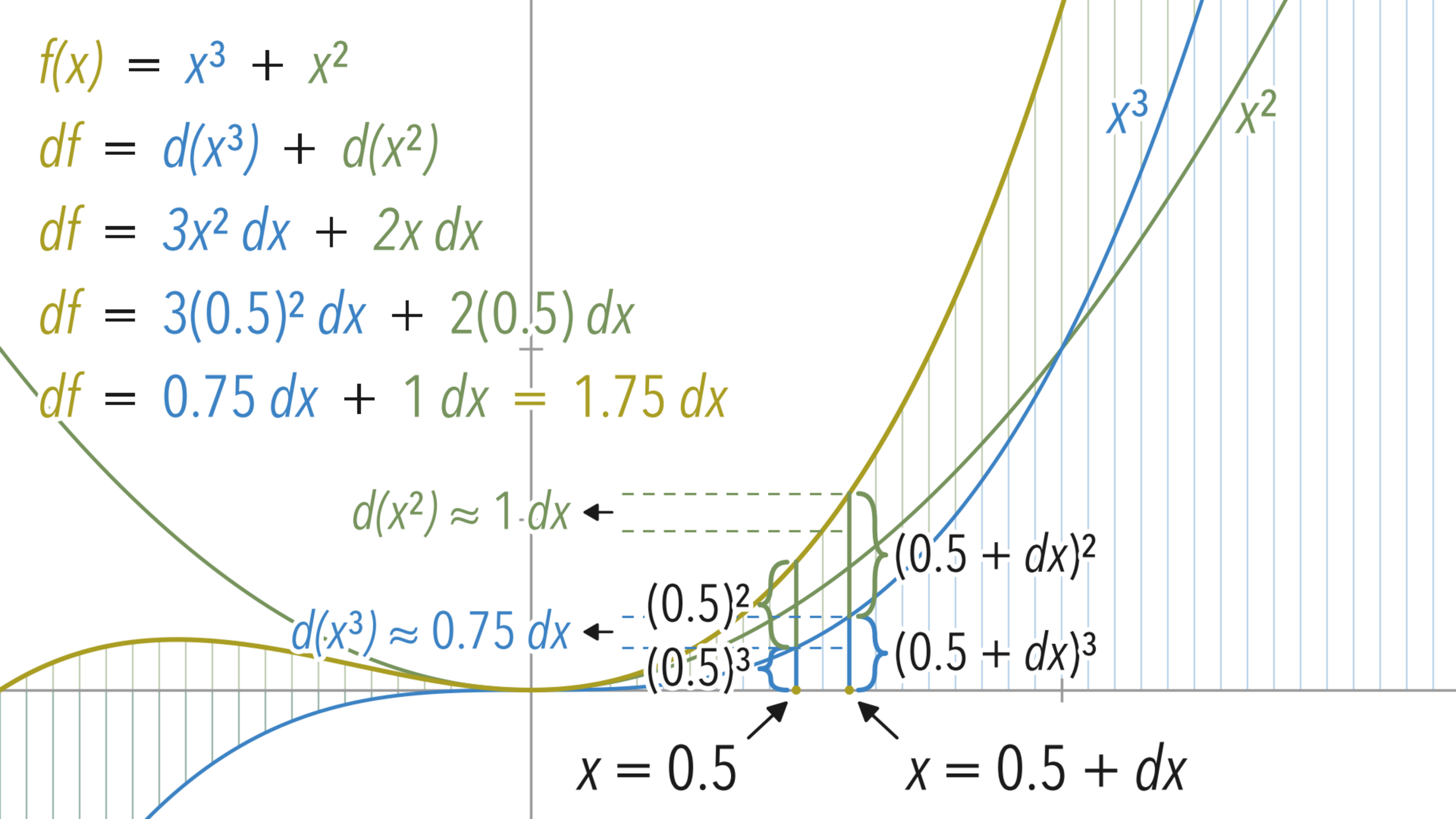
example

Differentiate the function $f(x)=x^3+x^4$.

$$\frac{d}{dx}\left(x^3+x^4\right)=\frac{d}{dx}\left(x^3\right)+\frac{d}{dx}\left(x^4\right)=3x^2+4x^3$$


$f(x) = x^3 + x^2$
x^3
x^2


$f(x) = x^3 + x^2$
$df = d(x^3) + d(x^2)$
$df = 3x^2\,dx + 2x\,dx$
x^3
x^2
$d(x^2)$
$d(x^3)$
$(0.5)^2$
$(0.5)^3$
$(0.5 + dx)^2$
$(0.5 + dx)^3$
$x = 0.5$
$x = 0.5 + dx$


$f(x) = x^3 + x^2$
$df = d(x^3) + d(x^2)$
$df = 3x^2\,dx + 2x\,dx$
$df = 3(0.5)^2\,dx + 2(0.5)\,dx$
$df = 0.75\,dx + 1\,dx = 1.75\,dx$
x^3
x^2
$d(x^2) \approx 1\,dx$
$d(x^3) \approx 0.75\,dx$
$(0.5)^2$
$(0.5)^3$
$(0.5 + dx)^2$
$(0.5 + dx)^3$
$x = 0.5$
$x = 0.5 + dx$

a special case

If $g(x) = c$ is constant then $\dfrac{d}{dx}\left(cf(x)\right) = c\dfrac{d}{dx}\left(f(x)\right) = cf'(x)$

The sum rule can then be generalized as:

If $f_1, f_2, \cdots, f_n$ are differentiable and $a_1, a_2, \cdots, a_n$ are constants, then

$$\frac{d}{dx}\left(a_1 f_1(x) + a_2 f_2(x) + \cdots + a_n f_n(x)\right) = a_1 f'(x) + a_2 f_2'(x) + \cdots + a_n f_n'(x)$$

example

Differentiating polynomial of degree n with constant coefficients given by

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \cdots + a_n x^n$$

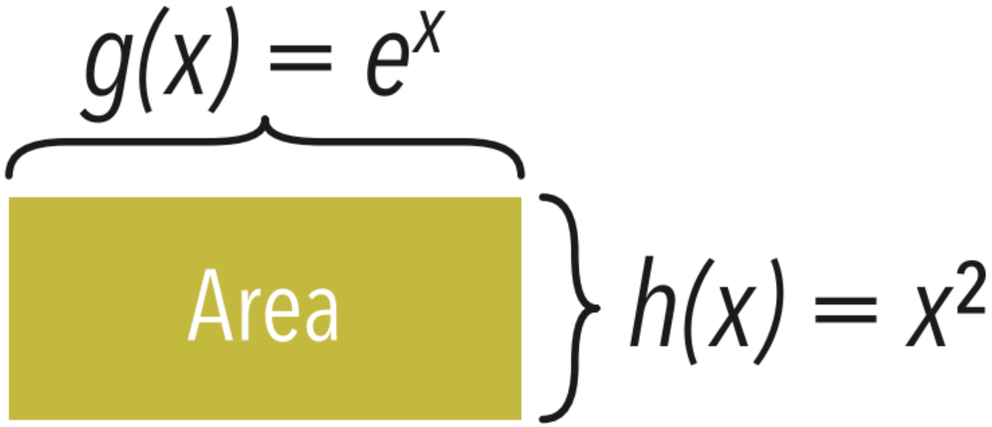
yields a polynomial of degree $n - 1$ with constant coefficients given by

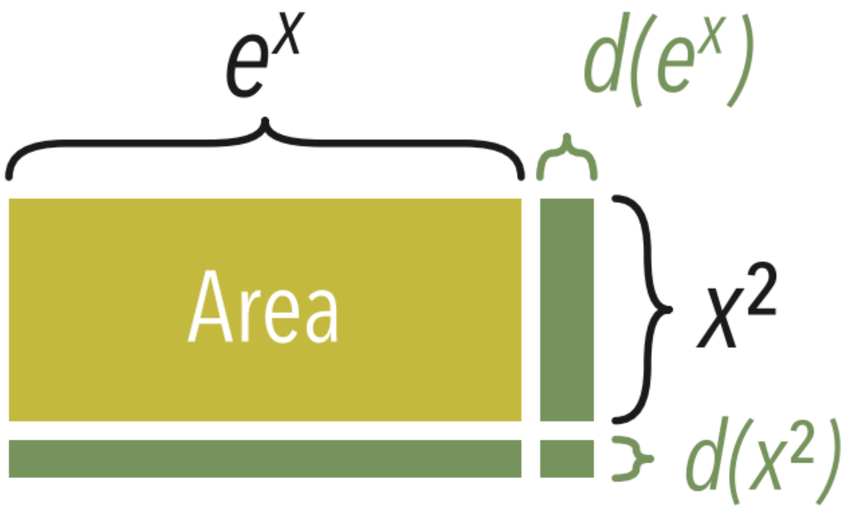
$$f'(x) = a_1 + 2a_2 x + 3a_3 x^2 + \cdots + n a_n x^{n-1}$$

the product rule

If g and h are differentiable, then

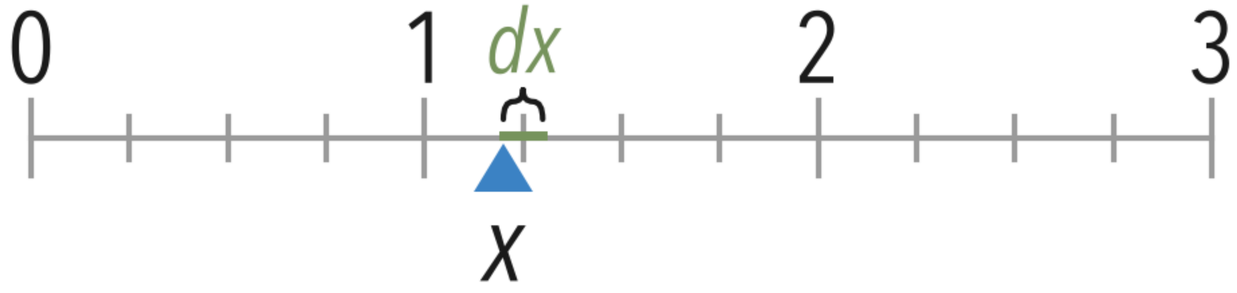
$$\frac{d}{dx}\Big(g(x)h(x)\Big) = g'(x)h(x) + g(x)h'(x)$$

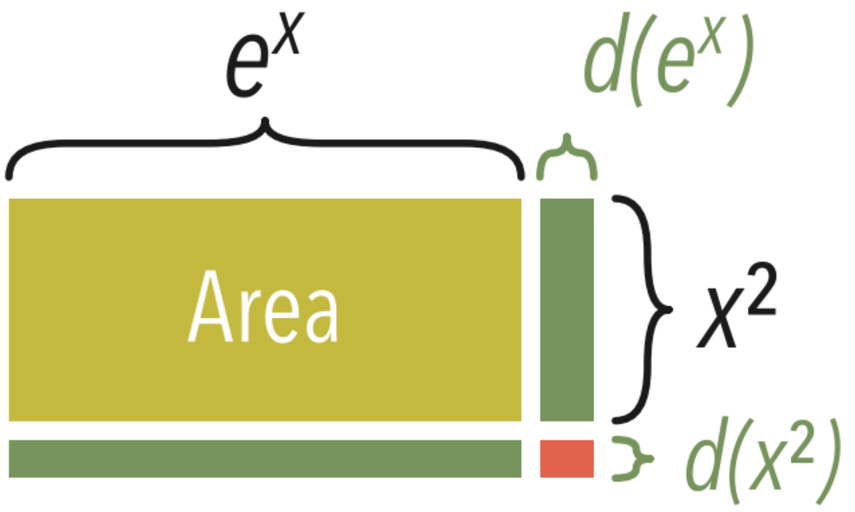

$g(x) = e^x$
Area
$h(x) = x^2$



$$f(x) = e^x x^2 = \text{Area}$$

$$df =$$



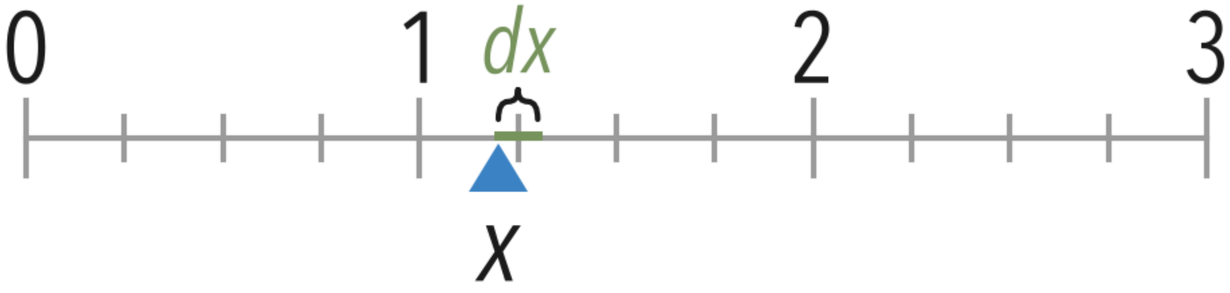


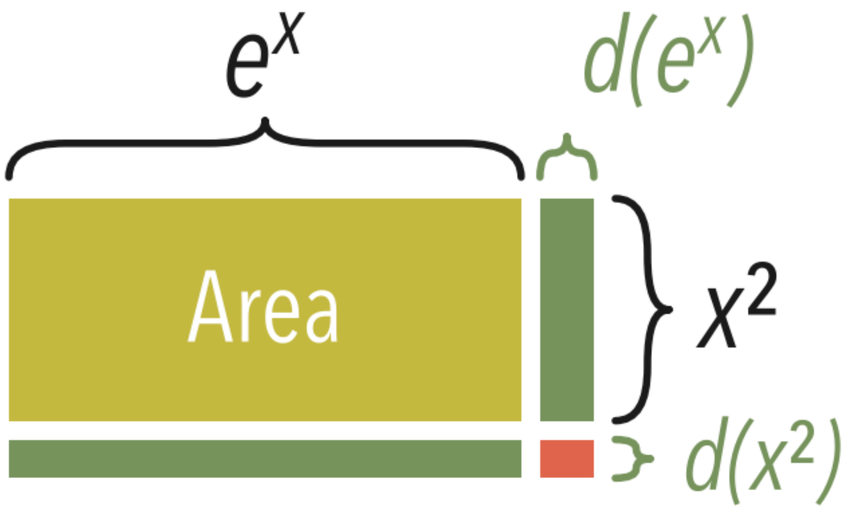
$$f(x) = e^x x^2 = \text{Area}$$

$$df = e^x\, d(x^2) + x^2\, d(e^x)$$

Ignore → + ■

$(\ldots)(dx)^2$

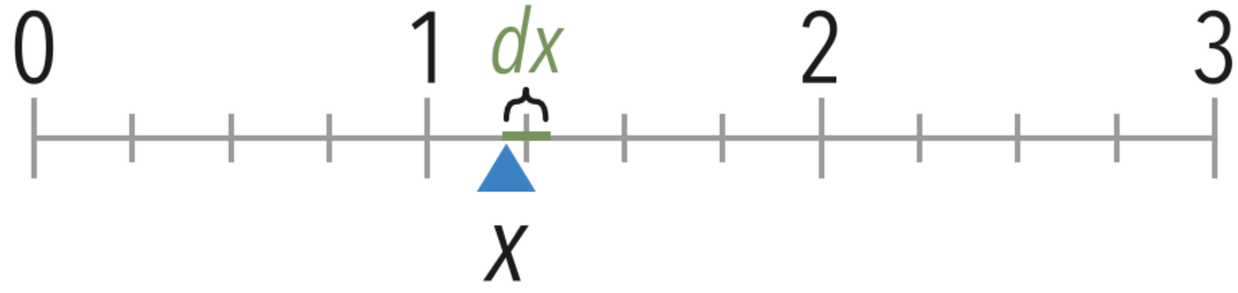


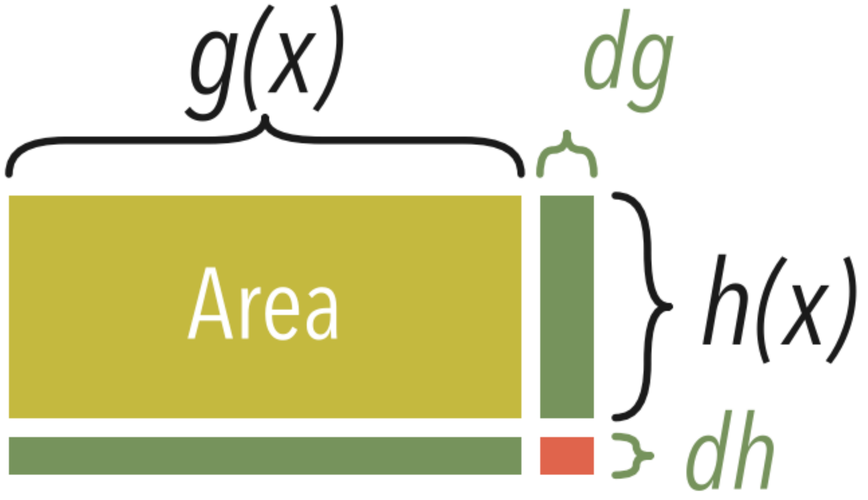


$$f(x) = e^x x^2 = \text{Area}$$

$$df = e^x\, d(x^2) + x^2\, d(e^x)$$

$$\frac{df}{dx} = e^x \cdot 2x + x^2 \cdot e^x$$



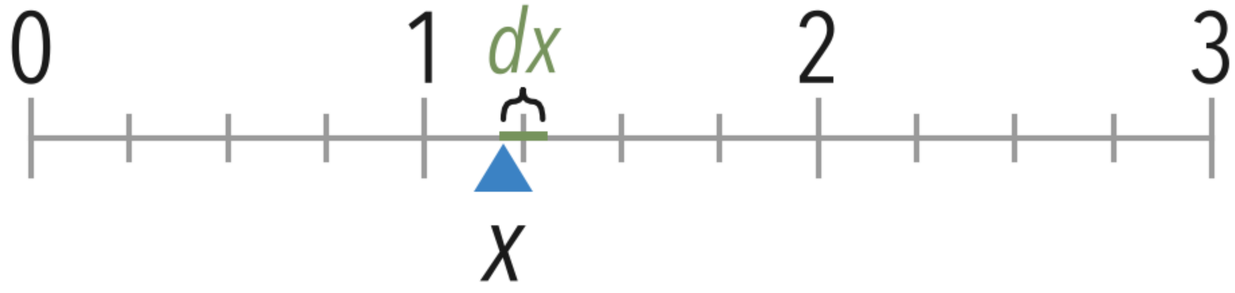


$$f(x) = g(x)\,h(x) = \text{Area}$$

$$df = g(x)\,dh + h(x)\,dg$$

$$\frac{df}{dx} = g(x)\frac{dh}{dx} + h(x)\frac{dg}{dx}$$

"Left d(Right) + Right d(Left)"



the product rule

If g and h are differentiable, then

$$\frac{d}{dx}\Big(g(x)h(x)\Big) = g'(x)h(x) + g(x)h'(x)$$

example

Differentiate the function $f(x) = (x^2 + 1)(x^3 - 1)$.

Here we set $g(x) = x^2 + 1 \implies g'(x) = 2x$

$h(x) = x^3 - 1 \implies h'(x) = 3x^2$

and use the product rule:

$$\begin{aligned}\frac{d}{dx}\Big((x^2 + 1)(x^3 - 1)\Big) &= 2x(x^3 - 1) + (x^2 + 1)3x^2 \\ &= 2x^4 - 2x + 3x^4 + 3x^2 \\ &= 5x^4 + 3x^2 - 2x\end{aligned}$$

the chain rule

Let's now consider differentiating "compositions" of functions:

$$g \circ h(x) = g(h(x)) \quad \text{"do h then g"}$$

or

$$g \circ h \circ q(x) = g(h(q(x))) \quad \text{"do q then h then g"}$$

If g and h are differentiable, then

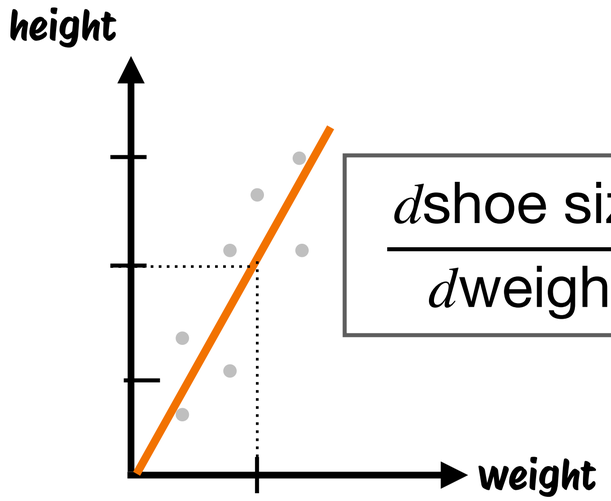
$$\frac{d}{dx}\Big(g(h(x))\Big) = g'(h(x)) \cdot h'(x)$$

the chain rule

a change in weight leads to a change in shoe size

for every one unit increase in **weight** there is a 2 unit increase in **height**

$$\frac{\Delta\text{height}}{\Delta\text{weight}} = \frac{d\text{height}}{d\text{weight}} = \frac{2}{1} = 2 \implies \text{height} = 2 \cdot \text{weight}$$



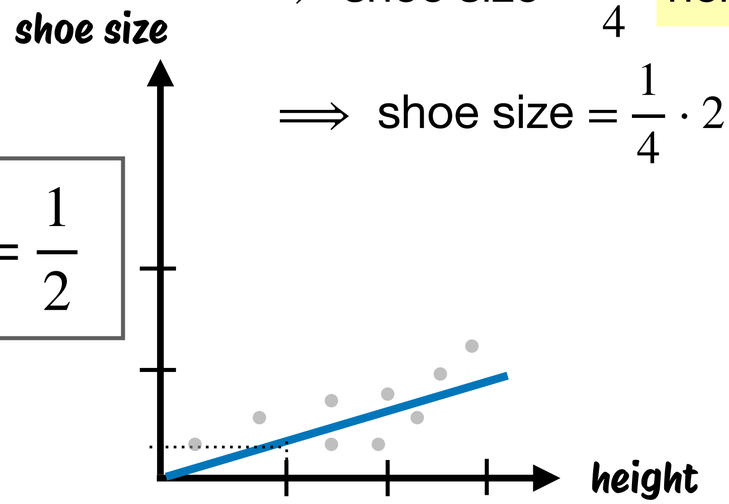
for every one unit increase in **height** there is a 1/4 unit increase in **shoe size**

$$\frac{\Delta\text{shoe size}}{\Delta\text{height}} = \frac{d\text{shoe size}}{d\text{height}} = \frac{1}{4}$$

$$\implies \text{shoe size} = \frac{1}{4} \cdot \text{height}$$

$$\implies \text{shoe size} = \frac{1}{4} \cdot 2 \cdot \text{weight}$$

$$\frac{d\text{shoe size}}{d\text{weight}} = \frac{d\text{shoe size}}{d\text{height}} \cdot \frac{d\text{height}}{d\text{weight}} = \frac{1}{2}$$



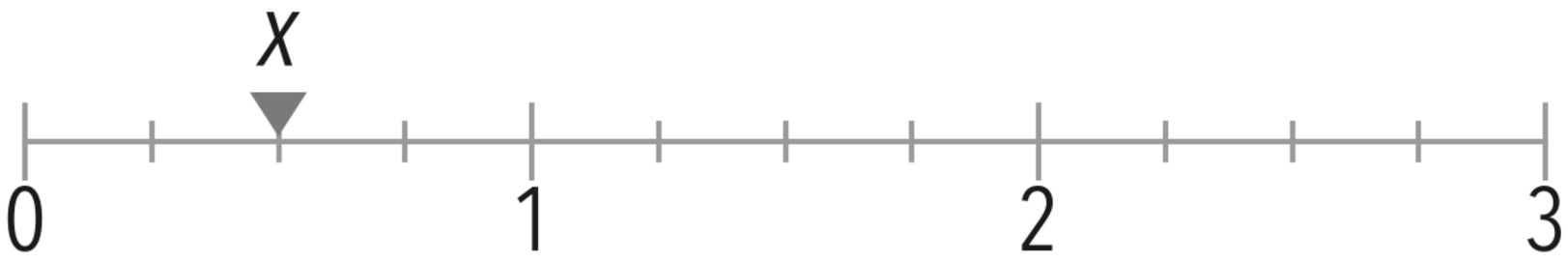
the chain rule

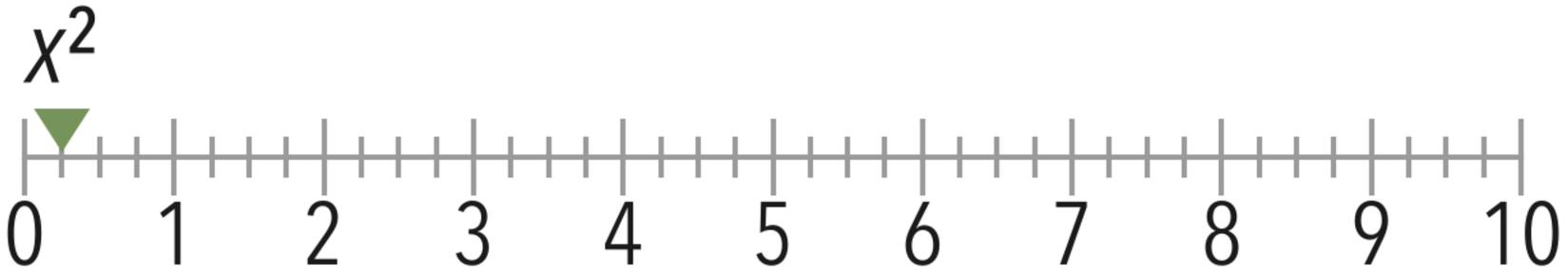
Function composition

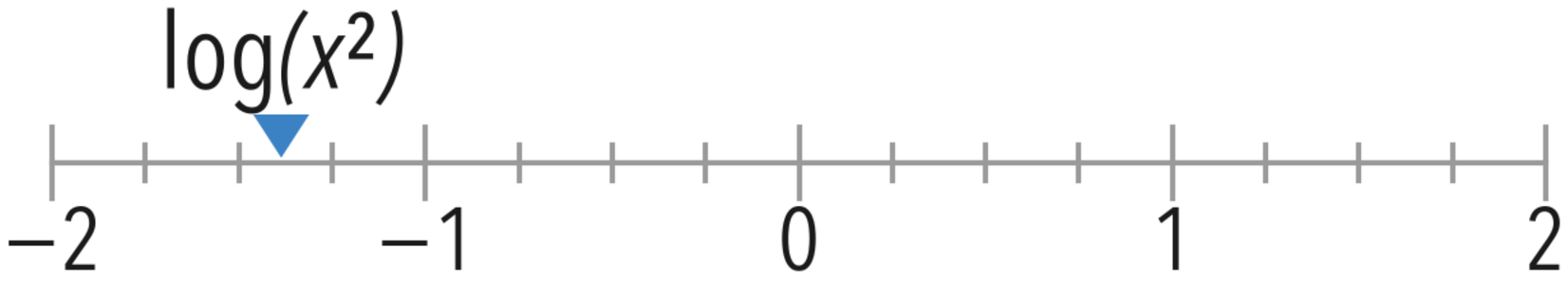
$g(x) = \log(x)$ $\quad$ $h(x) = x^2$

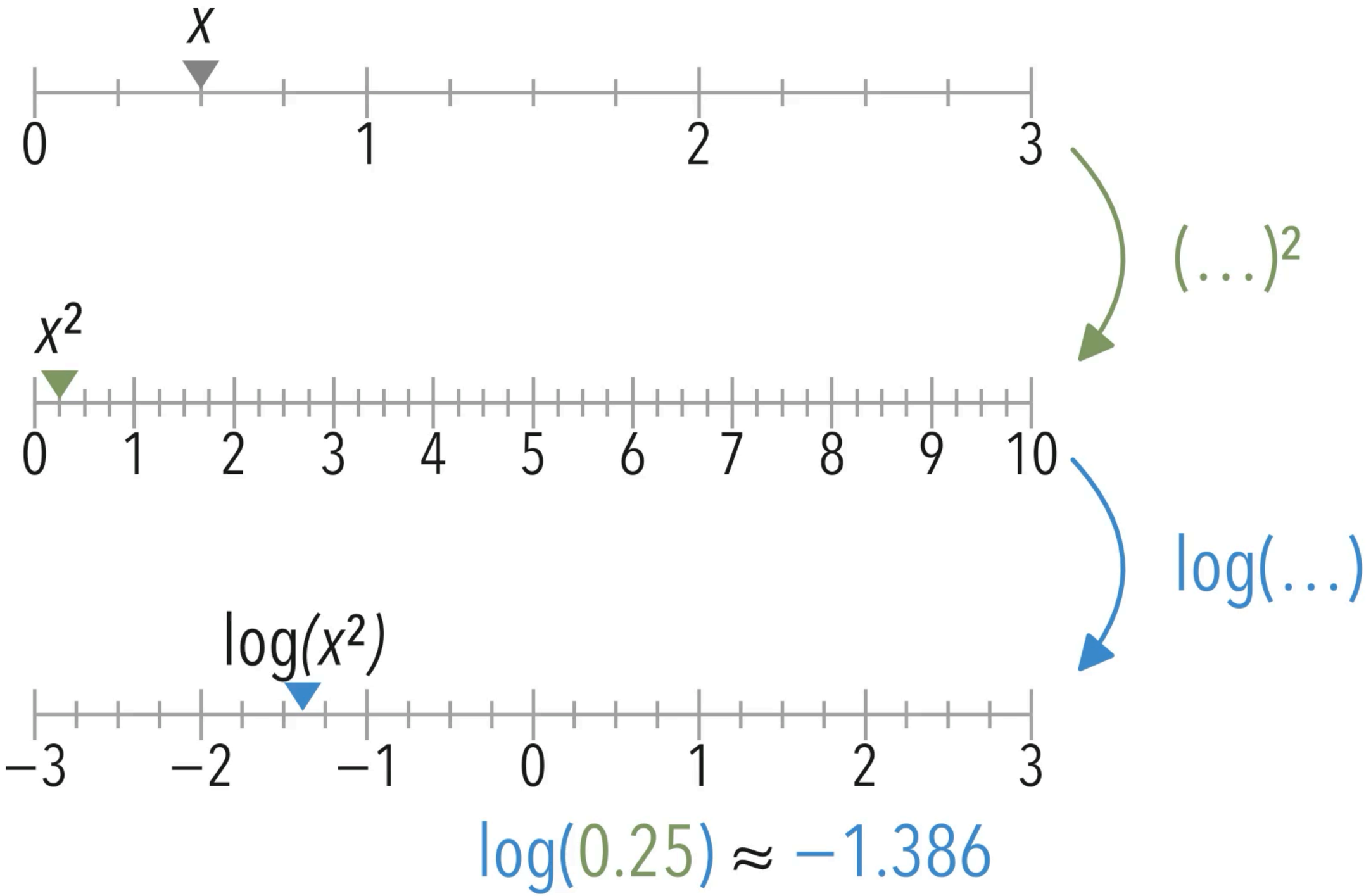
$$g(h(x)) = \underbrace{\log(x^2)}$$

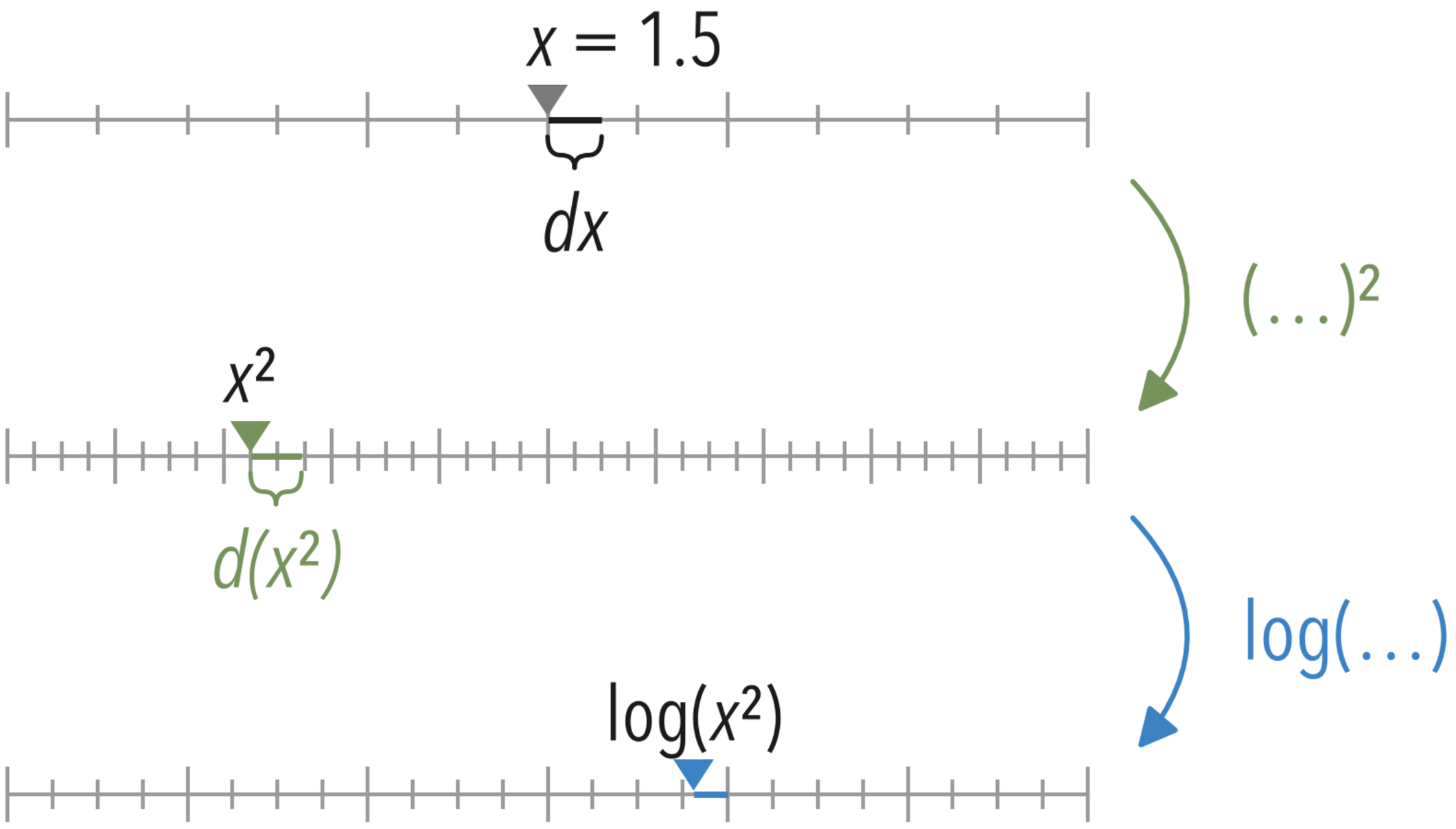
Derivative?

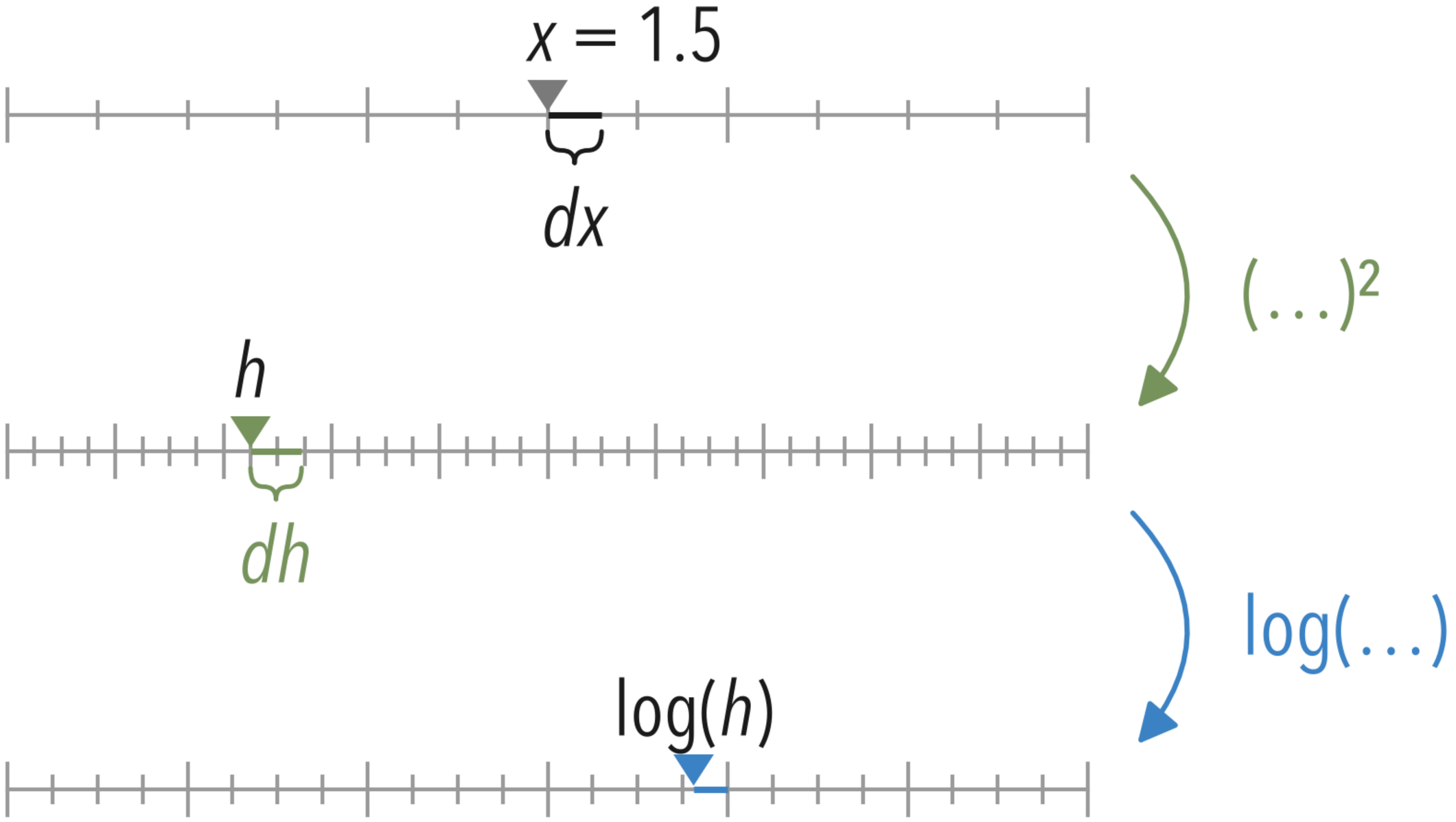

x
0
1
2
3

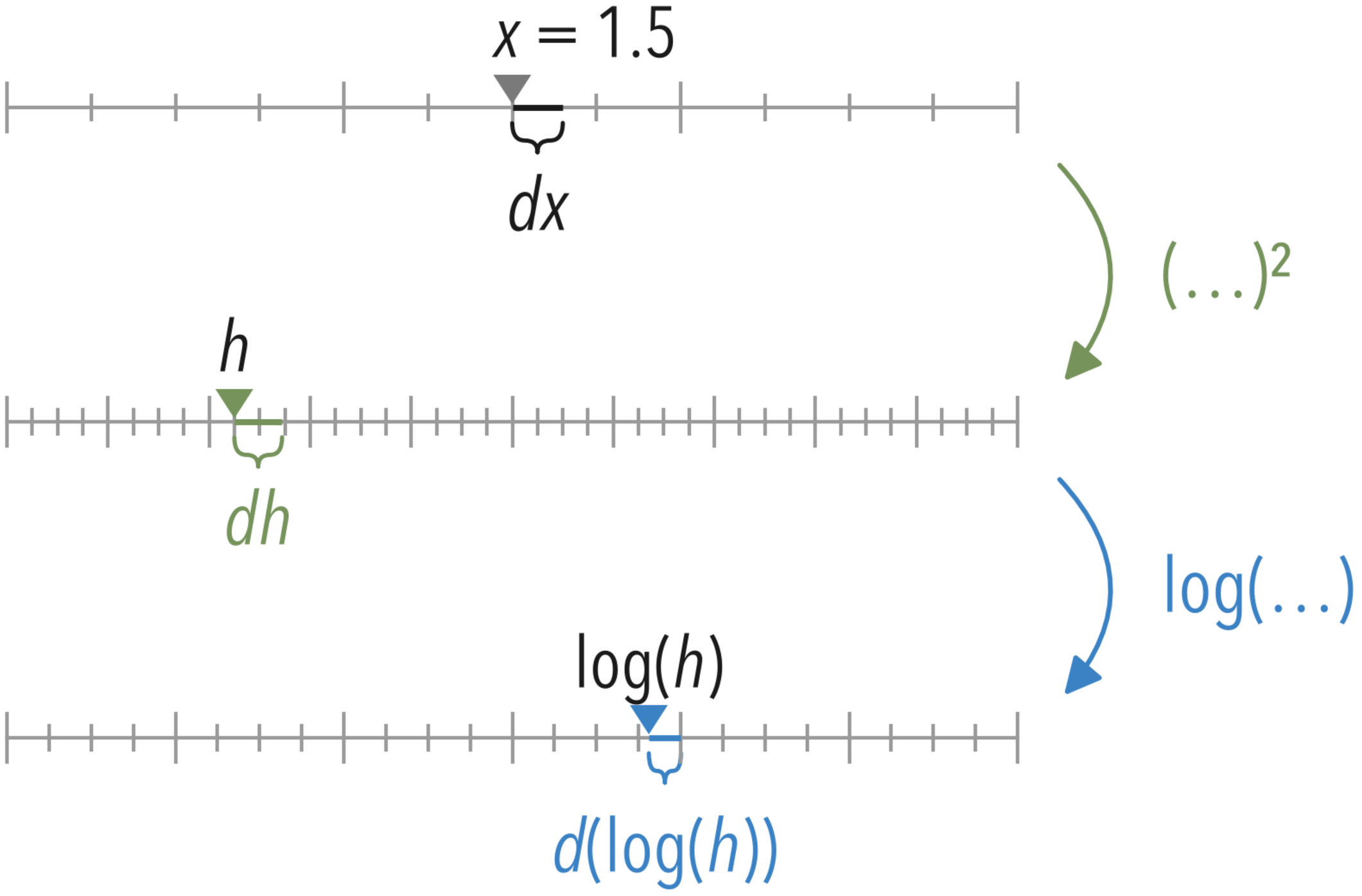

x²
0
1
2
3
4
5
6
7
8
9
10

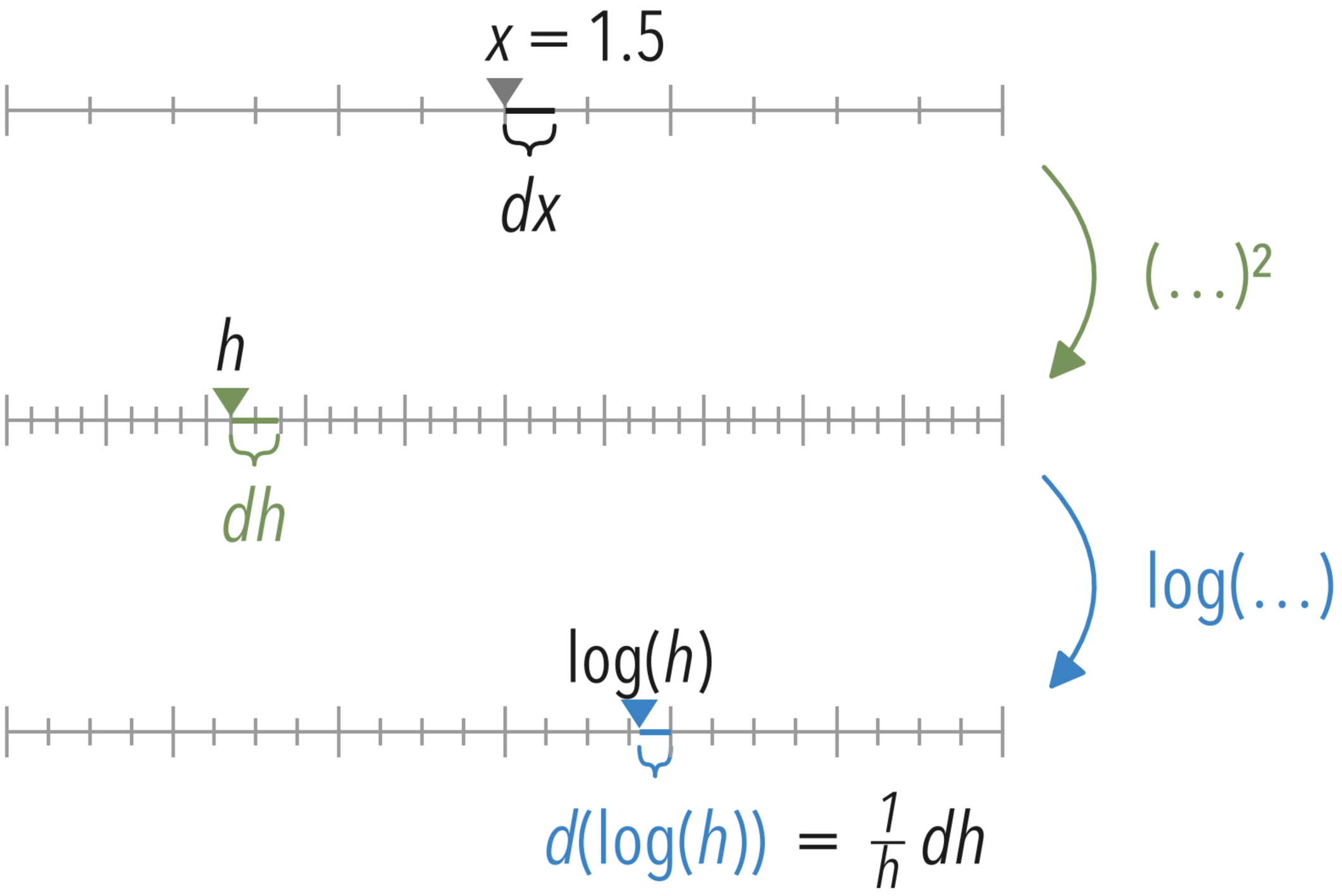

log(x²)
−2
−1
0
1
2

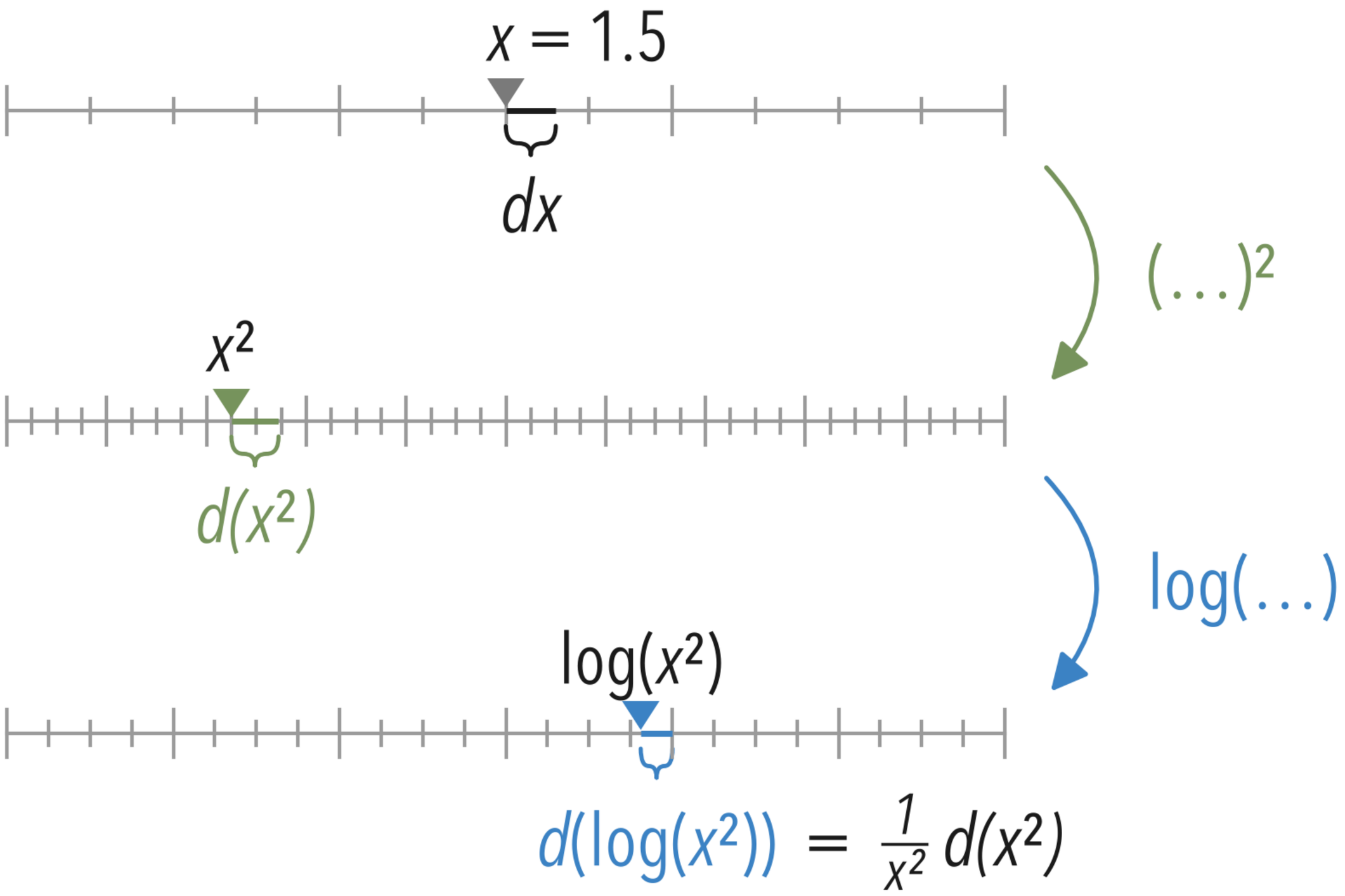

x
0
1
2
3
(…)²
x²
0
1
2
3
4
5
6
7
8
9
10
log(…)
log(x²)
−3
−2
−1
0
1
2
3
log(0.25) ≈ −1.386

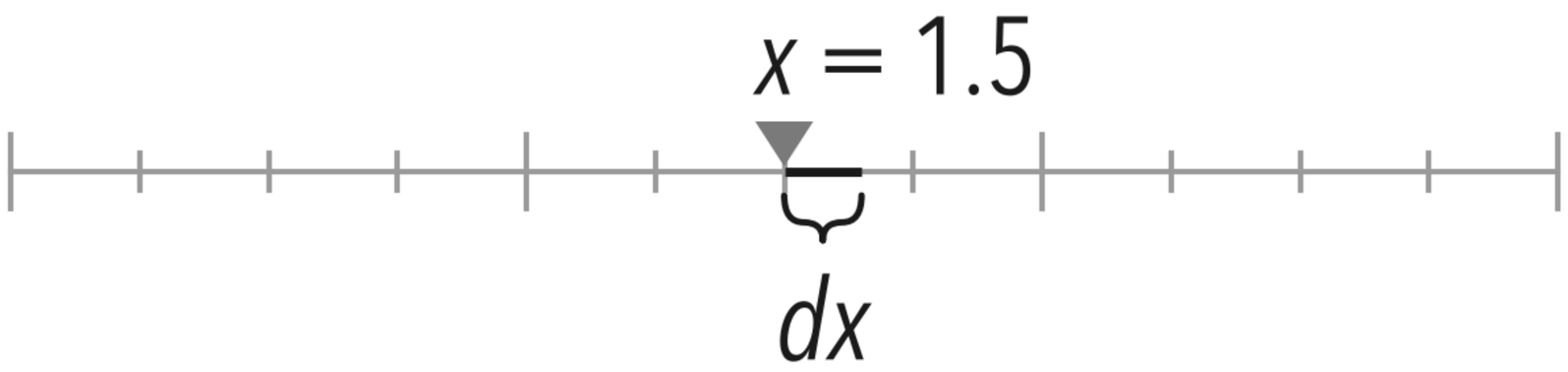

x = 1.5
dx
(…)²
x²
d(x²)
log(…)
log(x²)


x = 1.5
dx
(…)²
h
dh
log(…)
log(h)


x = 1.5
dx
(…)²
h
dh
log(…)
log(h)
d(log(h))


$x = 1.5$
dx
$(\ldots)^2$
h
dh
$\log(\ldots)$
$\log(h)$
$d(\log(h)) = \frac{1}{h}\,dh$


x = 1.5
dx
(…)²
x²
d(x²)
log(…)
log(x²)
d(log(x²)) = 1/x² d(x²)

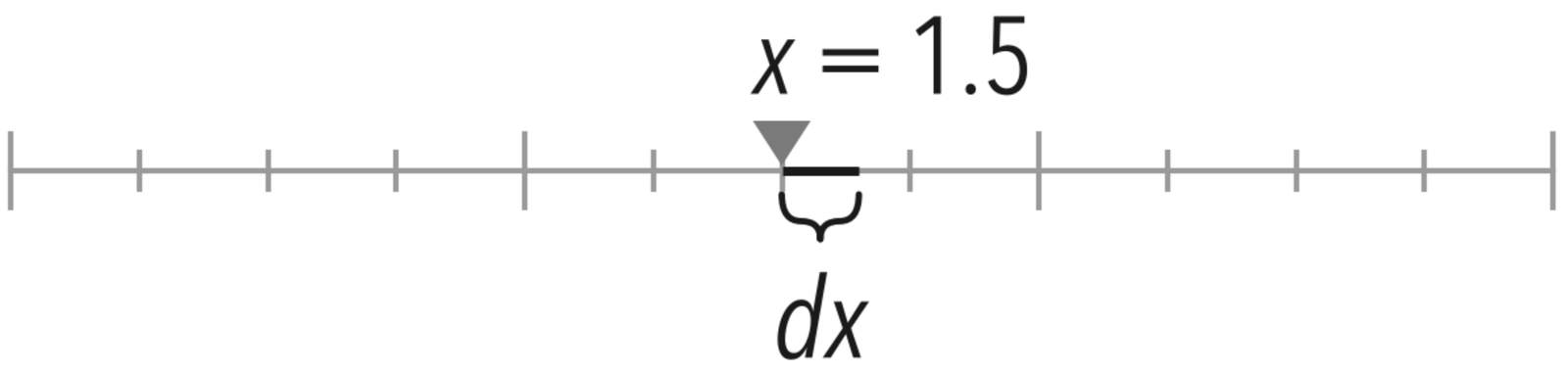




$$d(x^2) = 2x\,dx$$



$$d(\log(x^2)) = \frac{1}{x^2}\,2x\,dx$$





$$d(x^2) = 2x\,dx$$

For example,

$$\frac{1}{1.5^2} \cdot 2(1.5)\,dx$$



$$d(\log(x^2)) = \frac{1}{x^2} 2x\,dx$$

the chain rule

$$\frac{d}{dx}\log(x^2) = \frac{1}{x^2}\cdot 2x$$

$$\frac{d}{dx}\underbrace{g(\overbrace{h(x)}^{\text{Inner}})}_{\text{Outer}} = \underbrace{\frac{dg}{dh}(h(x))}_{\text{d(Outer)}}\overbrace{\frac{dh}{dx}(x)}^{\text{d(Inner)}}$$

the chain rule

$$\frac{d}{dx}\,g(h(x)) = \frac{dg}{dh} \cdot \frac{dh}{dx} = \frac{dg}{\cancel{dh}}\,\frac{\cancel{dh}}{dx} = \frac{dg}{dx}$$

derivative of the outer function, evaluated at $h(x)$

derivative of the inner function

$$\frac{d}{dx}\,g(h(x)) = g'(h(x)) \cdot h'(x)$$

Example: $g(x) = \log(x)$, $g'(x) = \frac{1}{x}$ $\quad h(x) = x^2$, $h'(x) = 2x$

$$\frac{d}{dx}\log(x^2) = \frac{1}{x^2} \cdot 2x = \frac{2}{x}$$

the chain rule

example

To differentiate the function $y(x) = (x^3 + 2x)^{10}$ we can set $f(z) = z^{10}$ and $g(x) = x^3 + 2x$.

We then have that $f'(z) = 10z^9$ and $g'(x) = 3x^2 + 2$. Then

$$\begin{aligned}\frac{d}{dx}(y(x)) &= \frac{d}{dx}(f(g(x)) \\ &= \frac{d}{dx}(x^3 + 2x)^{10} \\ &= f'(g(x))g'(x) \\ &= 10(x^3 + 2x)^9 \cdot (3x^2 + 2)\end{aligned}$$

exercise 3

Differentiate the function $y(x) = \dfrac{1}{x^3}$

a special case: quotient rule

If g and h are differentiable, then

$$\frac{d}{dx}\left[\frac{g(x)}{h(x)}\right] = \frac{g'(x)\,h(x) - g(x)\,h'(x)}{[h(x)]^2}$$

This is simply the product rule for rational functions!

exercise 4

Write

$$\frac{g(x)}{h(x)} = g(x)\,[h(x)]^{-1}$$

and show that the quotient rule is the product rule for rational functions.

relating back to the epsilon-delta definitionn of limits

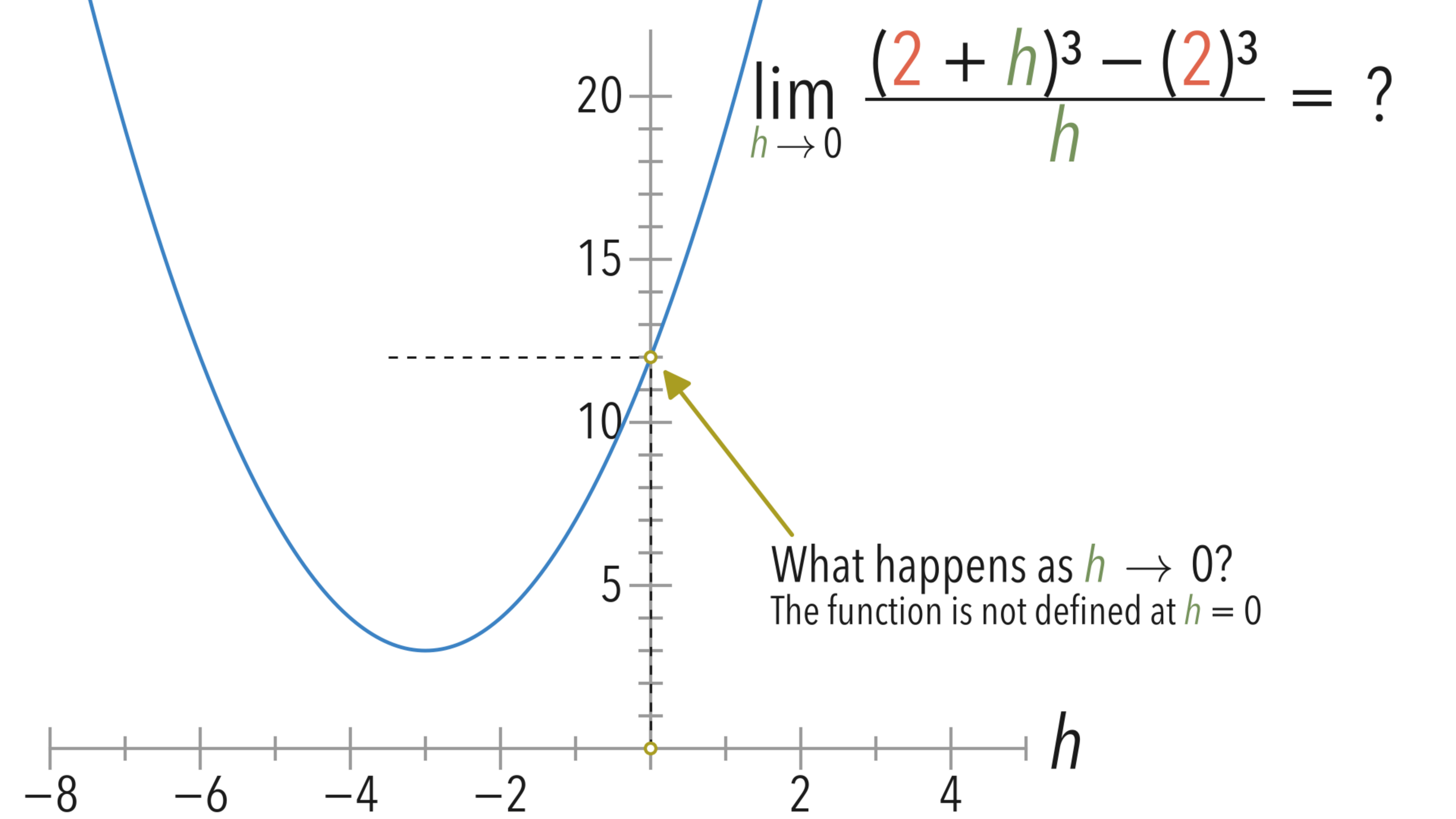
derivative of a function

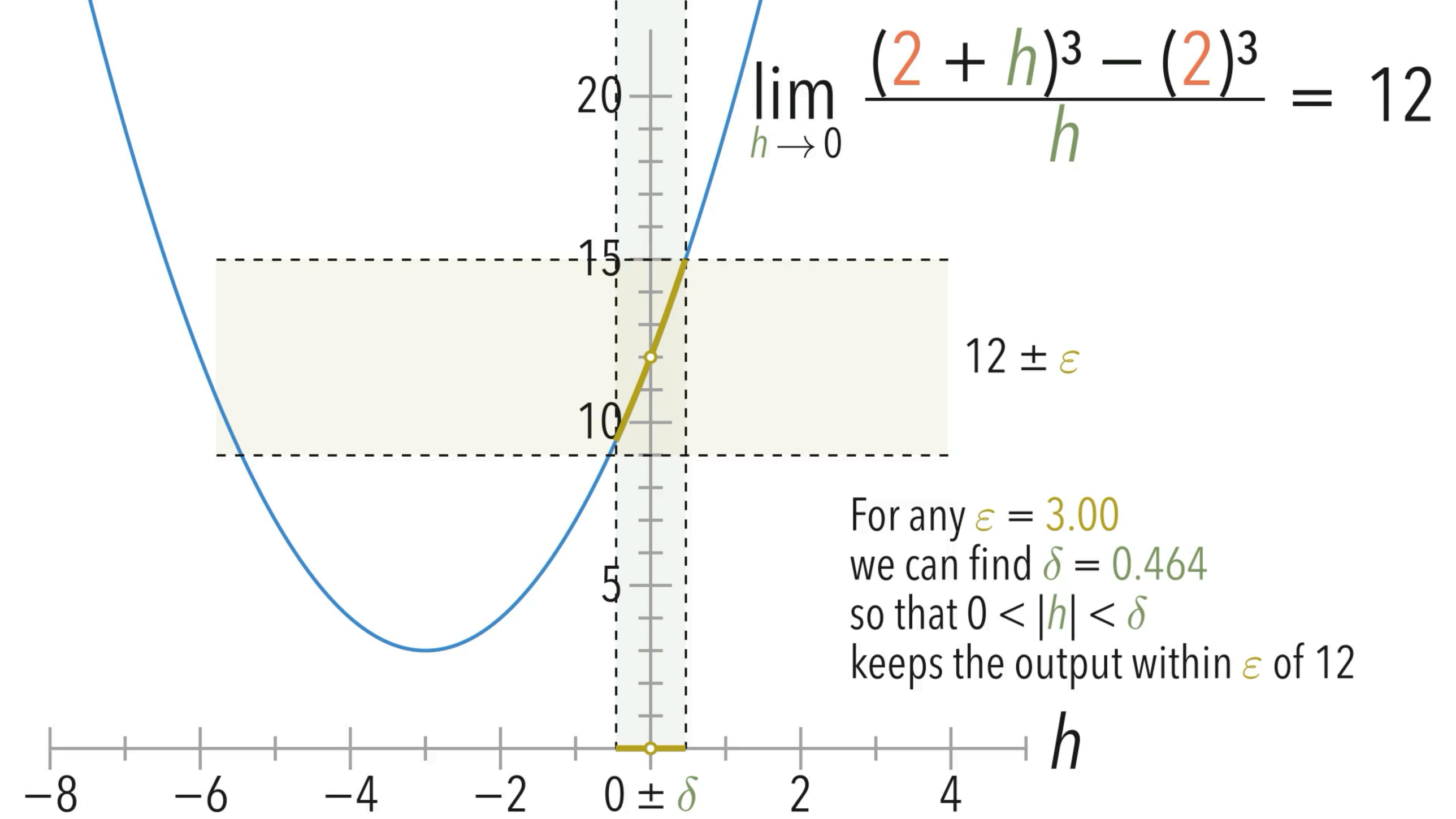
We define the derivative of a function f at x as

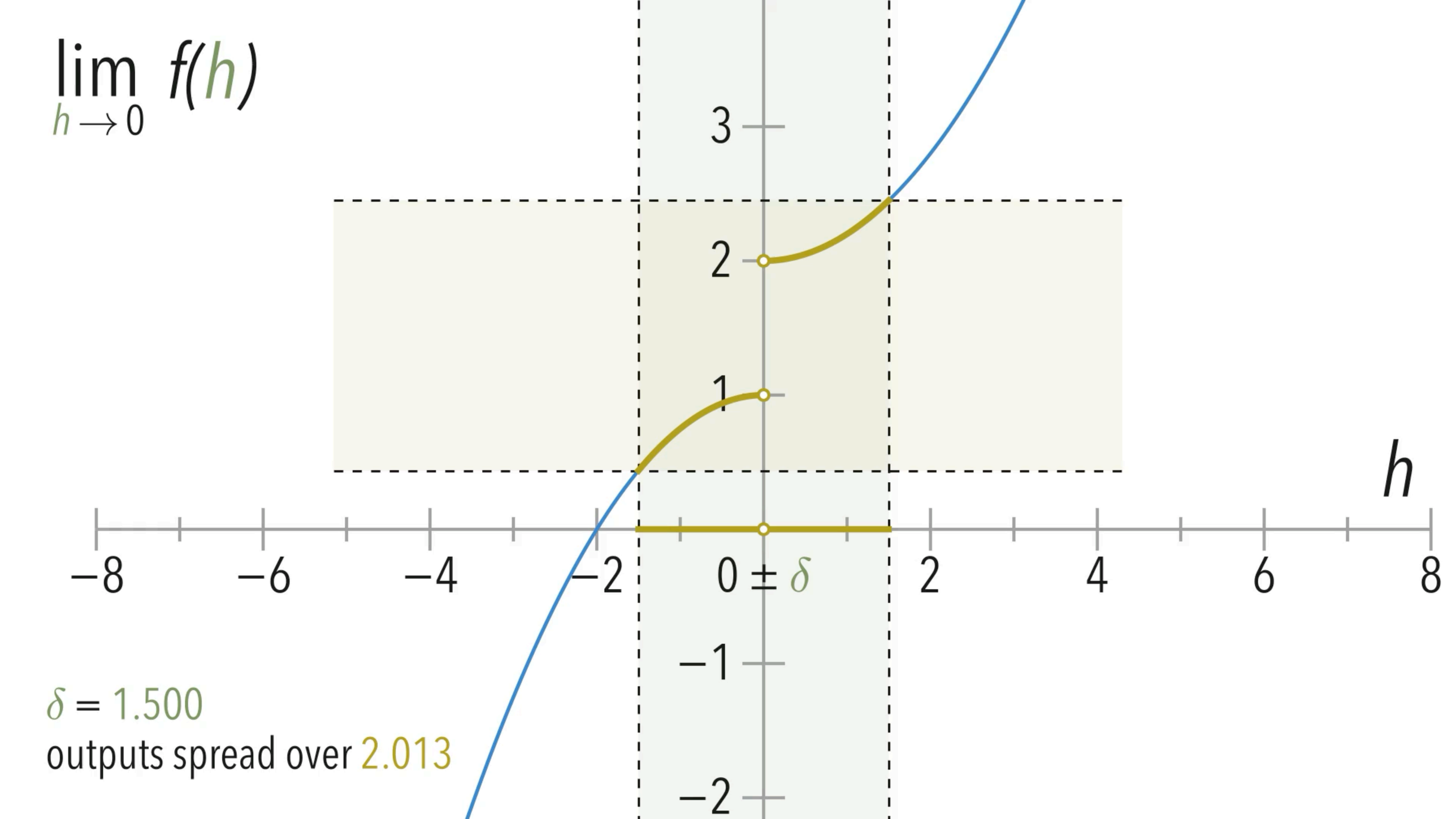
$$f'(x) = \frac{d}{dx} f(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

provided that the limit exists. If the limit exists, we say f is differentiable at x.

If we simply say f is differentiable, we mean f is differentiable at all values of x.
In this case, $f'(x)$ is also a function of x.


$\lim_{h \to 0} \frac{(2+h)^3 - (2)^3}{h} = ?$
What happens as $h \to 0$?
The function is not defined at $h = 0$
20
15
10
5
−8
−6
−4
−2
2
4
h


$\lim_{h \to 0} \frac{(2+h)^3 - (2)^3}{h} = 12$
20
15
10
5
$12 \pm \varepsilon$
For any $\varepsilon = 3.00$
we can find $\delta = 0.464$
so that $0 < |h| < \delta$
keeps the output within ε of 12
h
−8
−6
−4
−2
$0 \pm \delta$
2
4


lim h→0 f(h)
3
2
1
−1
−2
h
−8
−6
−4
−2
0 ± δ
2
4
6
8
δ = 1.500
outputs spread over 2.013

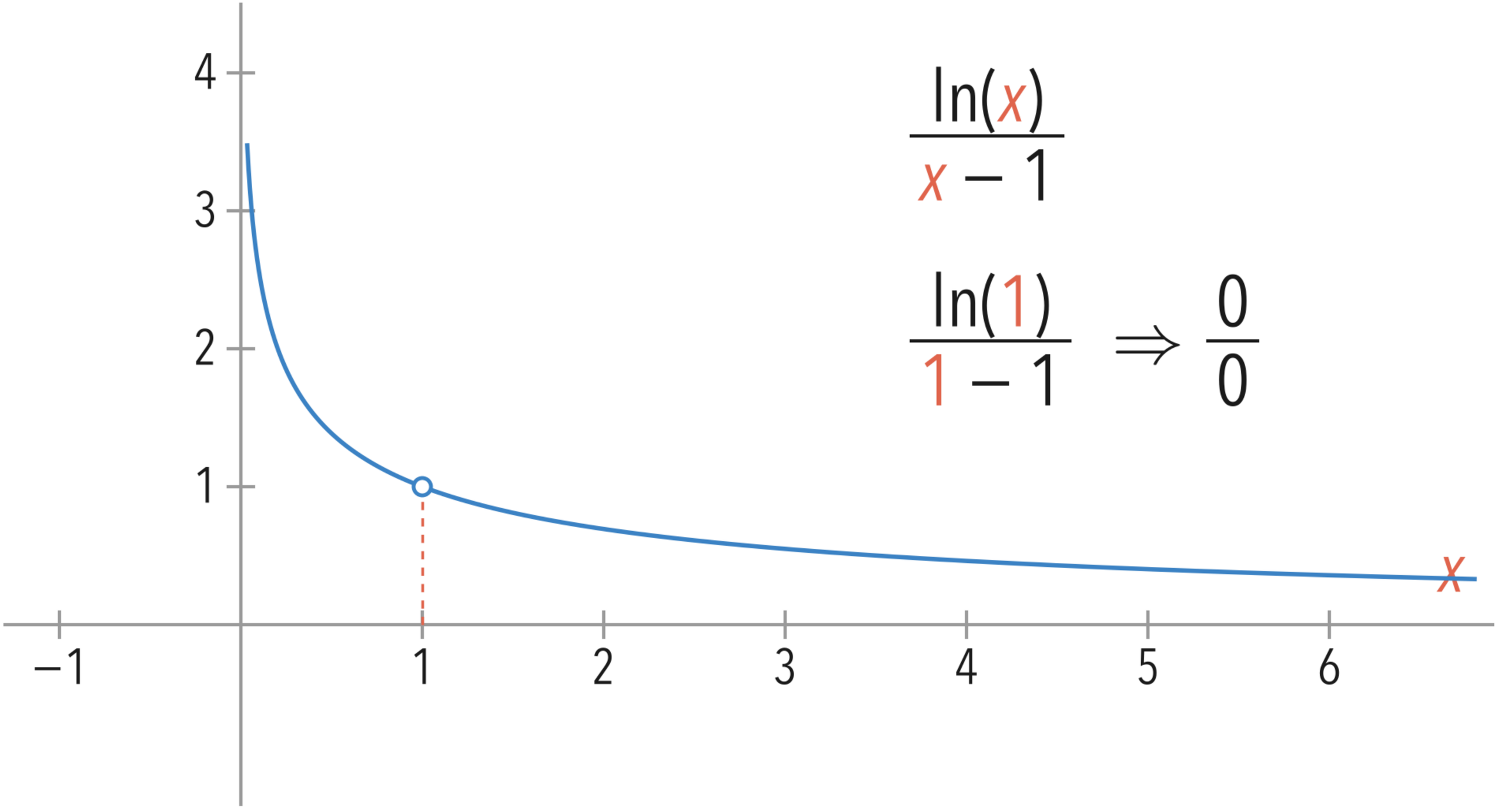
$$(\varepsilon, \delta)$$



$$\frac{df}{dx}(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

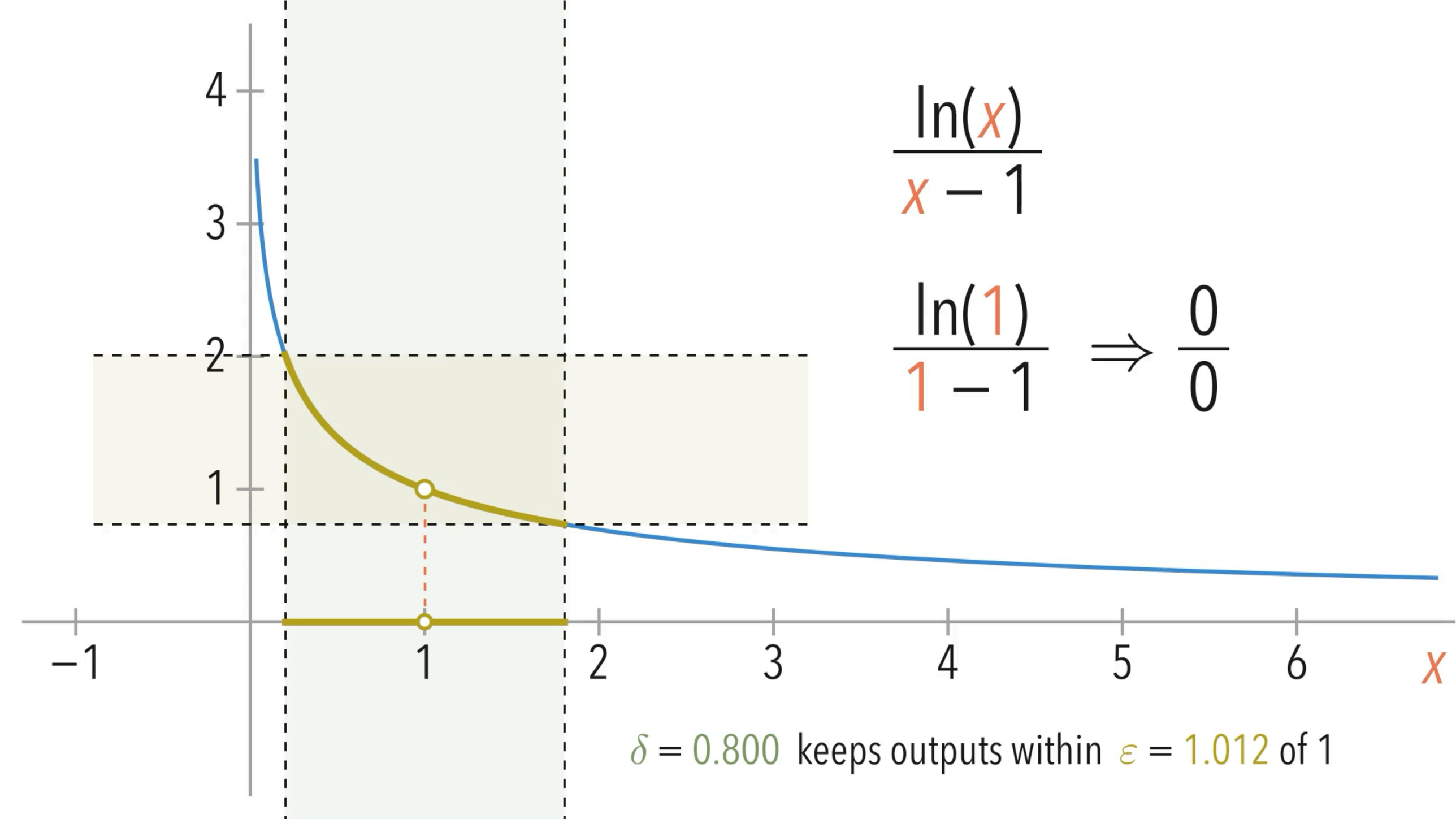
can we use this to compute limits?

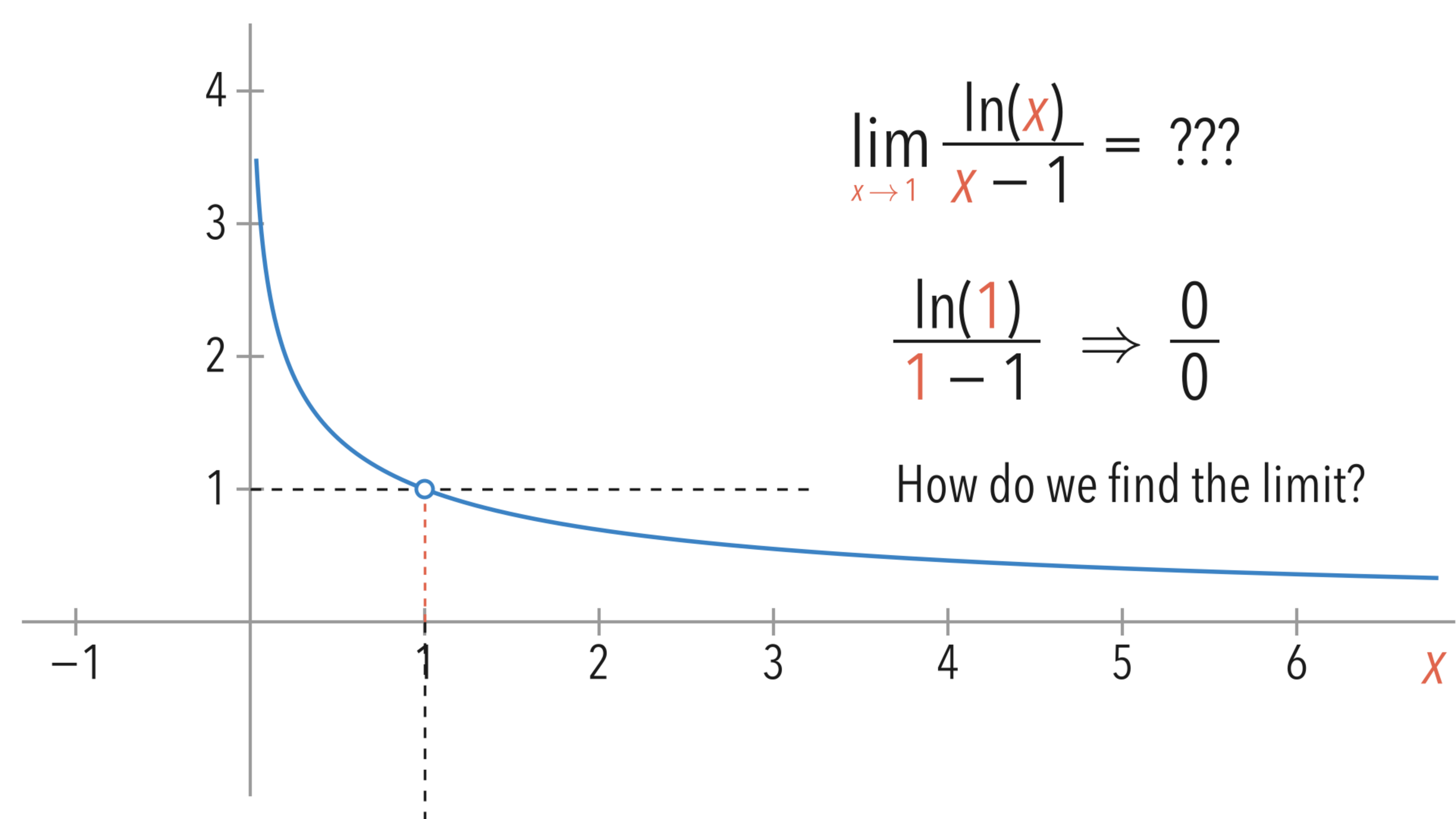
$$\lim_{h \to 0} \rightleftarrows \frac{df}{dx}$$



$$\frac{\ln(x)}{x-1}$$

$$\frac{\ln(1)}{1-1} \Rightarrow \frac{0}{0}$$

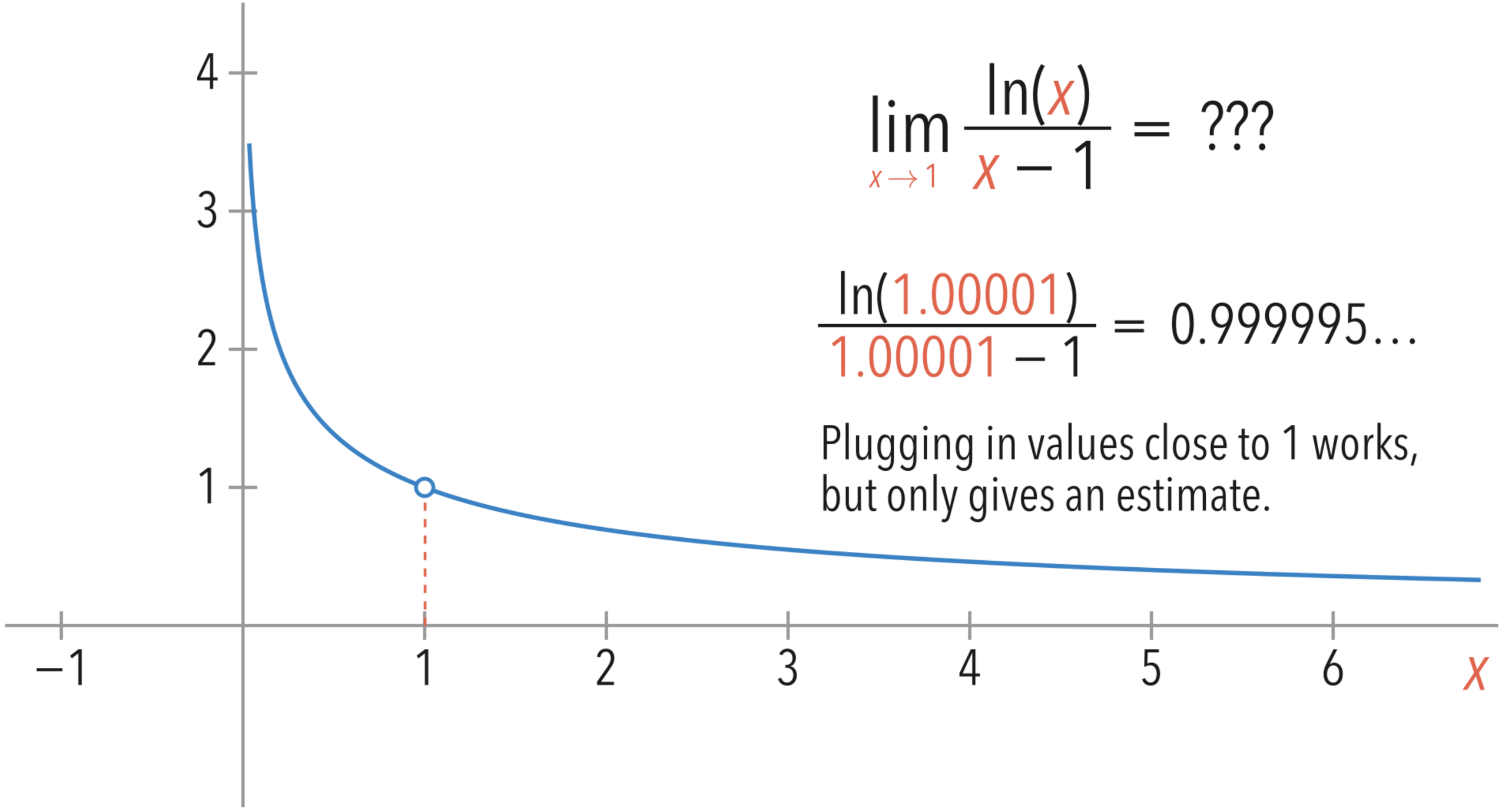

$\frac{\ln(x)}{x-1}$
$\frac{\ln(1)}{1-1} \Rightarrow \frac{0}{0}$
4
3
2
1
−1
1
2
3
4
5
6
x
$\delta = 0.800$ keeps outputs within $\varepsilon = 1.012$ of 1



$$\lim_{x \to 1} \frac{\ln(x)}{x - 1} = \text{???}$$

$$\frac{\ln(1)}{1 - 1} \Rightarrow \frac{0}{0}$$

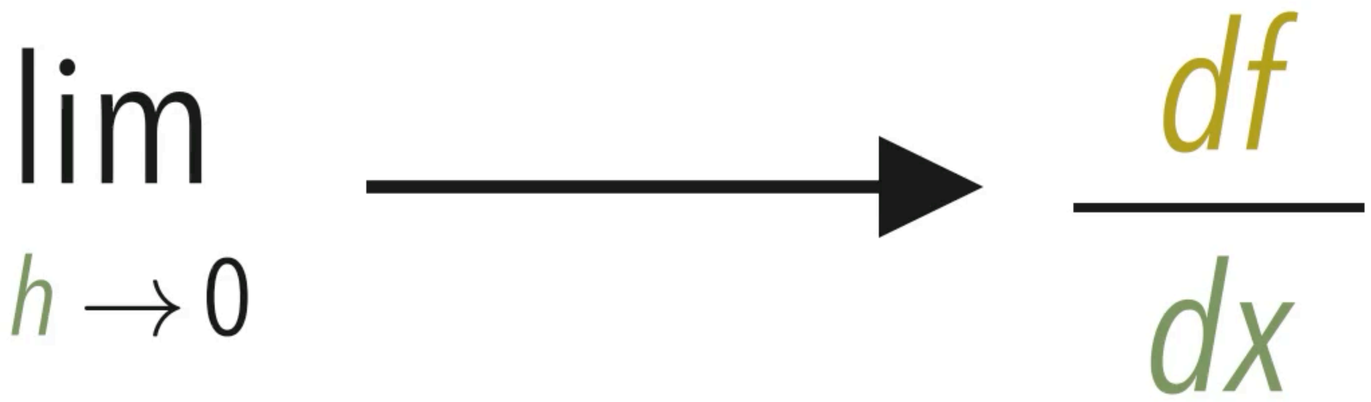
How do we find the limit?

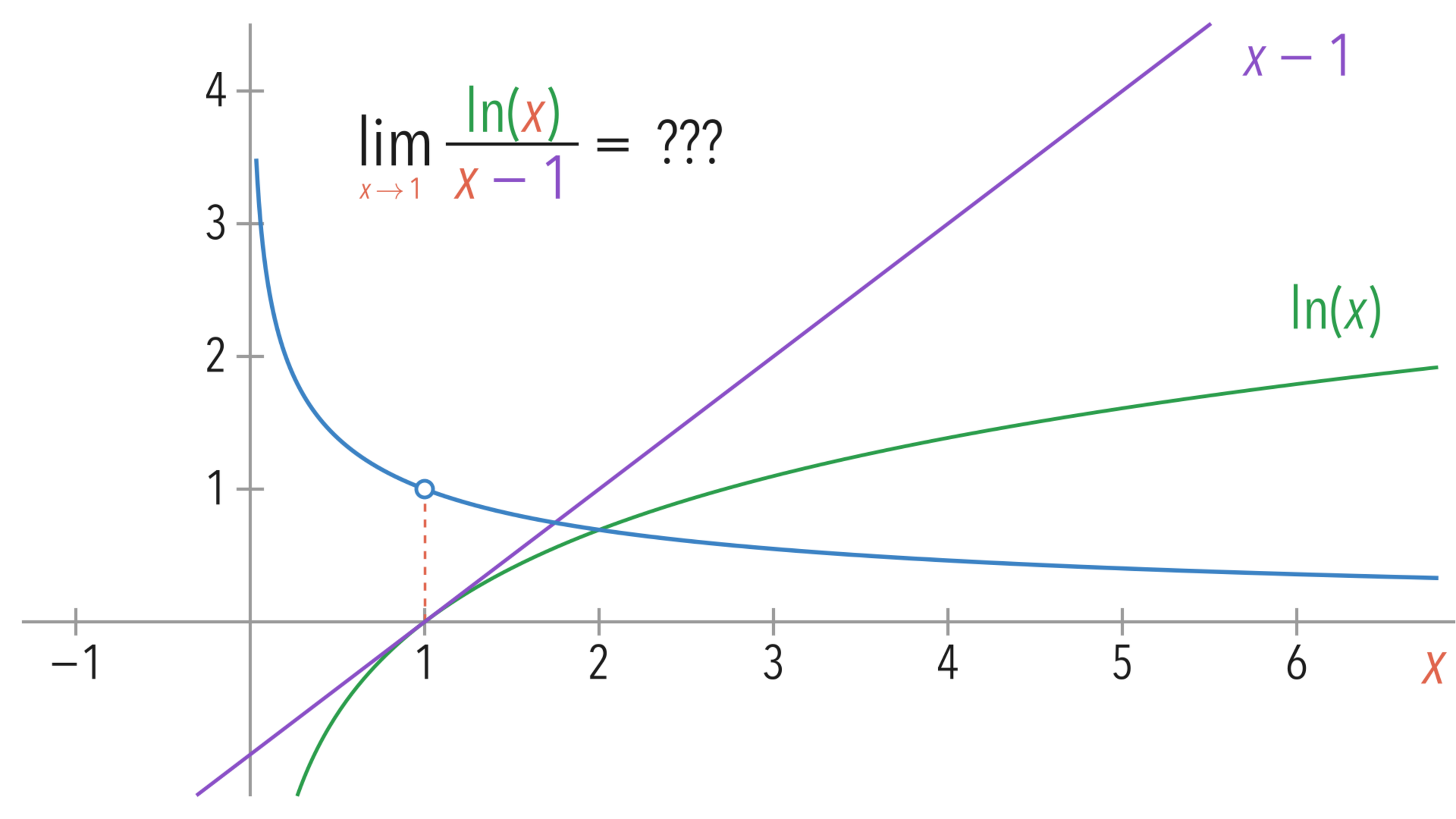


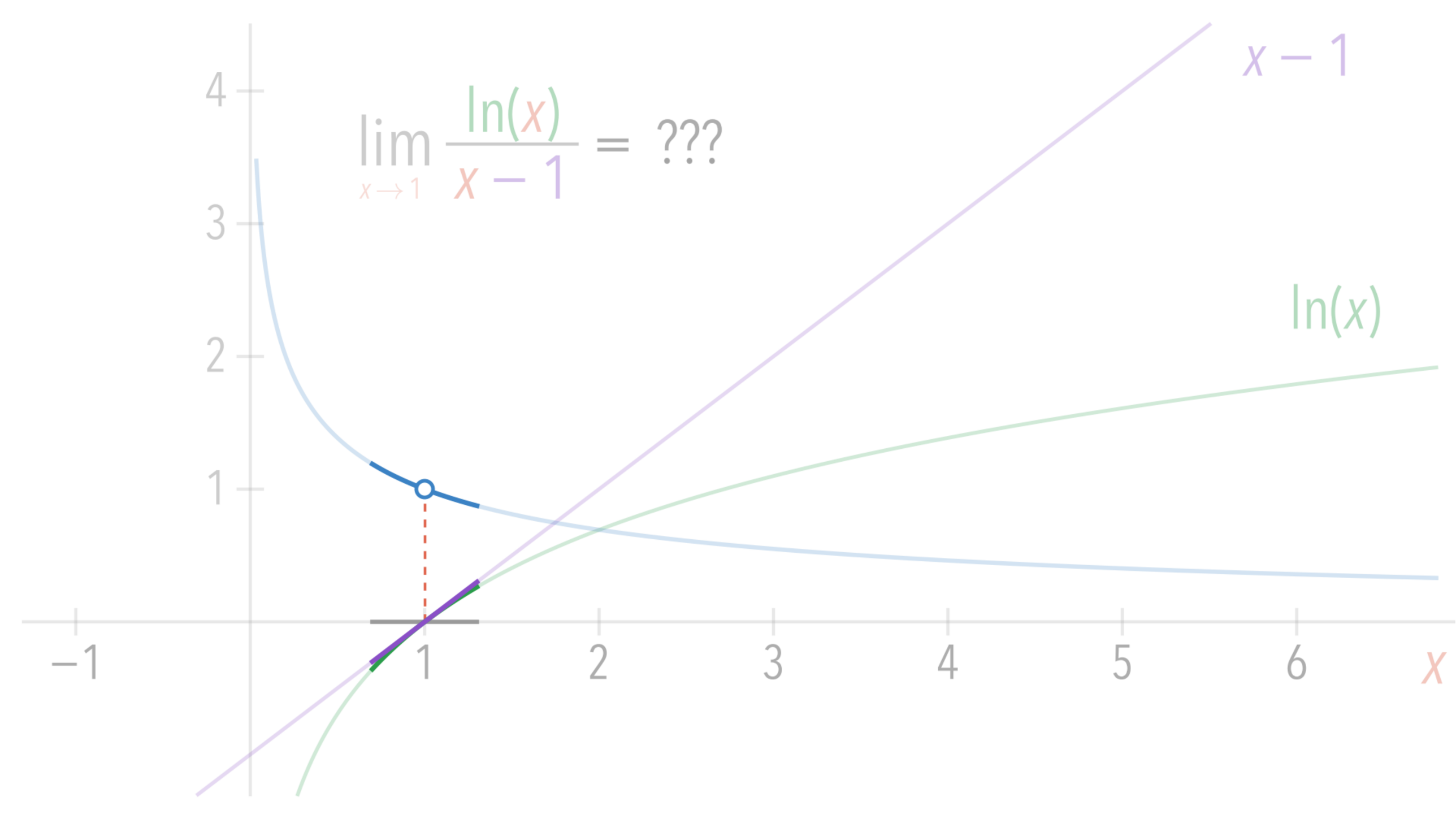
$$\lim_{x \to 1} \frac{\ln(x)}{x - 1} = \text{???}$$

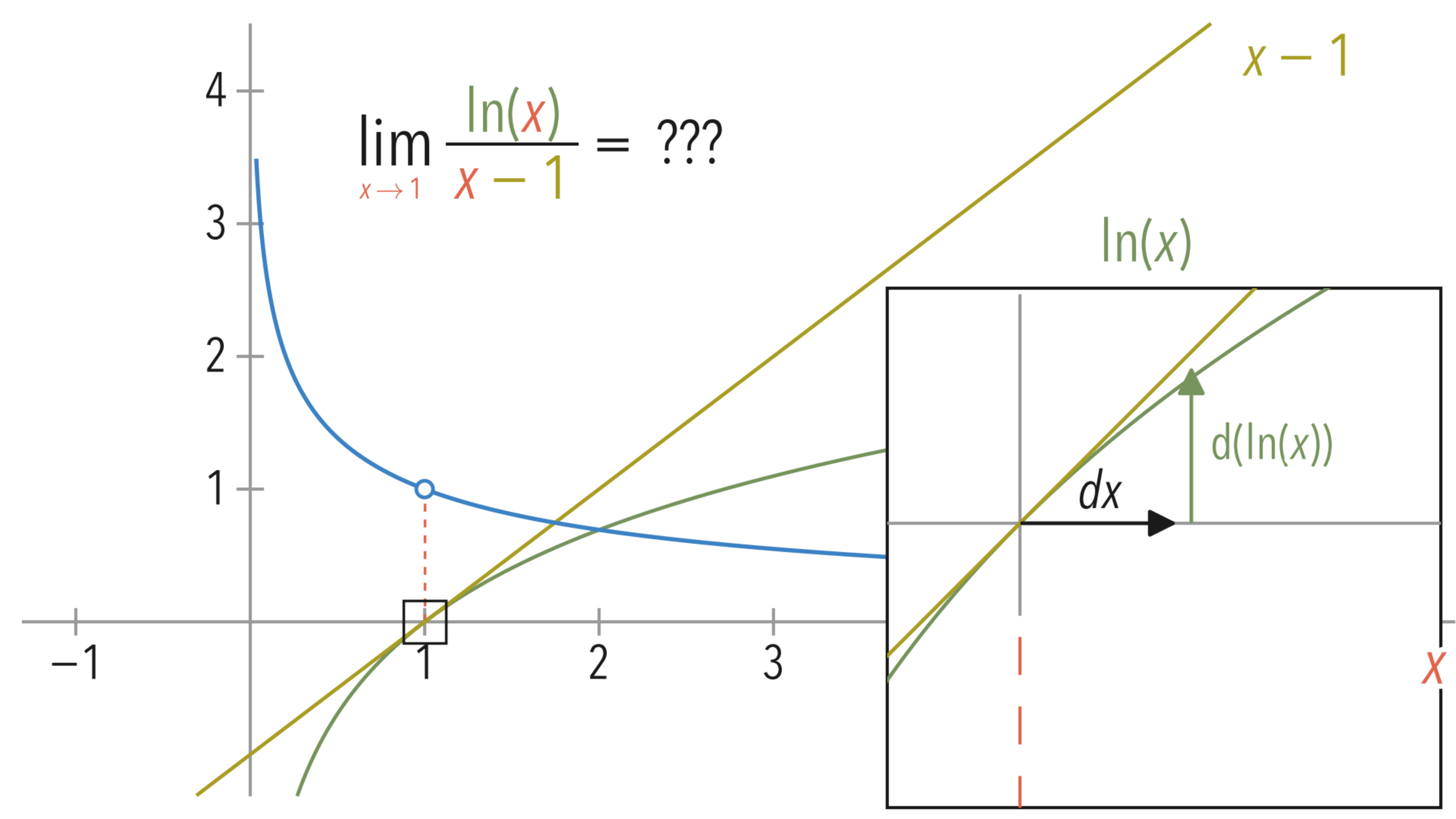
$$\frac{\ln(1.00001)}{1.00001 - 1} = 0.999995\ldots$$

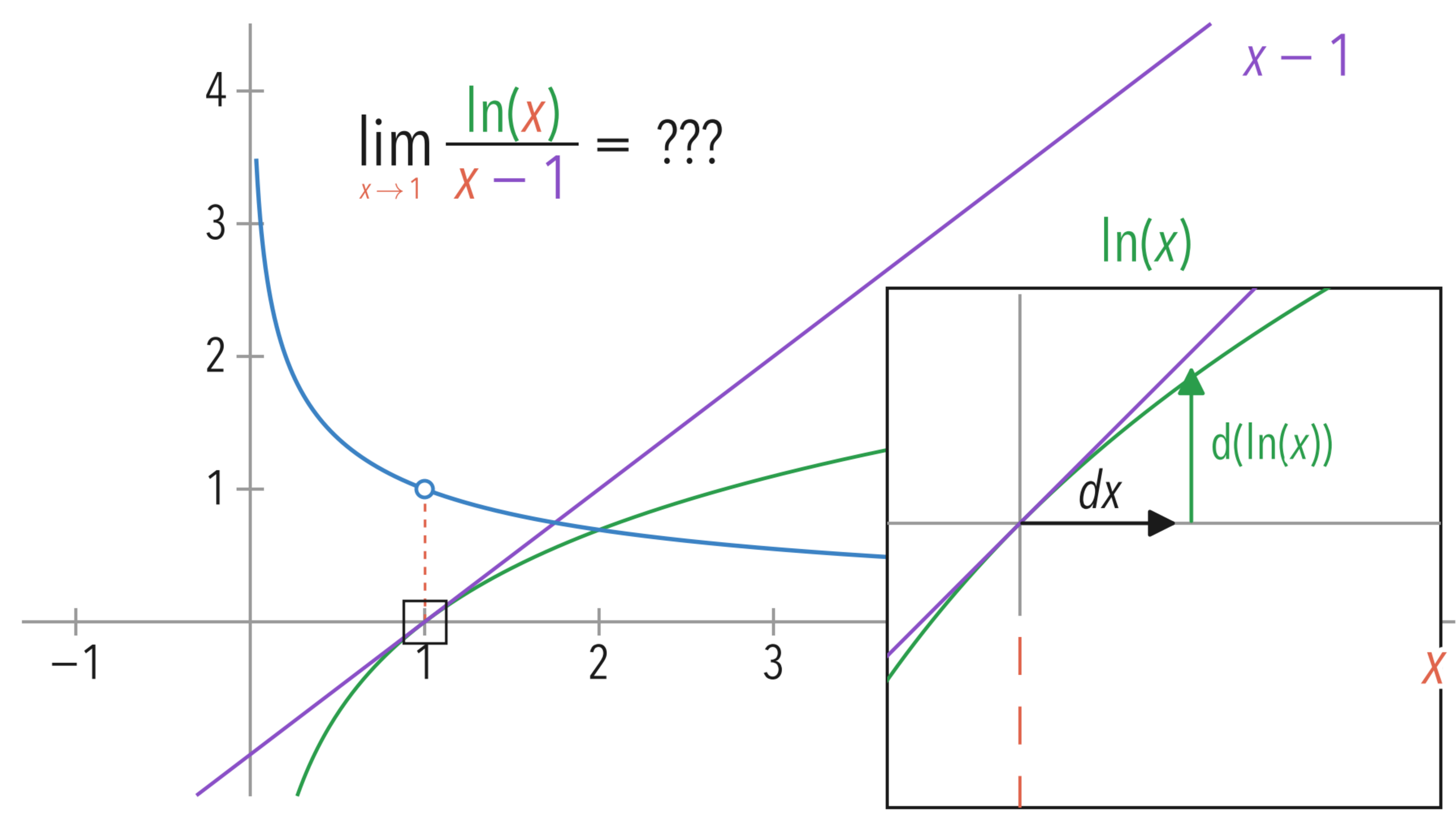
Plugging in values close to 1 works, but only gives an estimate.

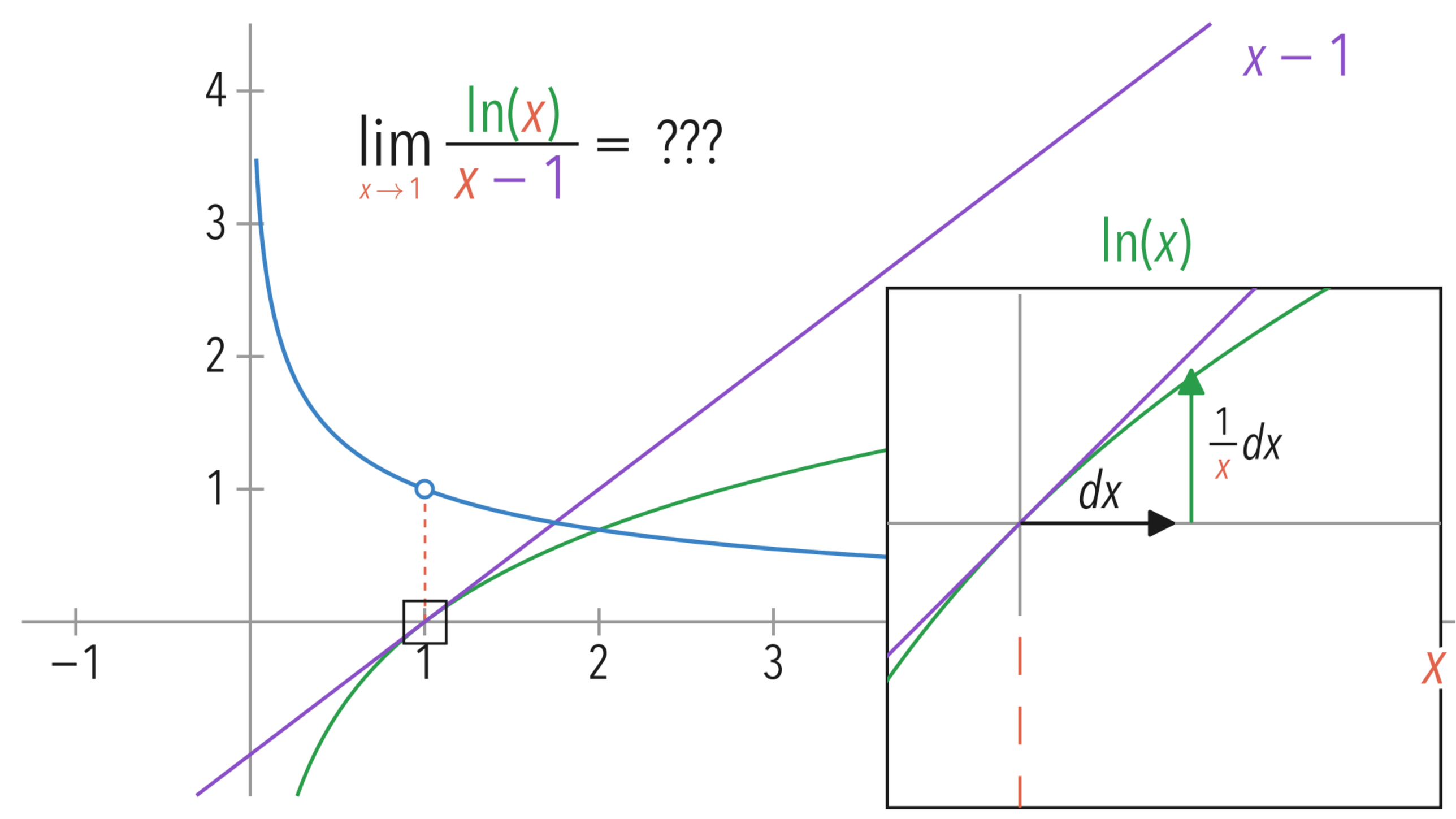

lim
h → 0
df
dx

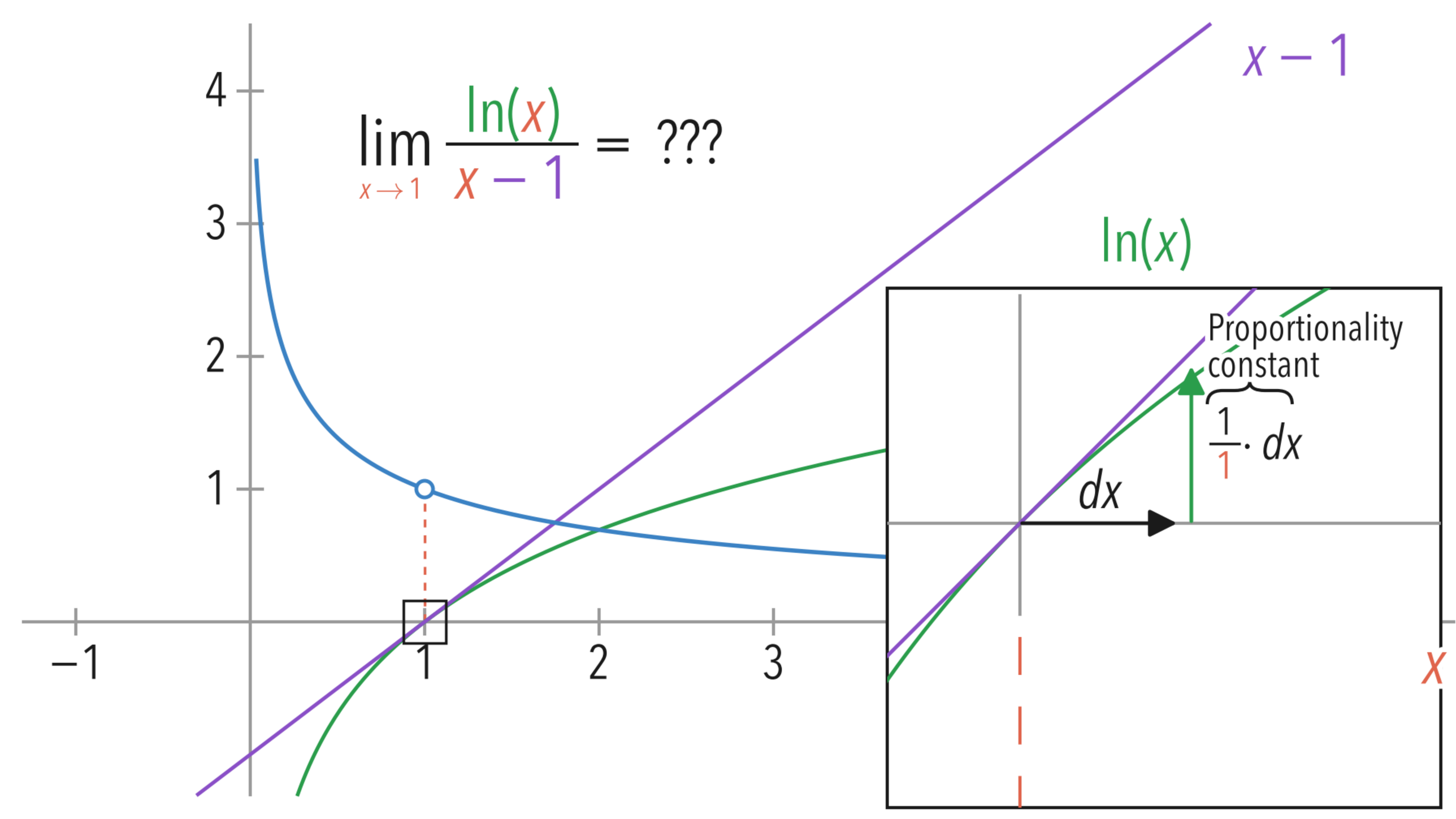

$\lim_{x \to 1} \frac{\ln(x)}{x-1} = ???$
$x - 1$
$\ln(x)$
4
3
2
1
−1
1
2
3
4
5
6
x

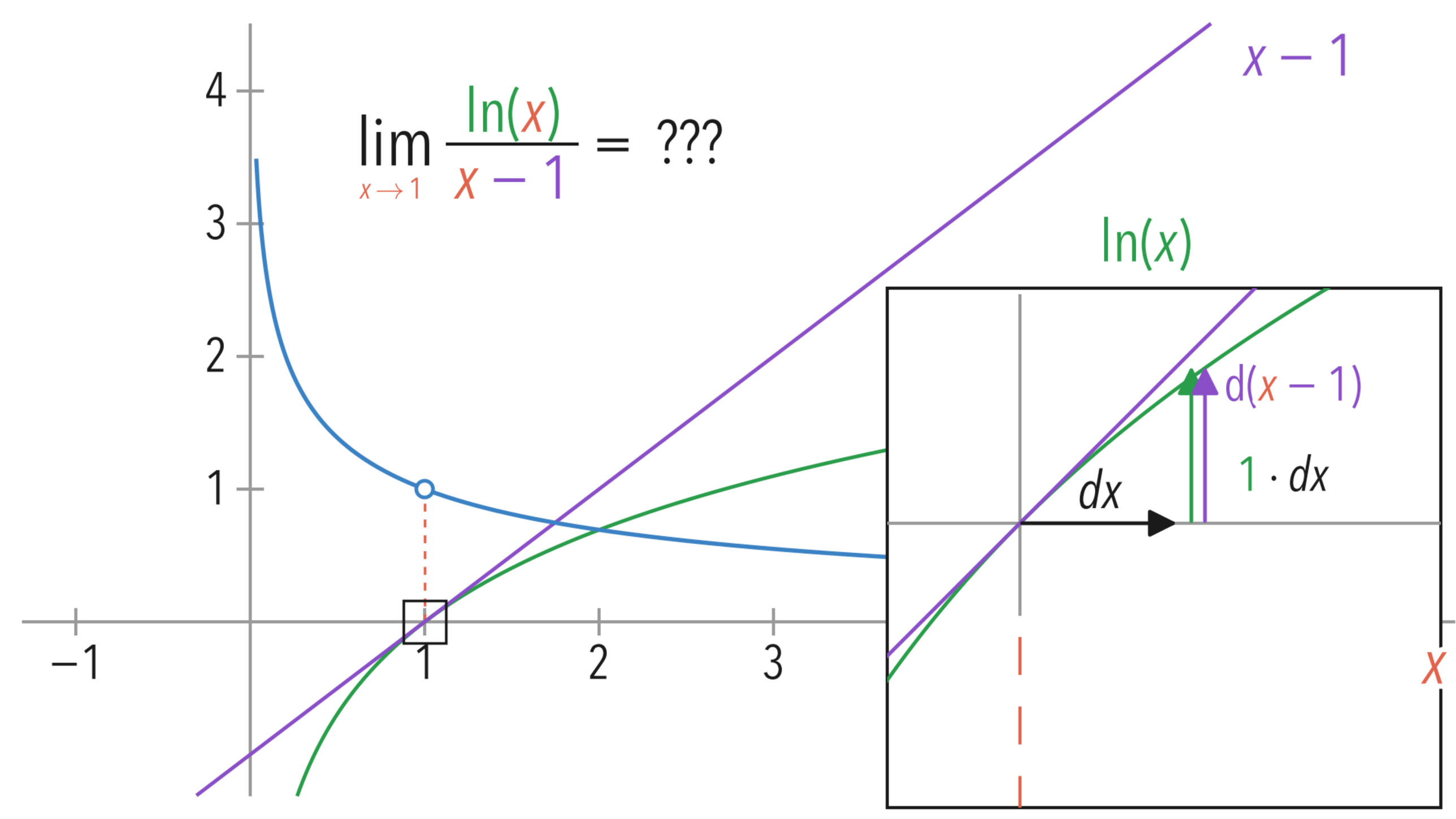

$\lim_{x \to 1} \frac{\ln(x)}{x-1} = ???$
$x-1$
$\ln(x)$
4
3
2
1
−1
1
2
3
4
5
6
x

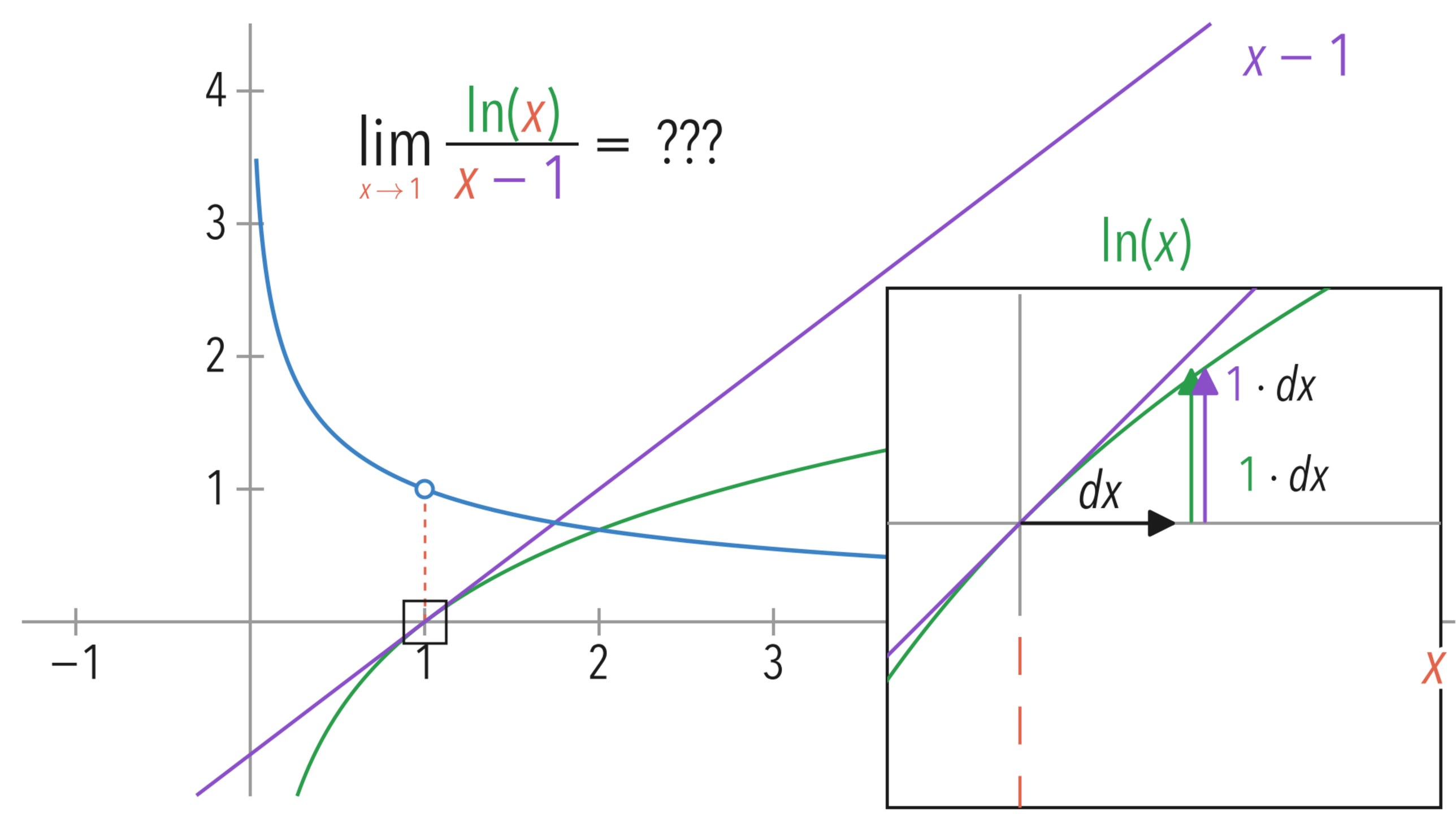

$\lim_{x \to 1} \frac{\ln(x)}{x-1} = ???$
$x - 1$
$\ln(x)$
$d(\ln(x))$
dx
x
4
3
2
1
−1
1
2
3

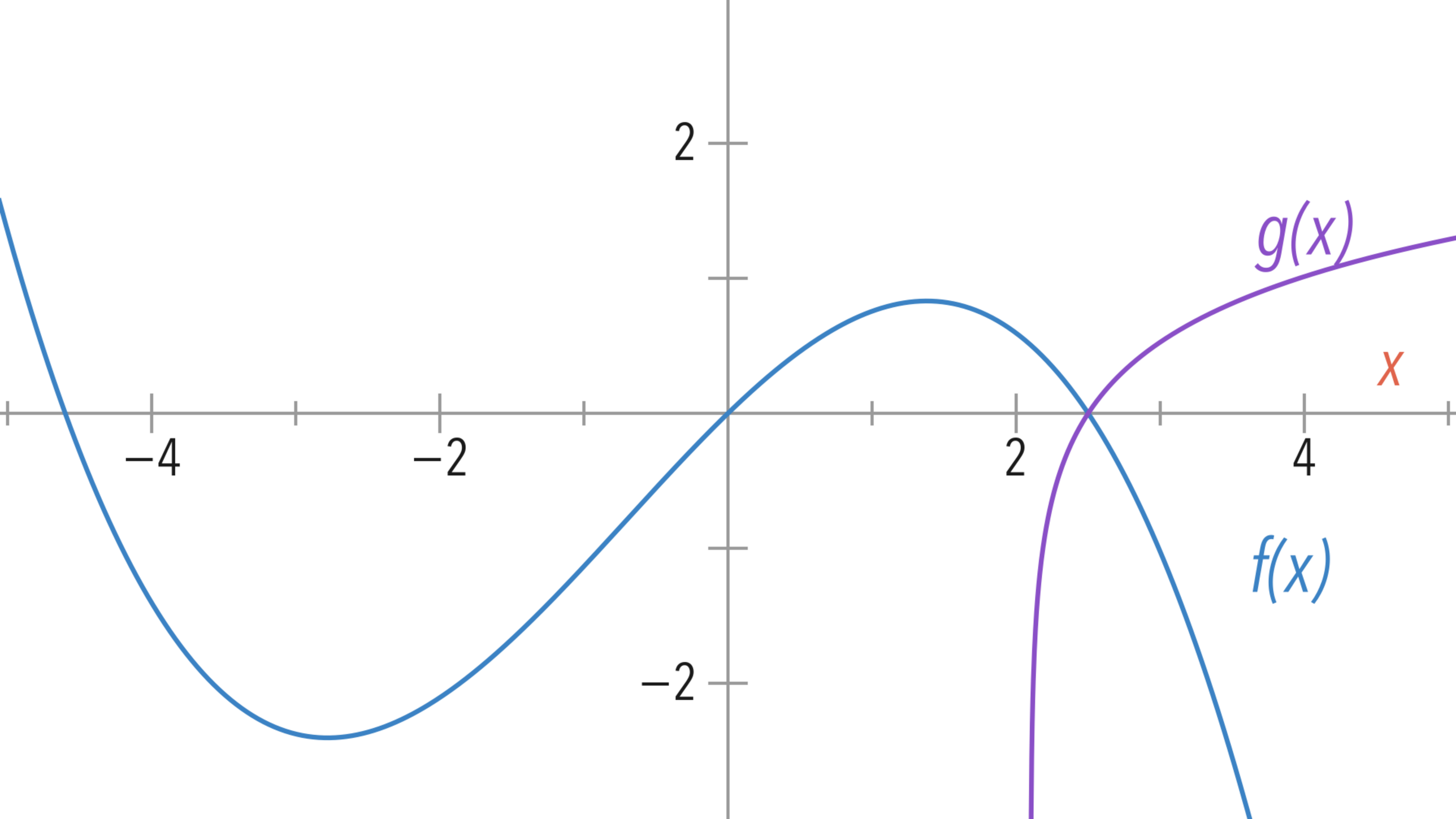

$\lim_{x \to 1} \frac{\ln(x)}{x-1} = ???$
$x - 1$
$\ln(x)$
$d(\ln(x))$
dx
x
4
3
2
1
−1
1
2
3

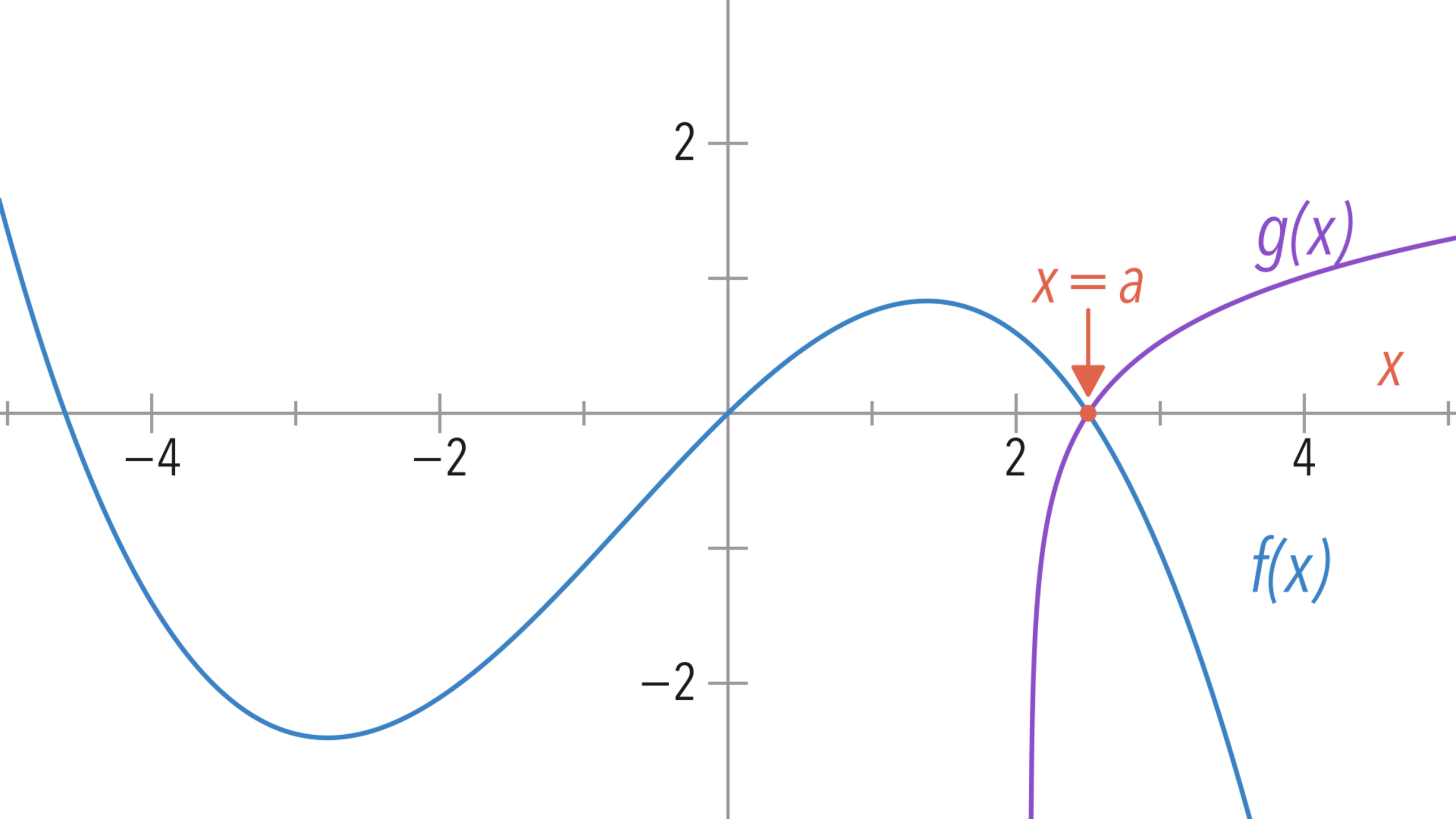

$\lim_{x \to 1} \frac{\ln(x)}{x-1} = ???$
$x - 1$
$\ln(x)$
$\frac{1}{x}dx$
dx
4
3
2
1
−1
1
2
3
x


$\lim_{x \to 1} \frac{\ln(x)}{x-1} = ???$
$x - 1$
$\ln(x)$
4
3
2
1
−1
1
2
3
Proportionality constant
$\frac{1}{1} \cdot dx$
dx
x

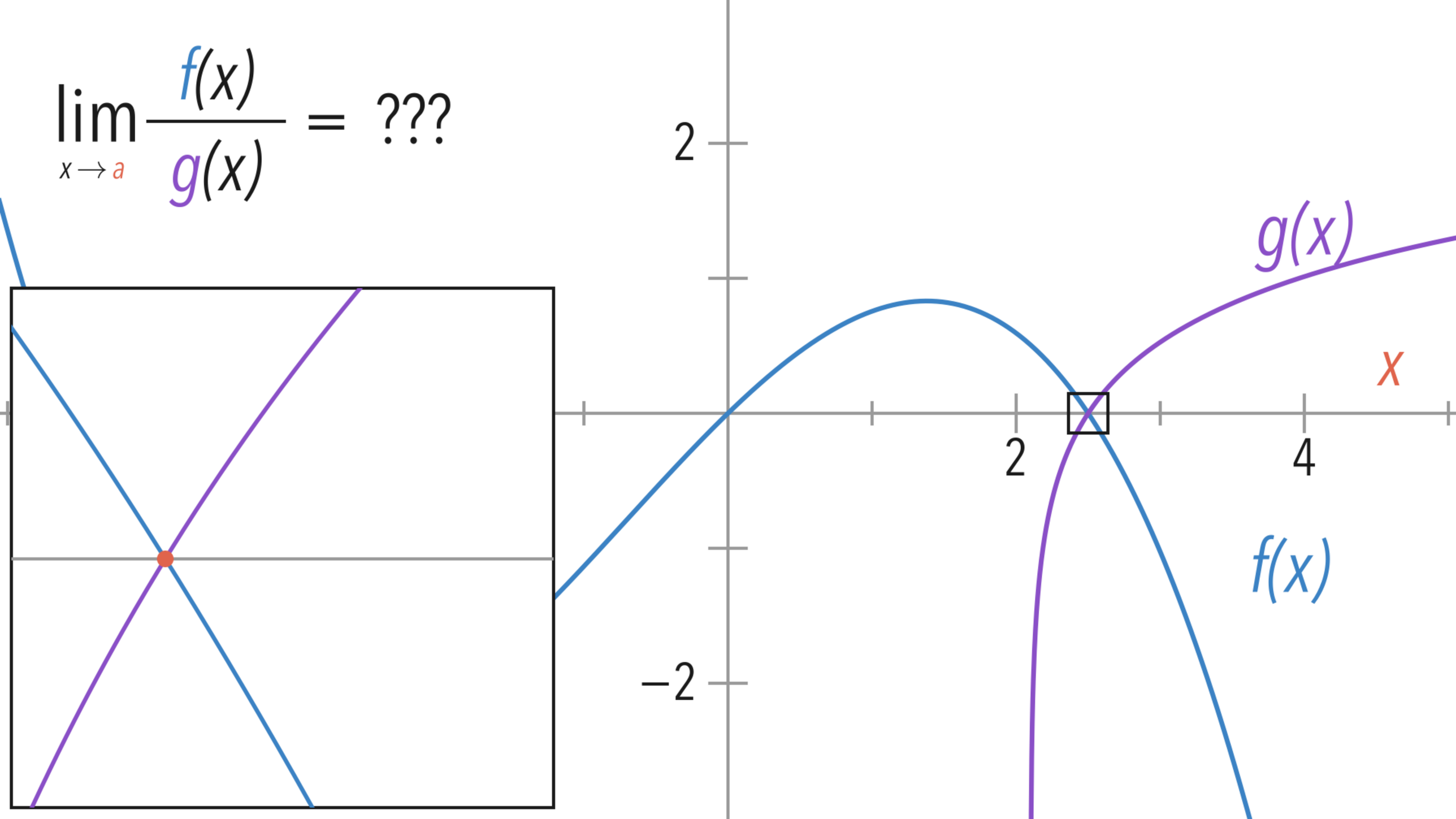

$\lim_{x \to 1} \frac{\ln(x)}{x-1} = ???$
$x - 1$
$\ln(x)$
4
3
2
1
−1
1
2
3
$d(x-1)$
$1 \cdot dx$
dx
x


$\lim_{x \to 1} \frac{\ln(x)}{x-1} = ???$
$x - 1$
$\ln(x)$
4
3
2
1
−1
1
2
3
$1 \cdot dx$
$1 \cdot dx$
dx
x

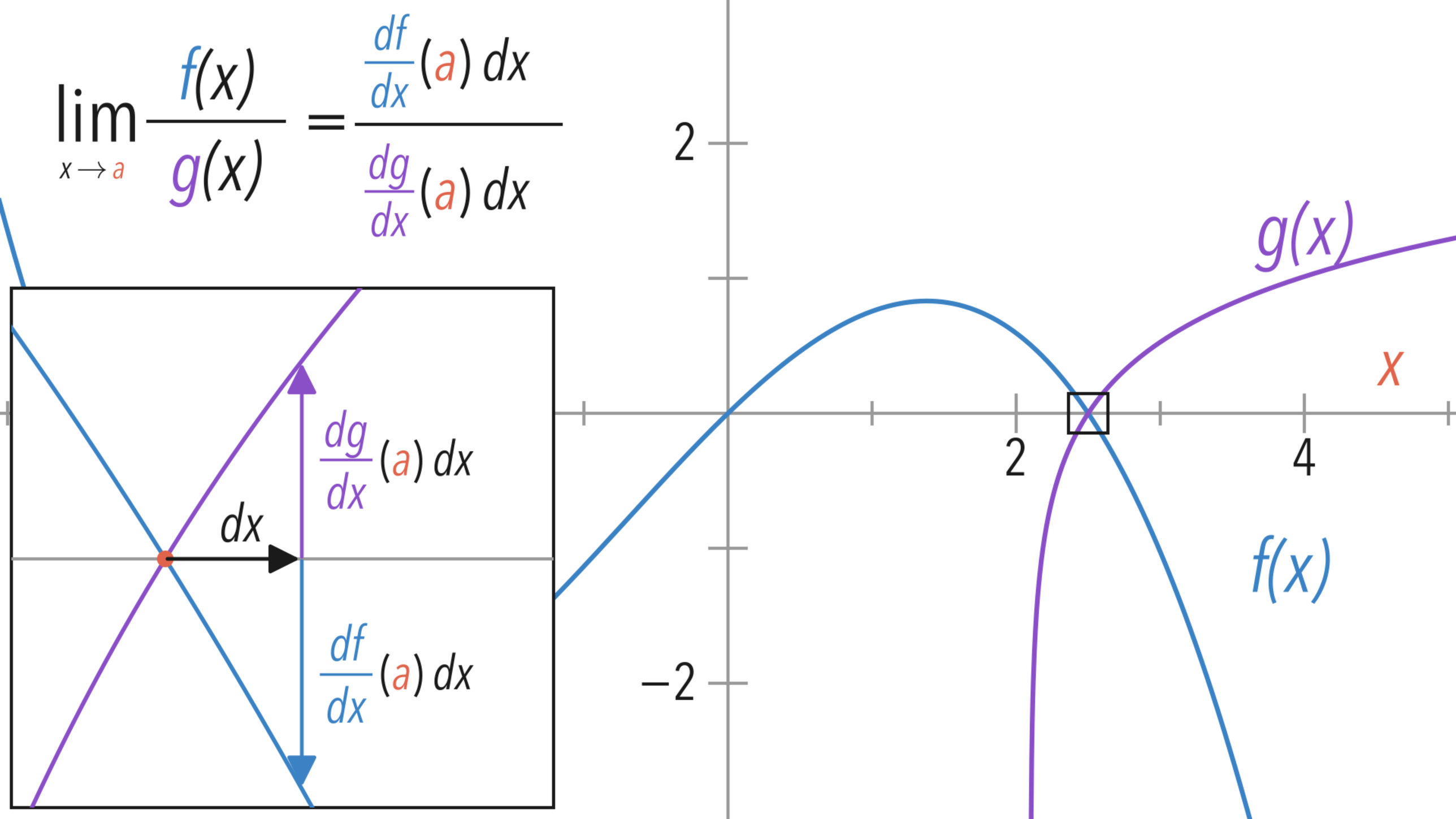

2
−4
−2
2
4
−2
g(x)
x
f(x)


2
−4
−2
2
4
−2
g(x)
x = a
x
f(x)

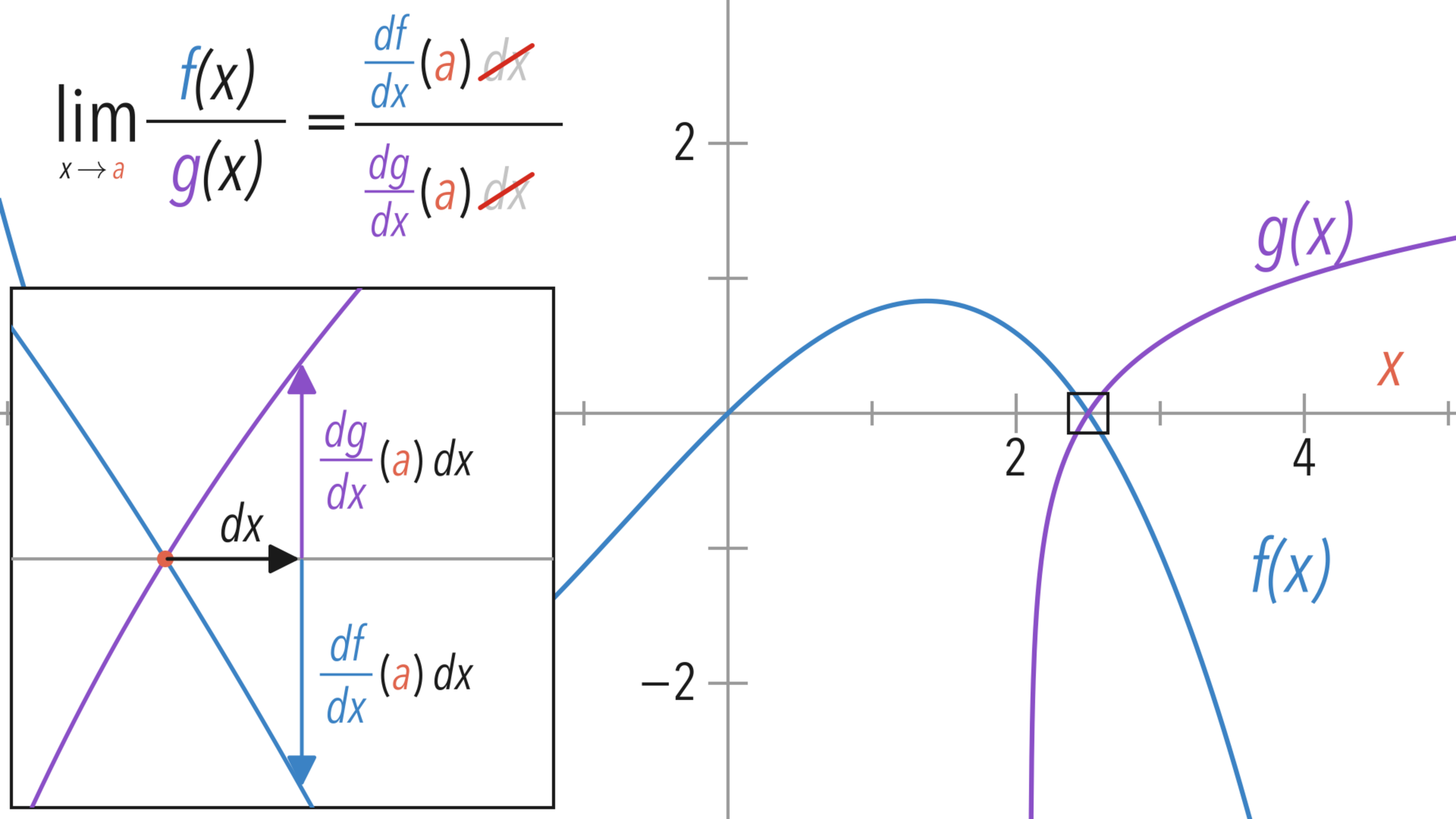
$$\lim_{x \to a} \frac{f(x)}{g(x)} = ???$$

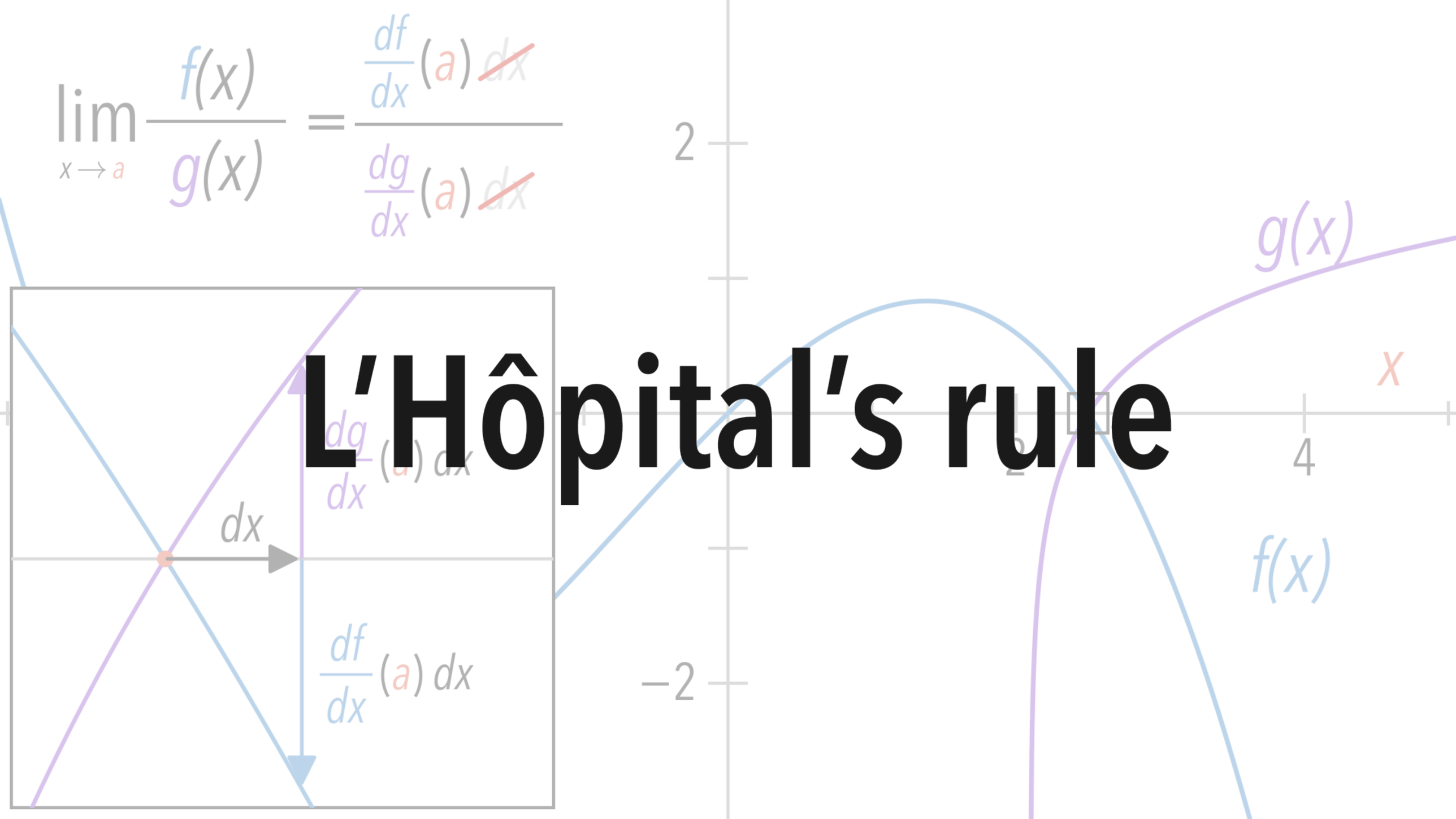


$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\frac{df}{dx}(a)\,dx}{\frac{dg}{dx}(a)\,dx}$$



$$\lim_{x \to a} \frac{f(x)}{g(x)} = \frac{\frac{df}{dx}(a)\,\cancel{dx}}{\frac{dg}{dx}(a)\,\cancel{dx}}$$



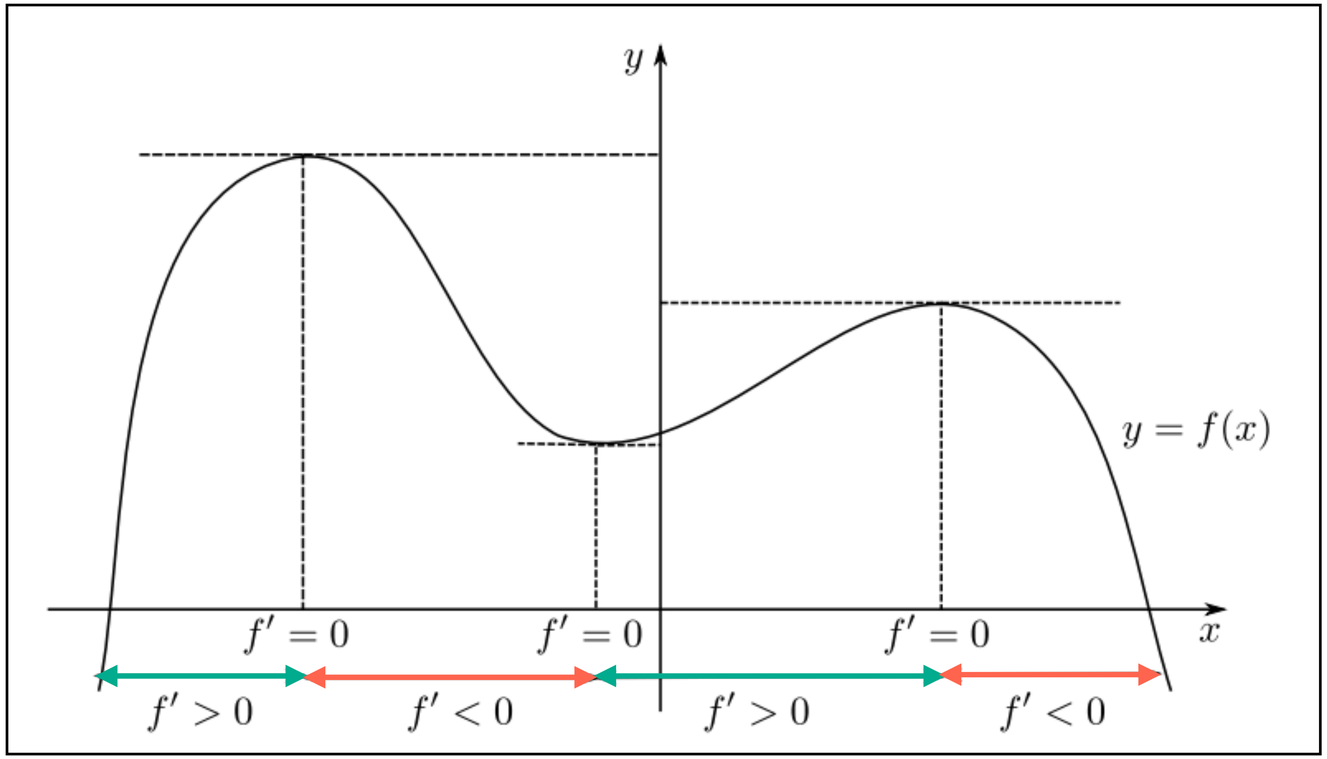


L'Hôpital's rule

maxima and minima

given some function f, where does it achieve its maximum or minimum values?

- if $f(x)$ is increasing, then $f'(x) > 0$
- if $f(x)$ is decreasing, then $f'(x) < 0$



- we have troughs and humps occur at places through which f' changes sign (where $f'(x) = 0$)
- The derivative gives us a way to look for maximum and minimum values of a function.

second derivative

- To describe how a graph **bends**, we need the second derivative.
- $f'(x)$ is itself a function of x, so we can differentiate it again:

$$\frac{d}{dx}(f'(x)) = \frac{d^2}{dx^2}f(x) = f''(x)$$

second derivative

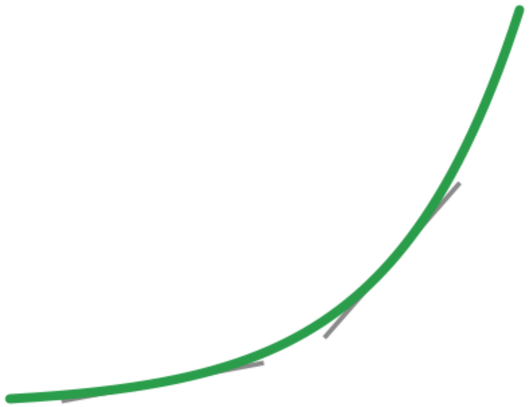
The geometric interpretation of f'':

1. If $f''(x) > 0$ on an interval, the slope $f'(x)$ is increasing: f is **concave up**

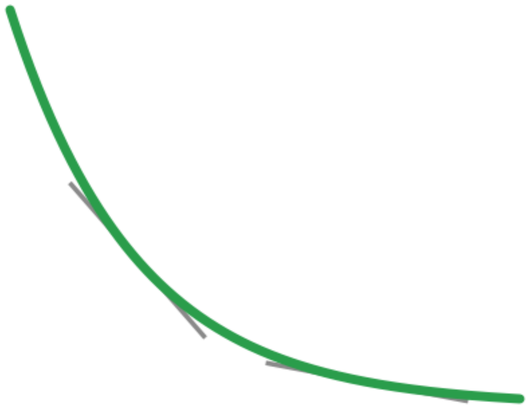
Second derivative test: if $f'(c) = 0$ and $f''(c) > 0$

$\Longrightarrow$ f has a **local minimum** at c

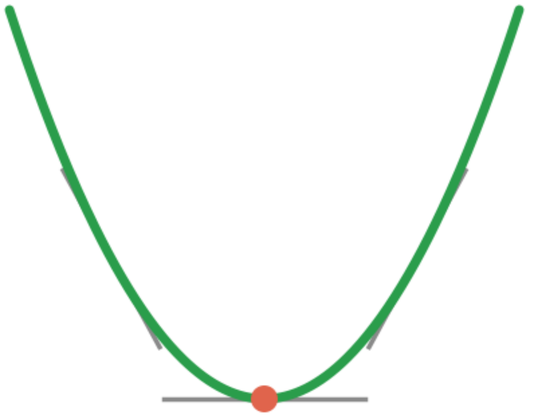
possible shapes:



increasing



decreasing



$f'(c) = 0$

second derivative

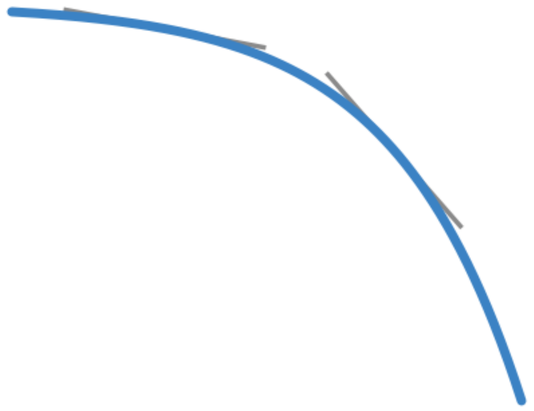
The geometric interpretation of f'':

2. If $f''(x) < 0$ on an interval, the slope $f'(x)$ is decreasing: f is **concave down**

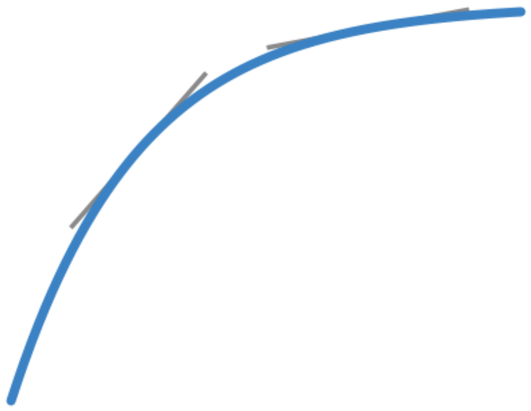
Second derivative test: if $f'(c) = 0$ and $f''(c) < 0$

$\Longrightarrow$ f has a **local maximum** at c

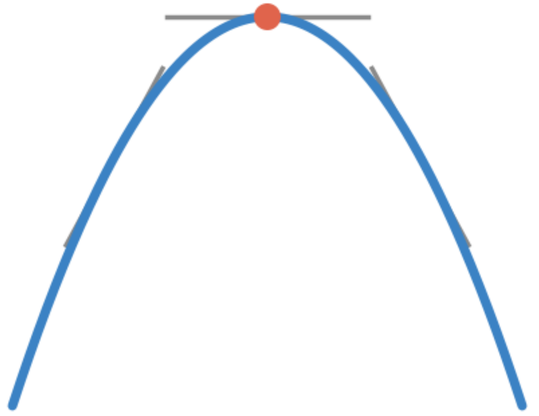
possible shapes:



decreasing



increasing



$f'(c) = 0$

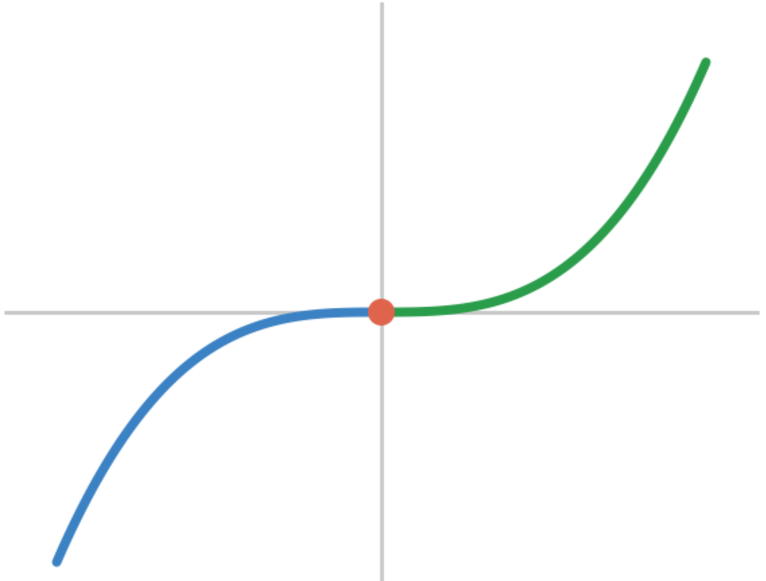
second derivative

The geometric interpretation of f'':

3. If $f''(x)$ changes sign at c, the graph switches between concave up and down
   $\Longrightarrow$ f has an **inflection point** at c (then $f''(c) = 0$)

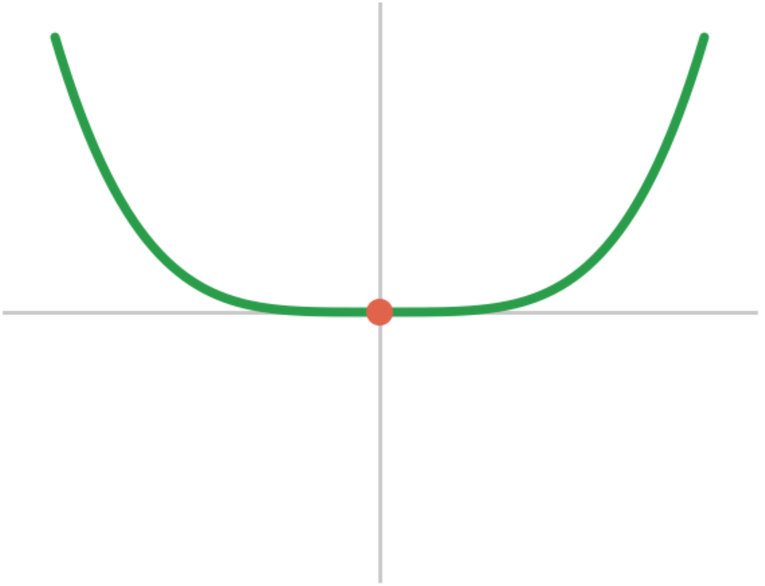
But $f''(c) = 0$ alone is not enough: the second derivative test is **inconclusive**.

$f(x) = x^3$



f'' changes sign at 0: inflection point

$f(x) = x^4$



$f''(0) = 0$, but no sign change: minimum